

Monash Qiskit Fall Fest 2025



Quantum Australia

IBM Quantum



Acknowledgement of Country

*Our event is being held on the lands of the Bunurong people
and we wish to acknowledge them as Traditional Owners.*

*We would also like to pay my respects to their Elders, past
and present, and Aboriginal Elders of other communities
who may be here today.*

The Aims of This Event

- Kick start your quantum journey
- Be able to explore quantum independently
- Get to know and introduce the quantum ecosystem at Monash



Agenda

10:00 – 10:10

Welcome / Opening Remarks

10:10 – 10:20

Introduction to Quantum Australia — Olivia Dennis

10:20 – 11:20

Quantum & Qiskit 101

Intro to qubits, gates, measurement, entanglement, and how Qiskit is used in practice

11:20 – 12:15

Qiskit Fundamentals Lab

Learn the basics of coding for a quantum computer

Hands-on notebook session: *run your first circuits on a real quantum computer via IBM Quantum*

12:15 – 1:00

Lunch Break

1:00 – 1:45

Guest Seminar — Dr Anna Phan

Asia Pacific Lead for IBM Quantum Innovation Centres

1:50 – 2:30

Guest Lecture — Nikai Jagganath

QKD and PQC protocols

2:30 – 3:00

Showcase: Quantum at Monash

Short presentations and demos of current research highlights

3:00 – 3:30

Mini Competition — *Test your understanding and **win a prize!***

House Keeping



Fill in the **Check-in form** to receive certificate of participation directly after sessions.

The materials for the workshops are in the Github repository.

Pictures being taken.



Monash Qiskit
29th October 2025

Acknowledgement of Country

Quantum Australia acknowledges the Aboriginal and Torres Strait Islander Peoples throughout Australia and their continuing connection to land, water, culture and community. We pay our respects to the Elders both past and present.



Quantum Australia: Who We Are

We're the national centre for quantum growth across industry, academia and the startup ecosystem.

Growing the ecosystem

We're building a national network to accelerate quantum adoption through events, workshops, and strategic industry engagement.

Transforming research

We connect industry and researchers to transform quantum breakthroughs into commercial solutions that drive economic growth.

Supporting start-ups

We provide mentorship, funding pathways, and industry connections to help innovators launch, scale, and succeed.

The Victorian Node

Joe Dodd
Partnership
Development Lead



Daniel Hunter
Partnership
Development
Manager



Adrian Gertler
Partnership
Development
Manager



Olivia Dennis
Ecosystem
Development
Coordinator



Partnership Development

- **Industry–Research Connection** – Match industry needs with research capabilities through workshops, partnerships, internships, and funding opportunities.
- **Showcase Success** – Highlight impactful collaborations and promote quantum innovation through media, events, and national capability map.
- **Unlock Funding** – Help partners navigate grants, connect with investors, and access trade and investment support.
- **Key Industry Sectors include** – Defence & Space, Medical, Energy, Finance, Agriculture, Resources, Transport, Advanced Technology



Ecosystem Development

- **Educate & Connect** – Build quantum awareness across industry and government, and strengthen collaboration with researchers.
- **Events & Engagement** – Fund local Node events, host targeted industry and diversity initiatives including Quantum Meets, Quantum Connect, and provide speakers and delegations for key conferences and other external targeted events.
- **Support Changemakers** – Advise of and offer (where applicable) travel funding and opportunities for researchers, intrapreneurs, and underrepresented groups to engage in the ecosystem.
- **Boost Visibility** – Promote quantum activities and partnerships through marketing, media, and a strong presence at key events.



Contact Us

Olivia Dennis, Ecosystem Development Coordinator

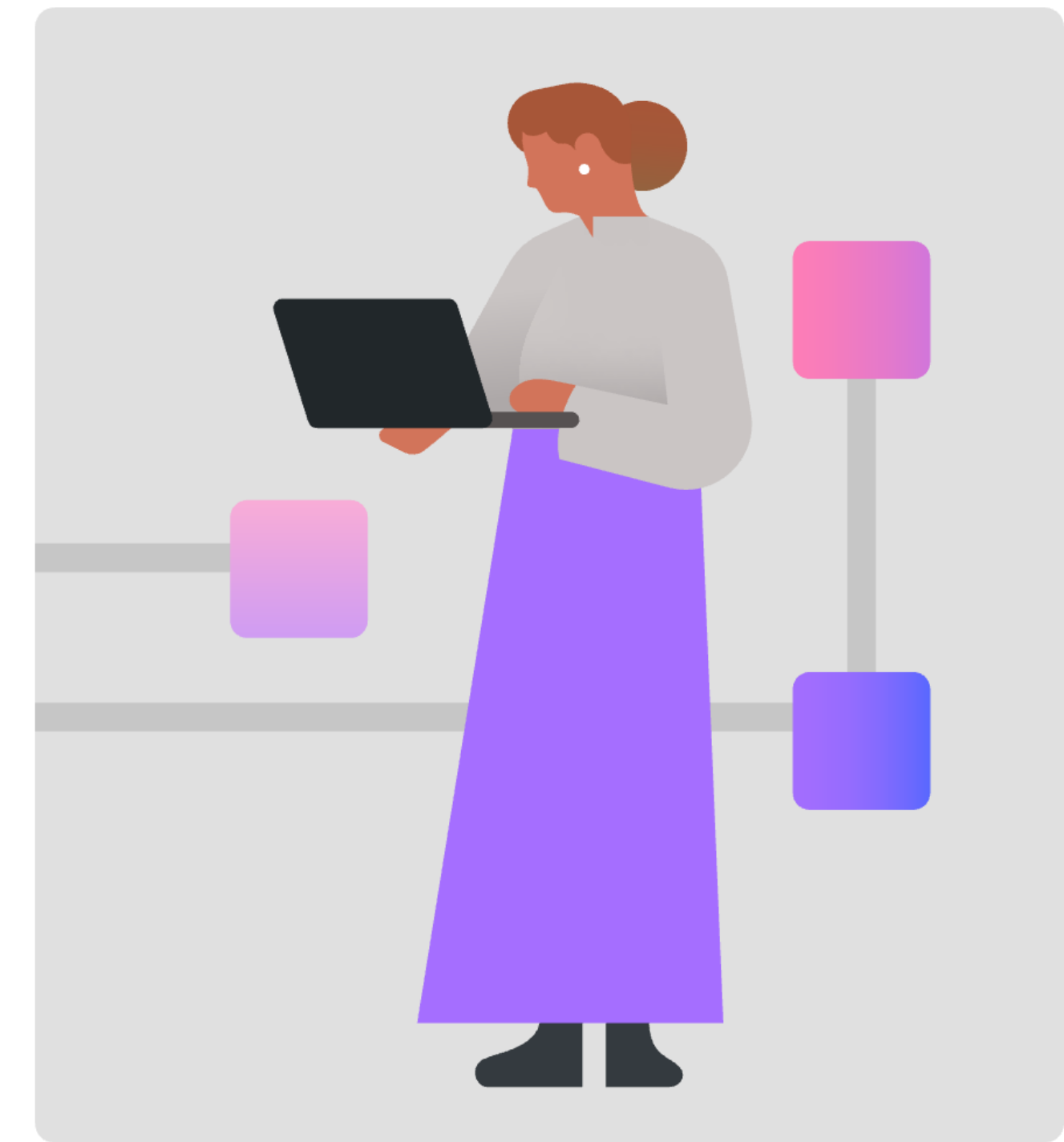
E: olivia.dennis@quantum-australia.com

Quantum Australia

Introduction to IBM Quantum and Qiskit

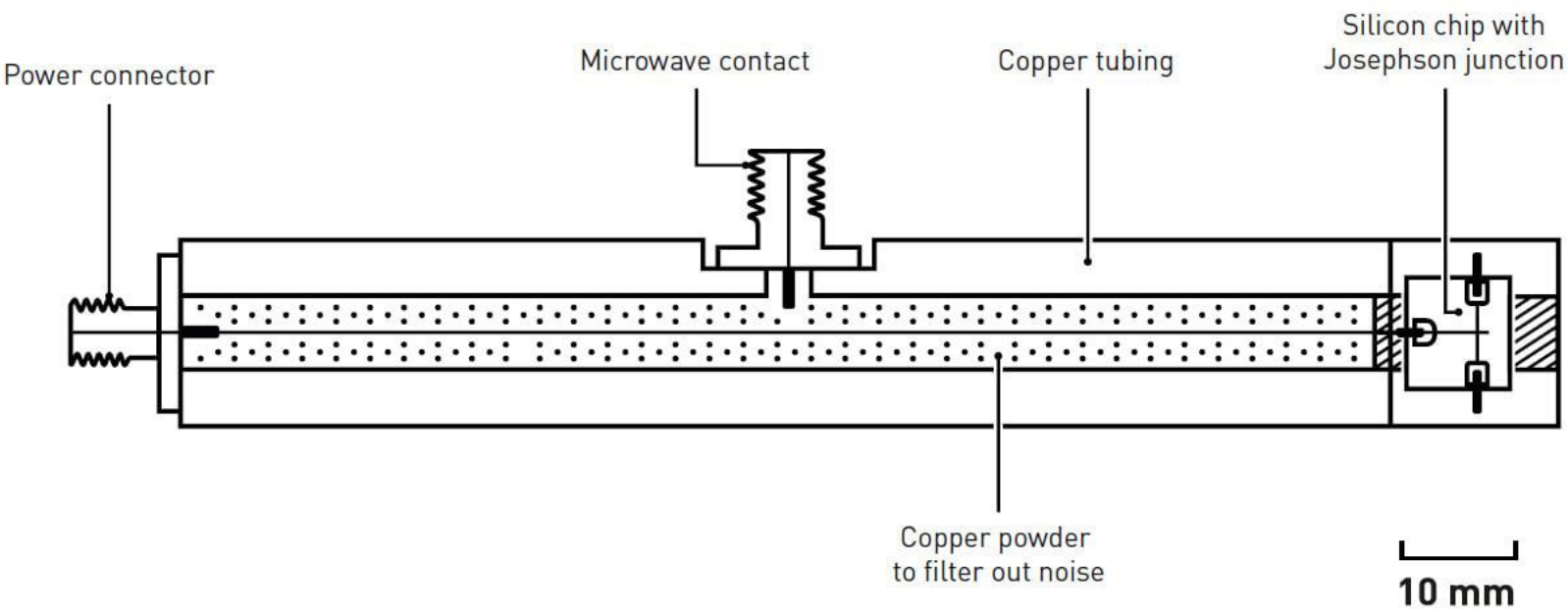
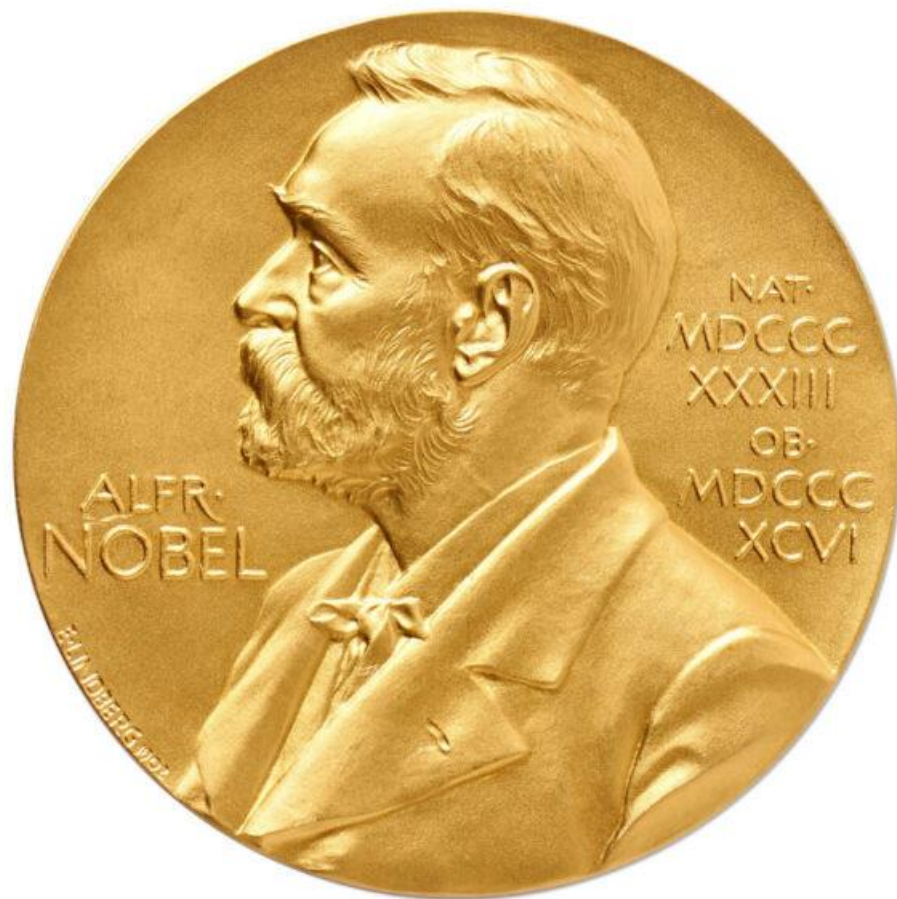
Jack Bray

Qiskit Advocate / Student at Monash University



Exciting news

Quantum Computing is on the rise!



John Clarke, Michel Devoret and John Martinis constructed an experiment using a superconducting electrical circuit. The chip that held this circuit was about a centimetre in size. Previously, tunnelling and energy quantisation had been studied in systems that had just a few particles; here, these phenomena appeared in a quantum mechanical system with billions of Cooper pairs that filled the entire superconductor on the chip. In this way, the experiment took quantum mechanical effects from a microscopic scale to a macroscopic one.

IBM Quantum mission

Bring useful
quantum
computing to
the world



What is quantum computing?

What is quantum computing?

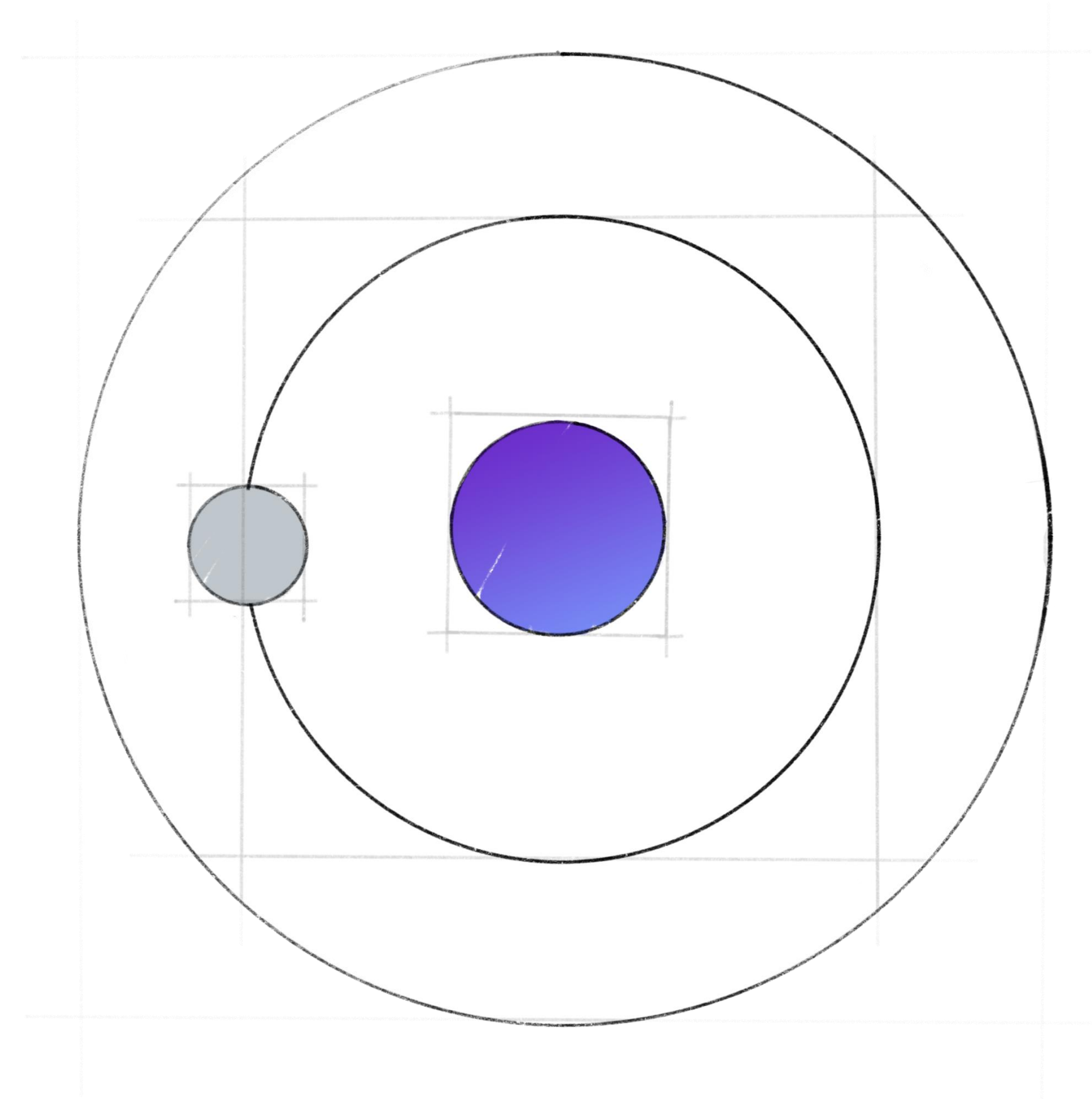
Quantum mechanics: the science of matter and energy on the atomic and subatomic levels.

Quantum computers:

- harness the laws of quantum mechanics to solve complex mathematical problems
- store information in qubits (quantum bits) as 0, 1, or a superposition of 0 and 1, and
- typically provide ranges of possible answers due to their nondeterministic (probabilistic) nature.

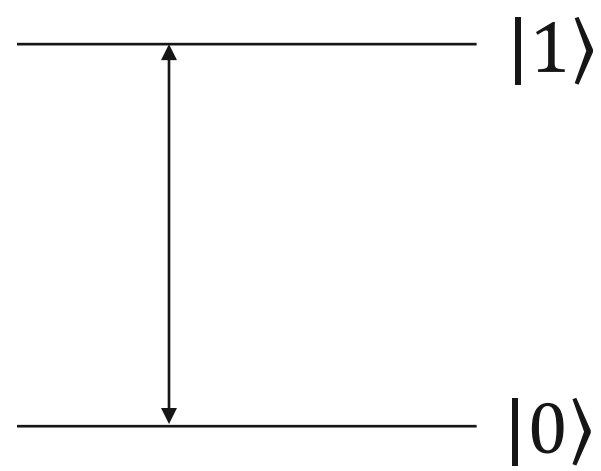
Classical computers (e.g. traditional computers like our laptops and phones):

- store information in bits with a discrete number of possible states, 0 or 1,
- process data logically and sequentially, and
- provide singular answers.

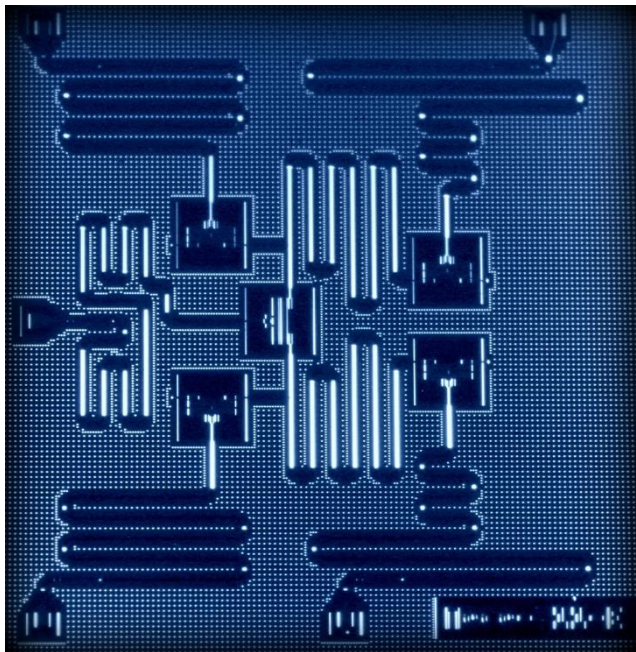


Qubit technology

Theoretically, any two-level quantum system can be used to make a qubit:

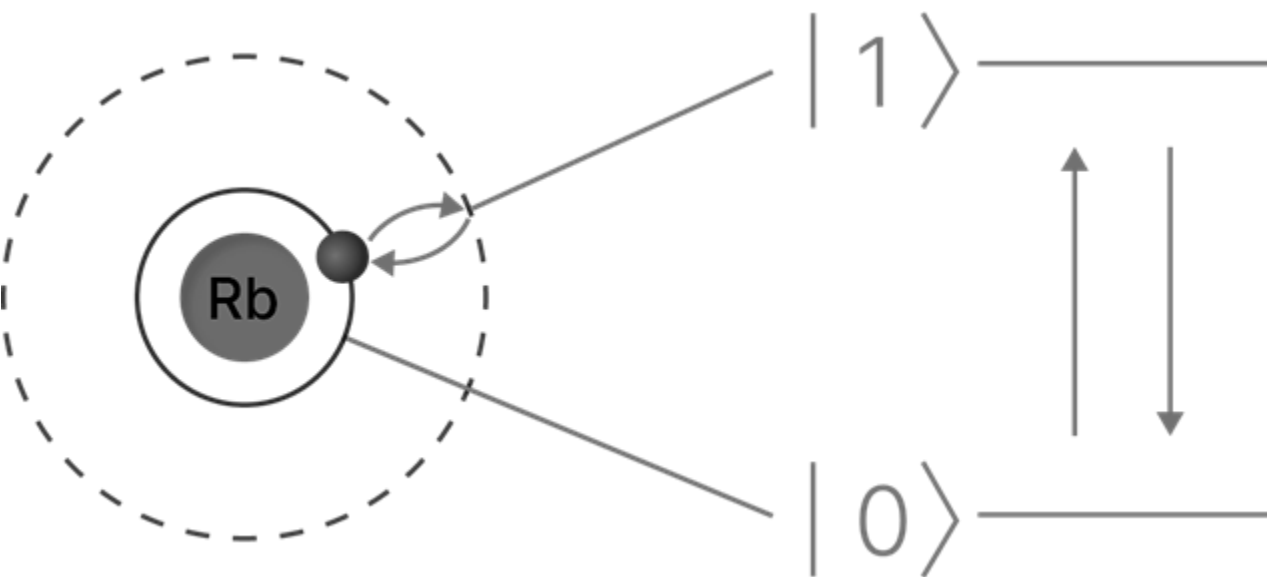


Superconducting qubits



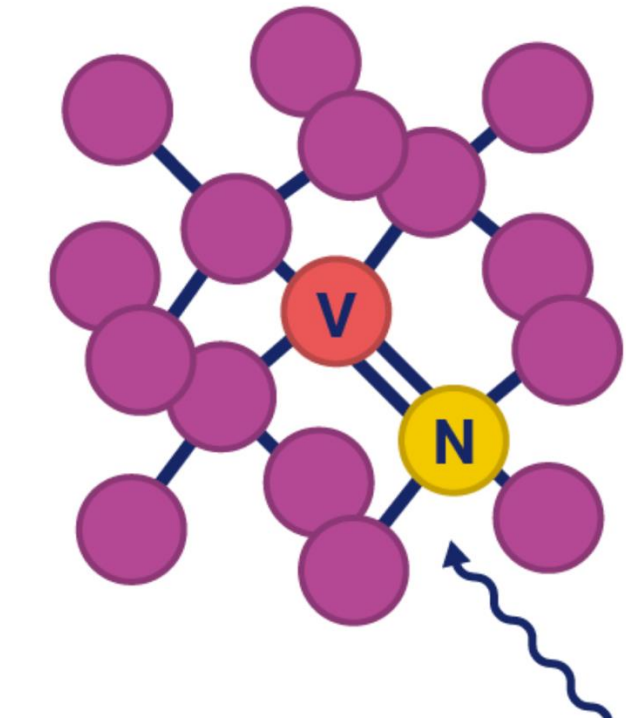
Source: IBM Quantum

Neutral atoms



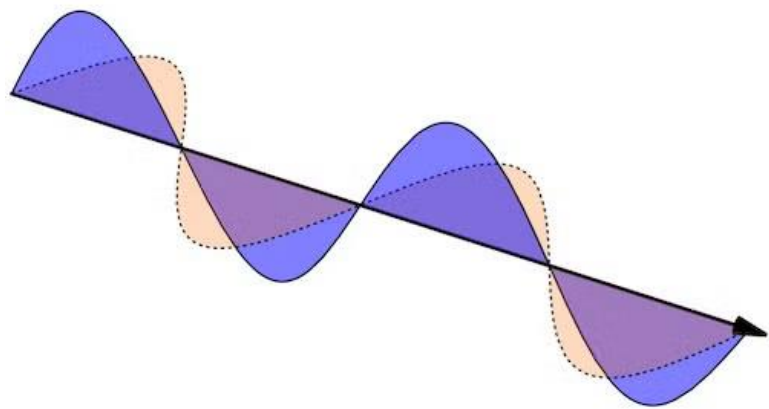
Source: QuEra

Nitrogen vacancy center qubits



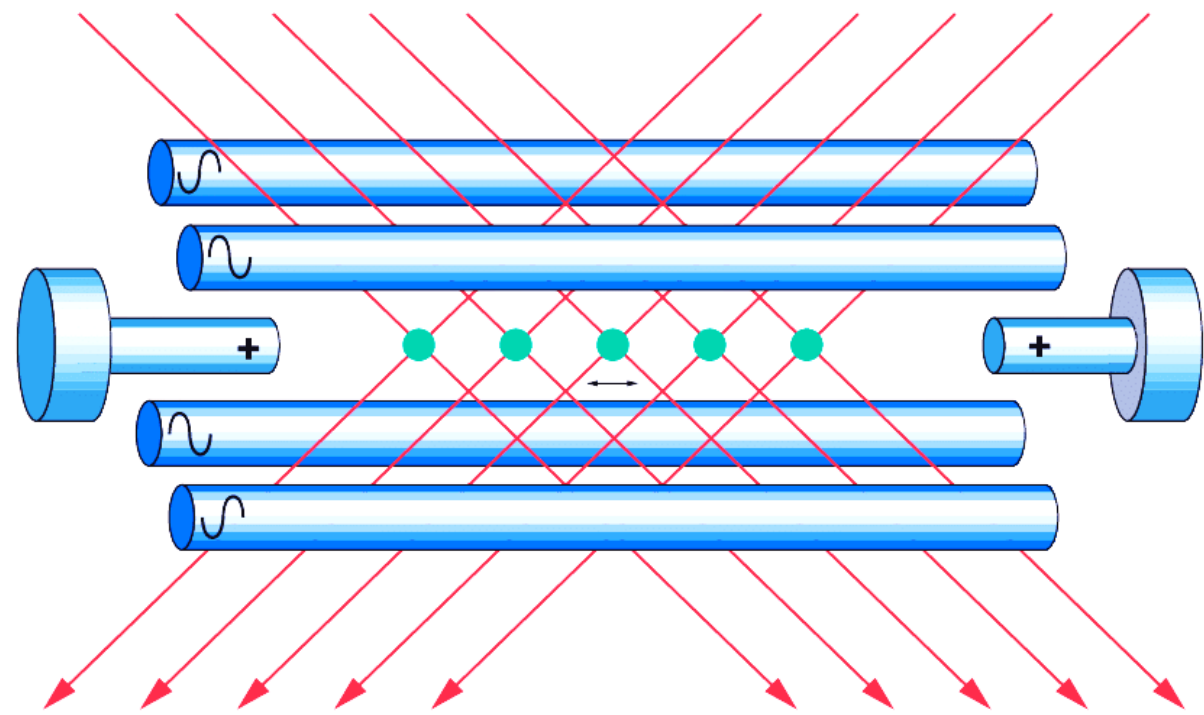
Source: IDTechEx

Photons



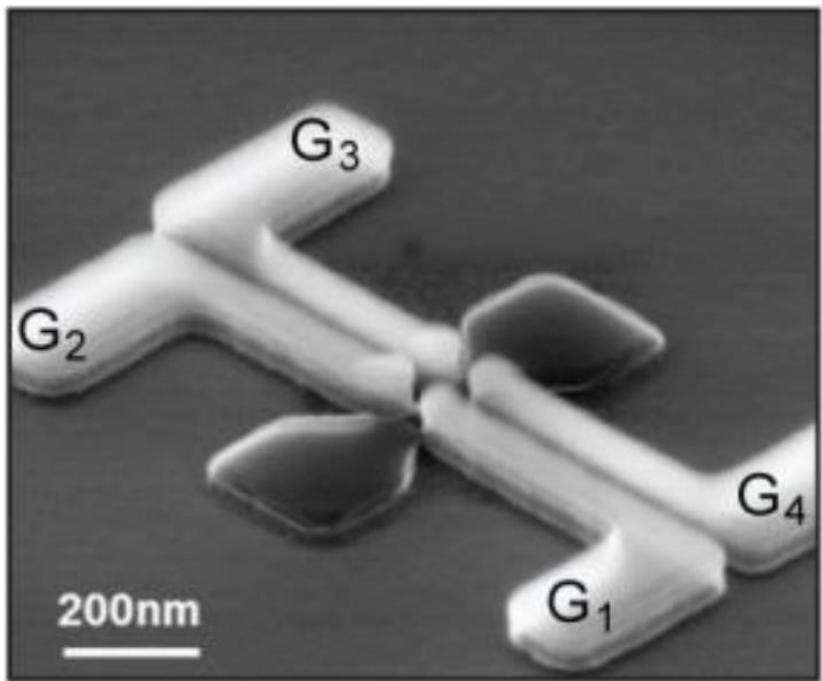
Source: University of Waterloo

Trapped ion qubits



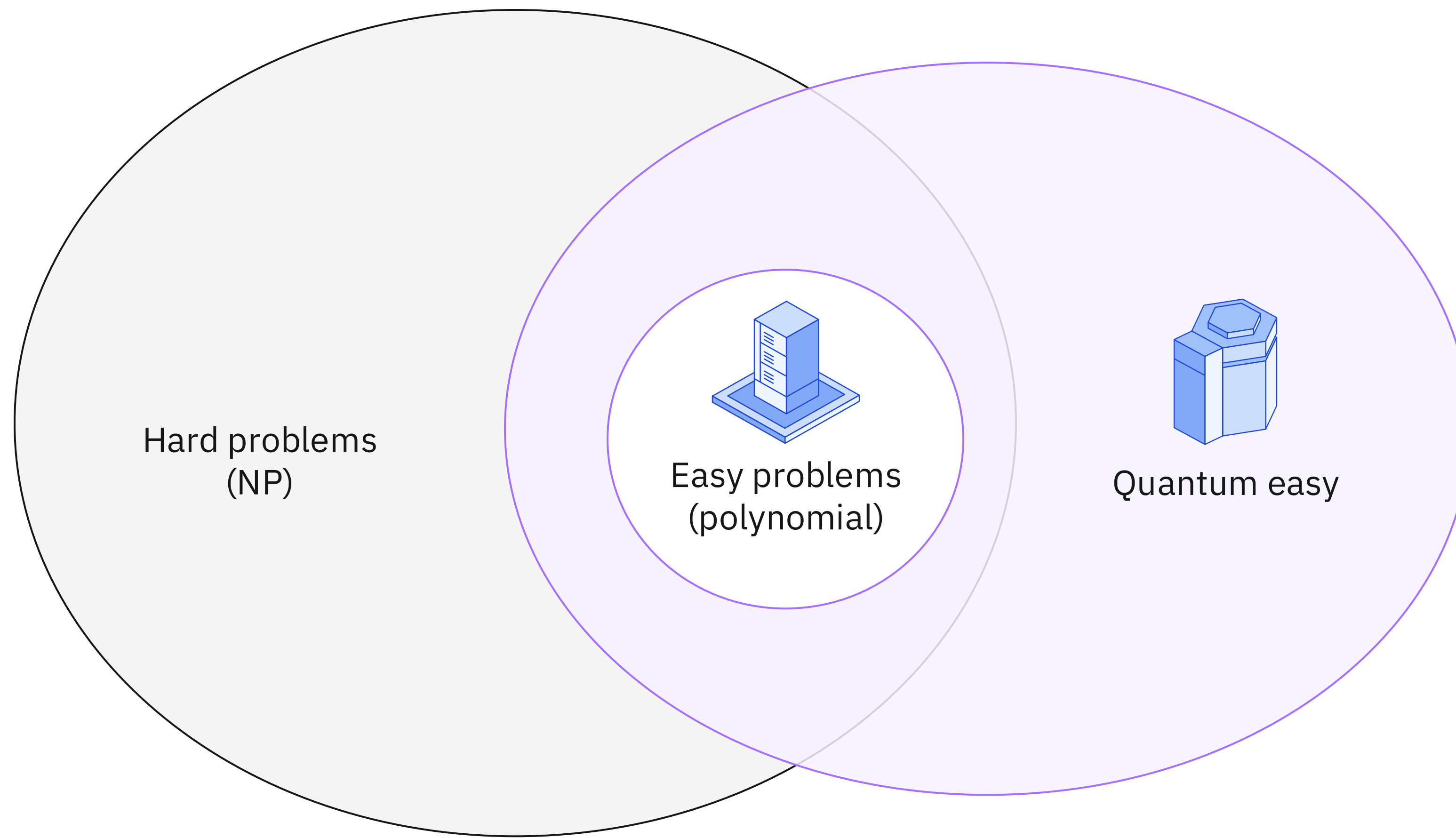
Source: Steane & Rieffel

Quantum dot qubits



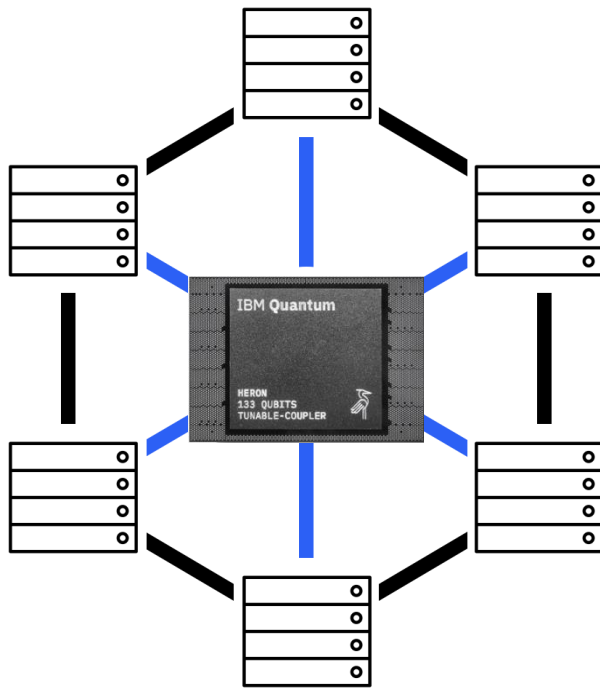
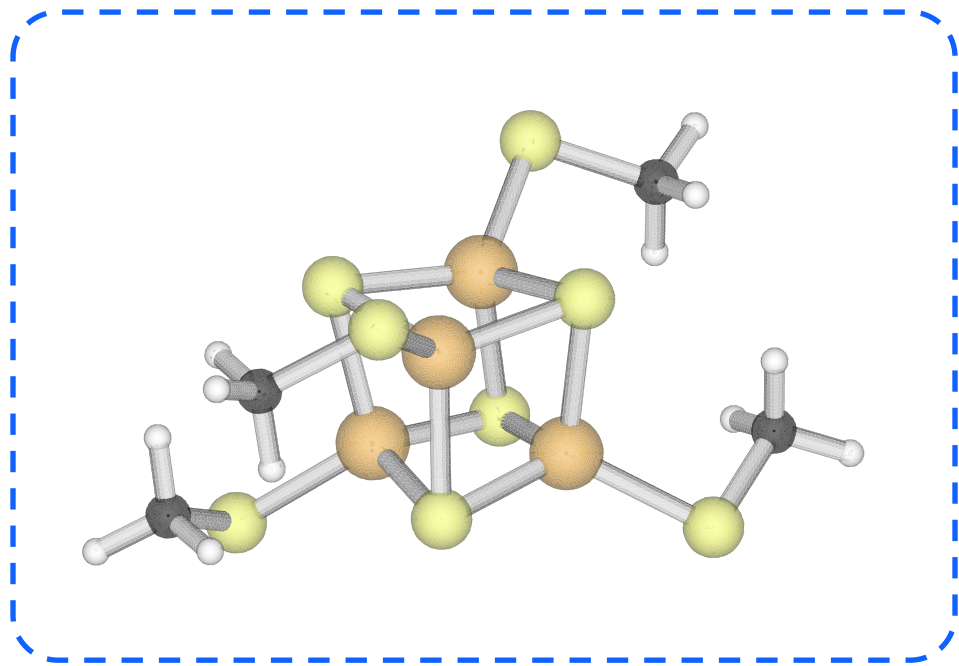
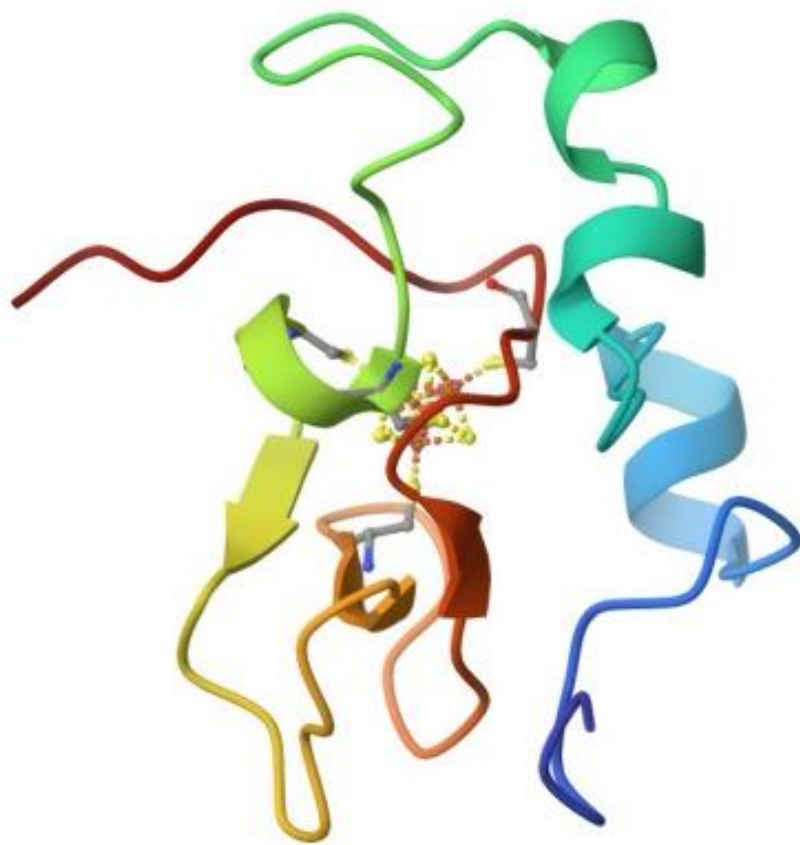
Source: Ansaloni, F

There is a rich seam of problems that cannot be solved by classical and AI supercomputing, and never will.
These are the trillion-dollar problems that quantum computing was designed to solve.



What are these problems?

Modeling molecules, atoms, electrons, and quarks with **unprecedented accuracy**



[4Fe-4S] using an active space of 54 electrons in 36 orbitals from the TZP-DKH basis set

$$H\Psi = E\Psi$$

Classical exact method	63 PiB of memory ¹
Classical approximate method (DMRG)	8 hours ^{2,3}
Quantum method (QCSC)	12 min using 77 qubits + 1.5 hours supercomputer time ¹

We are already using quantum (Heron, 77 qubits) and HPC (Supercomputer Fugaku) to achieve results comparable with the best classical approximate methods (DMRG) in accuracy and timing

1. Science Advances vol. 11, no. 25 (2025)
2. J. Chem. Theory Comp. 20 (2024): 775–786
3. J. Chem. Phys. 159, 234801 (2023)

What are these problems?

Modeling molecules, atoms, electrons, and quarks with **unprecedented accuracy**



Developing lighter, longer-lasting batteries for electric vehicles, electronics, and energy grid storage



Designing lighter, stronger materials to allow planes to be more efficient and to need less maintenance

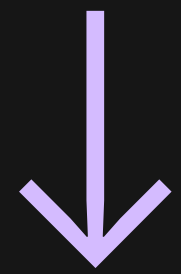


Discovering new classes of antibiotics to counter the emergence of multidrug-resistant bacterial strains

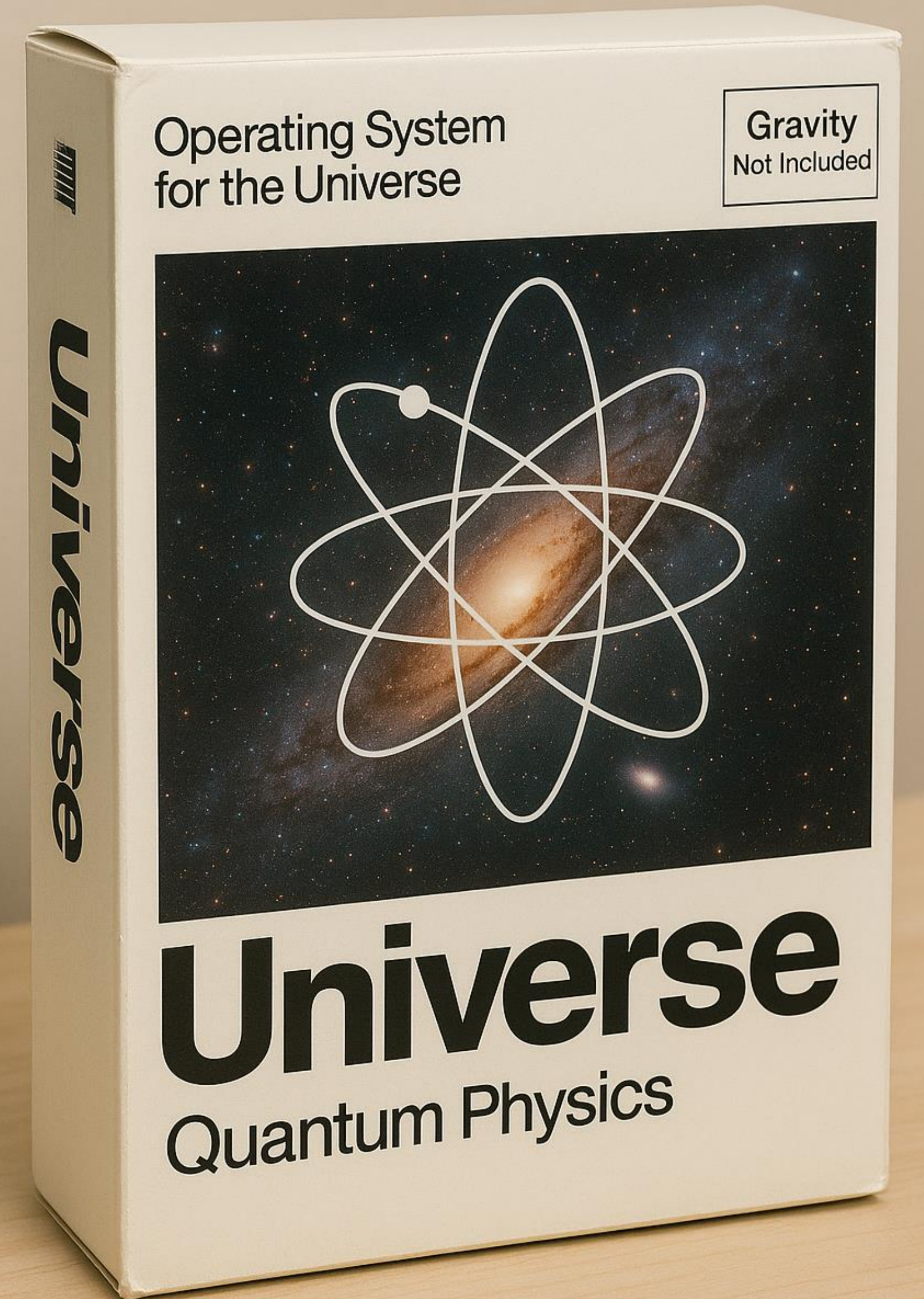


Designing optimal superconductors for MRI, electromobility, and renewable energies

Nature is
quantum.



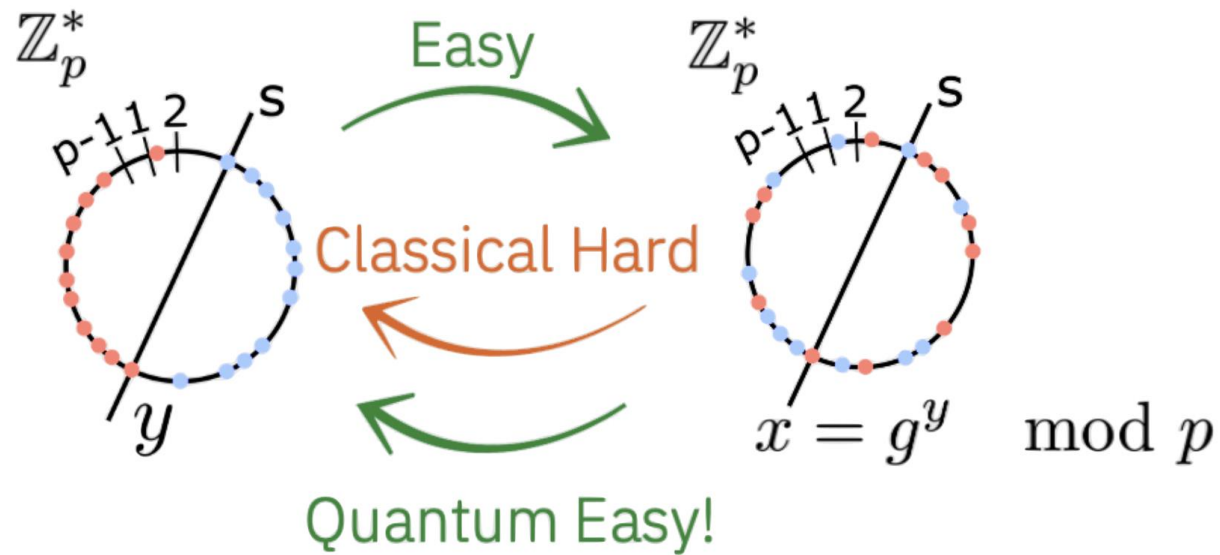
You need
quantum to
simulate nature.



What are these problems?

Solving algebra in [exponential] spaces.
Finding **hidden patterns** in structured problems.

Groups encode symmetries and transformations of data.



$$f(g\mathbf{b}) = f(g) \quad \forall \mathbf{b} \in B$$

=

$$\frac{1}{\sqrt{|G|}} \sum_{g \in G} |f(g)\rangle |g\rangle$$
$$\sqrt{\frac{|B|}{|G|}} \sum_{g_0 \in G/B} |f(g_0)\rangle \frac{1}{\sqrt{|B|}} \sum_{\mathbf{b} \in B} |g_0 \mathbf{b}\rangle$$

Quantum computers speak the mathematics of group theory and work in the large spaces needed to answer difficult computational problems inaccessible to classical methods.

What are these problems?

Solving algebra in [exponential] spaces.
Finding **hidden patterns** in structured problems.



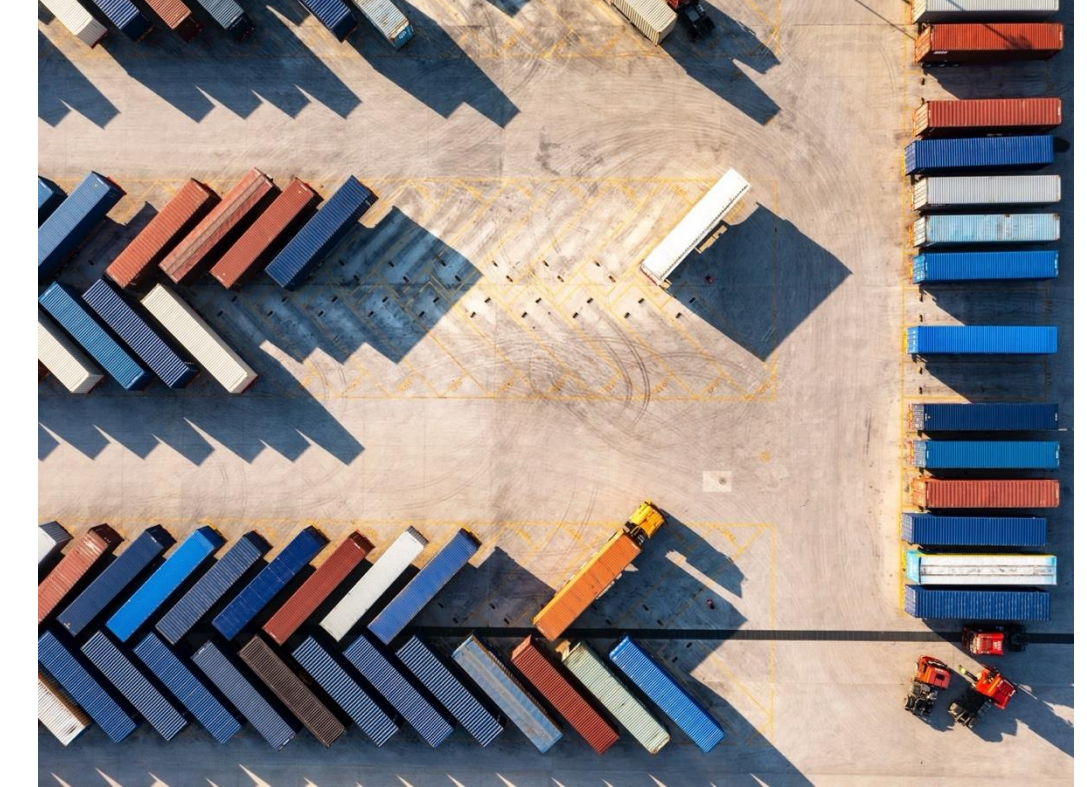
Improving anomaly detection, as for rare events detection and fraud detection



Improving patient outcomes by designing optimal cell-centric therapeutics



Strengthening risk management through better time series and sequence prediction



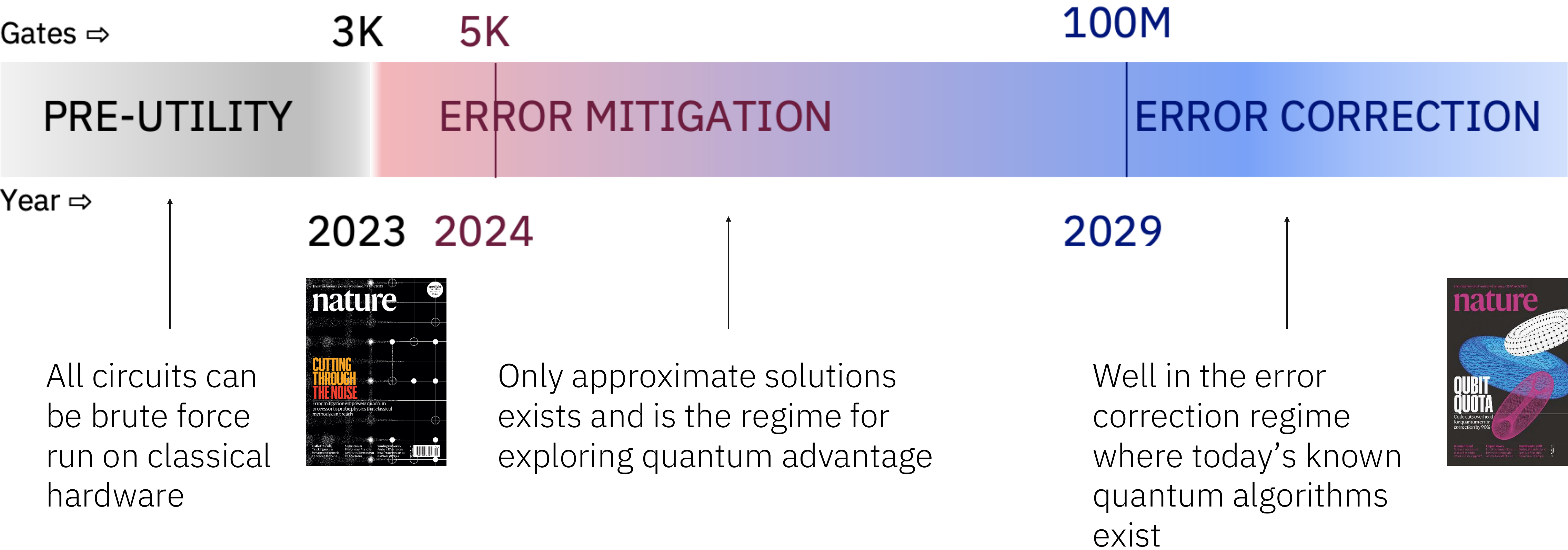
Optimizing vehicle routing and scheduling for large-scale logistics networks

And we know there are other
problems with a structure
that quantum can exploit.

↳ So we're also looking for value
in problems that are hard
classically, like optimization.

State of the technology

Quantum Utility

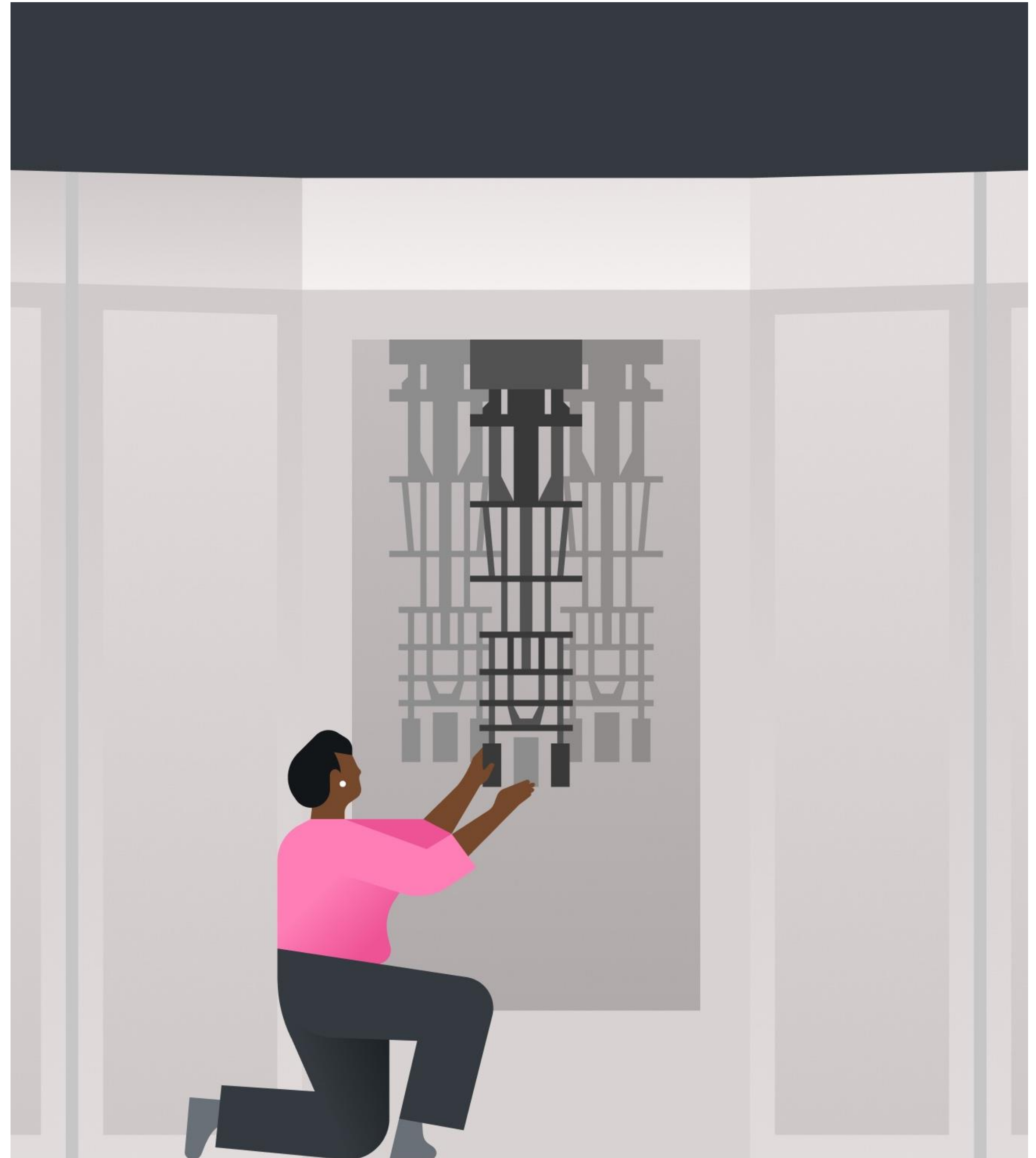


```
for _ in loop:
    qc,matrix,... = do_classical(param)
    A,... = SGESVD(matrix)
    job = sampler.run([qc])
    do_classical(job.result(), A, ...)
    ...
```

Thesis

The future of computing is mapping all problems to tensors and circuits

How does a quantum
computer work?



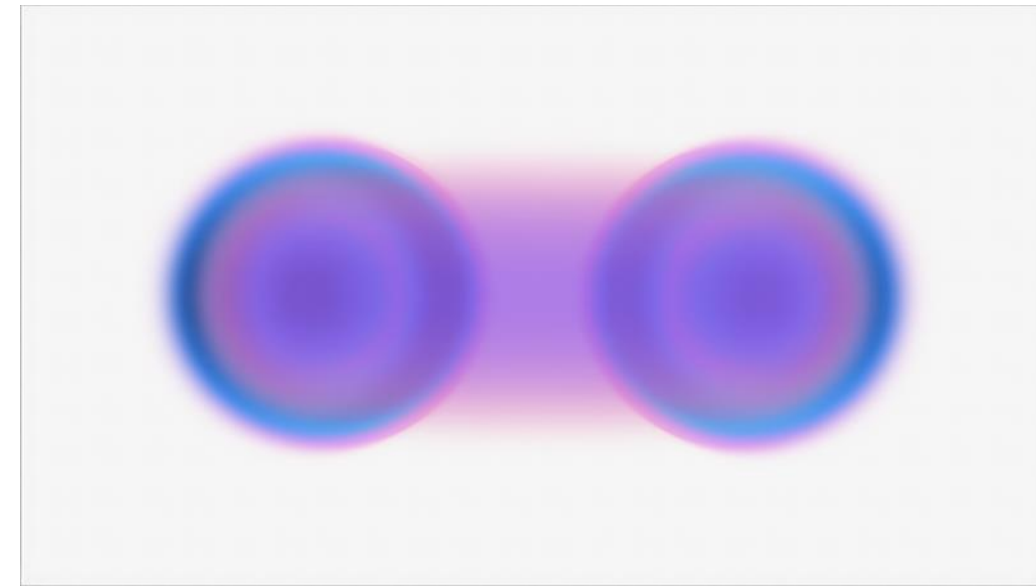
Uniquely quantum

Some problems are classically intractable and will never be solvable with traditional computers



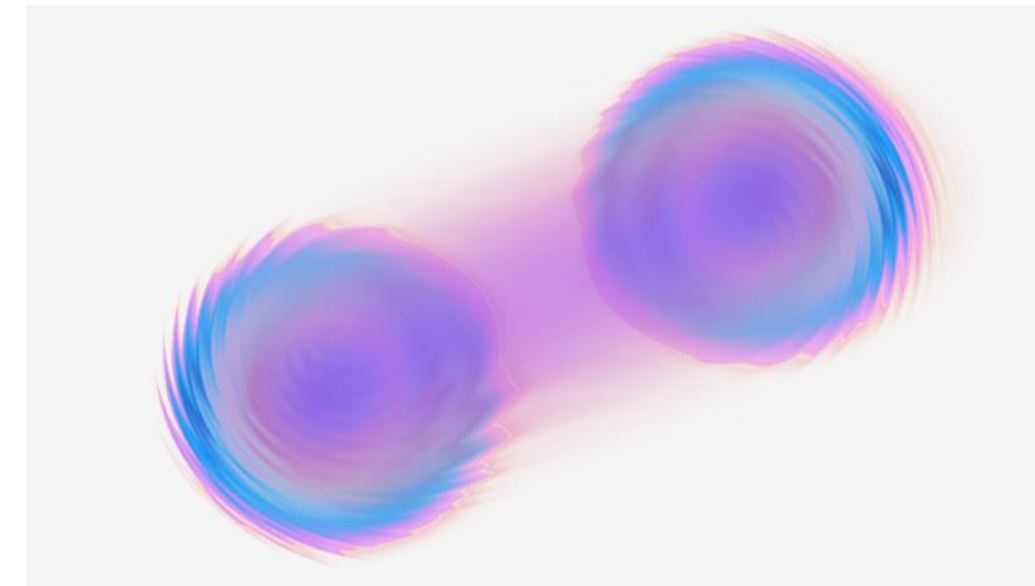
Superposition

A quantum system existing in a complex linear combination of two states until it is measured



Entanglement

Information shared jointly between entangled pairs or groups



Interference

Interaction that affects likelihood of solutions

Moore's law: the number of transistors in a classical integrated circuit doubles about every two years ... but we are approaching the end due to physical limitations

[Approaching the physical limit: IBM created the world's first 2 nm node chip in 2021, with transistors as small as 10 silicon atoms](#)

These Quantum concepts can reduce the number of computational steps required for certain algorithms & At a ~100 qubit scale, with sufficient circuit depth and complexity, classical computers can no longer simulate exactly

The limit of bits

For decades we've been simplifying nature into **1**s and **0**s because that was the only way we could **manage** to create a useful and scalable system of computation.

```
00100110111001001000100100100110010011100101110
01111100101001000111000100010010100010010010101
01001010101110010011011100100100010010010011001
00111001011100111110010100100011100010001001010
0010010010101010010101011101110011100101011110
```

The limit of bits

For decades we've been simplifying nature into **1**s and **0**s because that was the only way we could **manage** to create a useful and scalable system of computation.

But the future isn't
just **1**s and **0**s.

```
00100110111001001000100100100110010011100101110
01111100101001000111000100010010100010010010101
01001010101110010011011100100100010010010011001
00111001011100111110010100100011100010001001010
0010010010101010010101011101110011100101011110
```

Bits and classical logic circuits

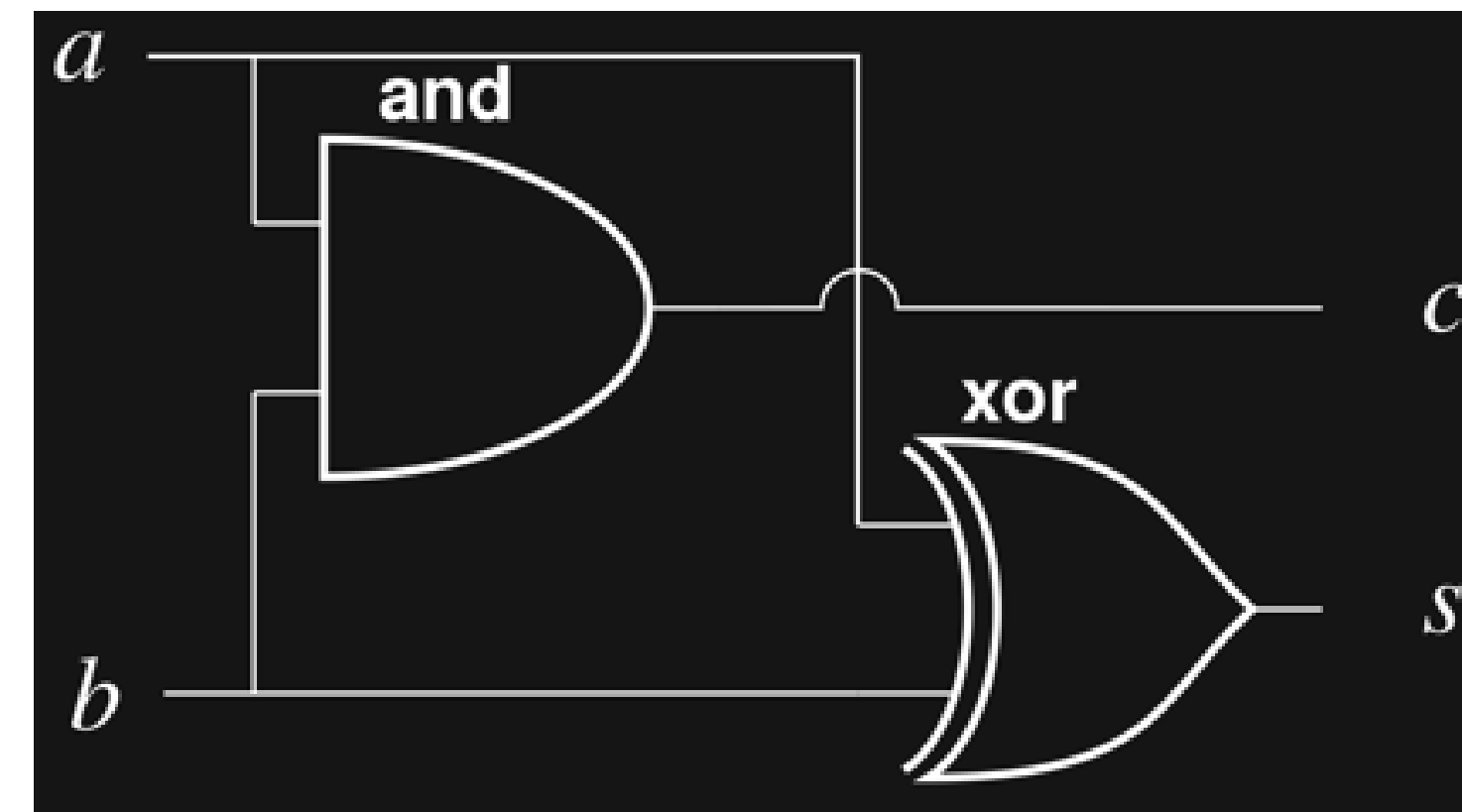
0

•

•

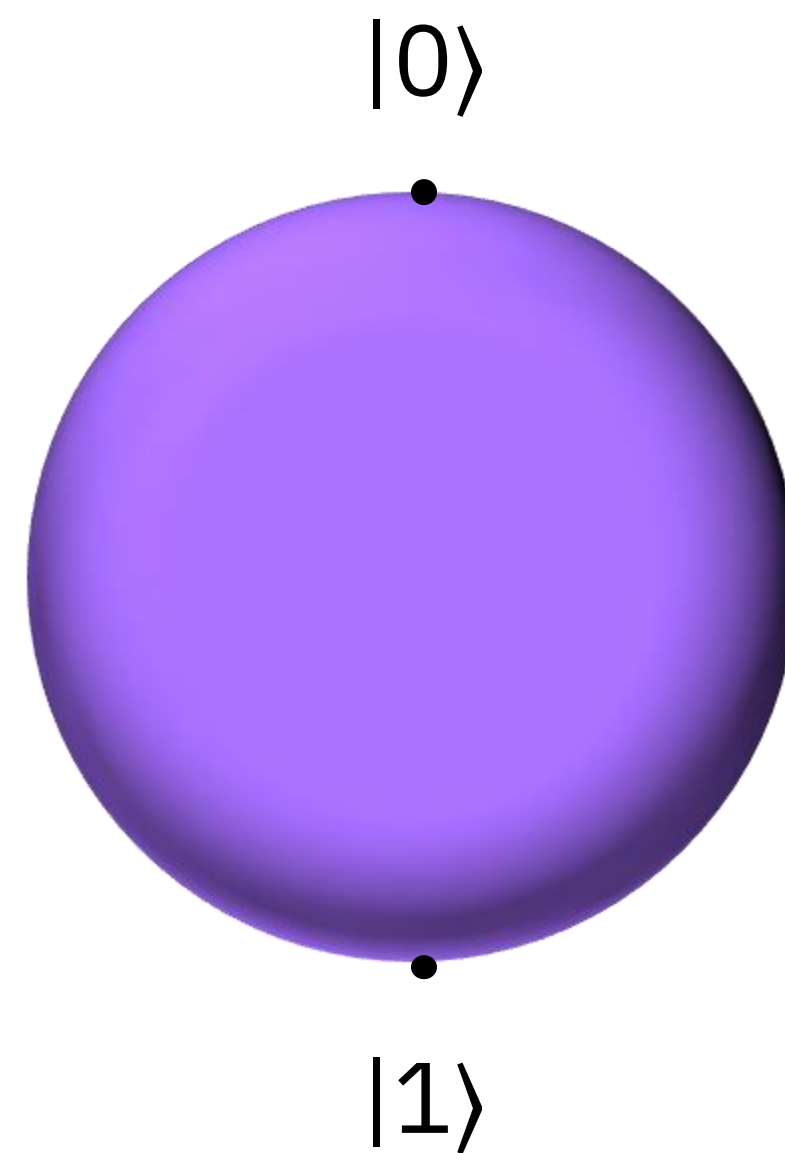
1

A **bit** is a controllable classical object that is the unit of information

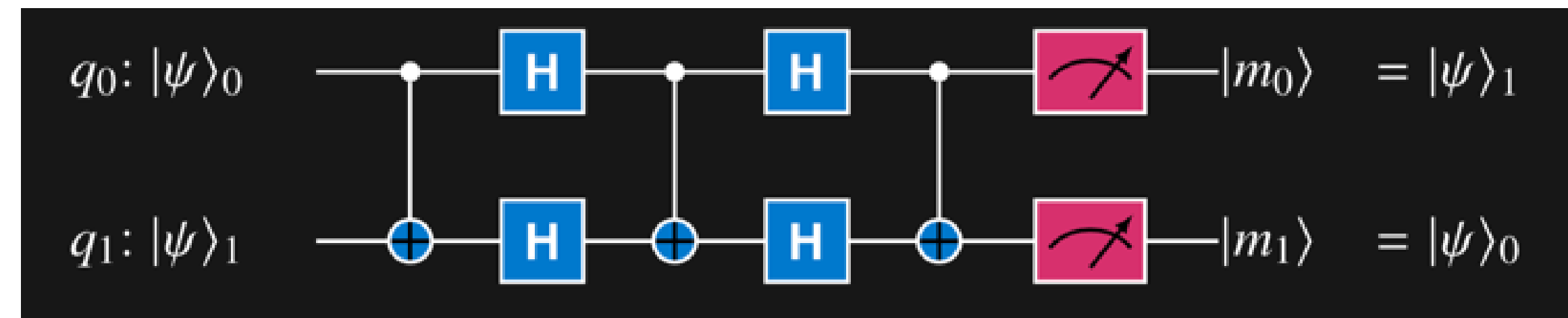


A **classical logic** circuit is a set of gate operations on bits and is the unit of computation

Quantum bits (qubits) and quantum circuits

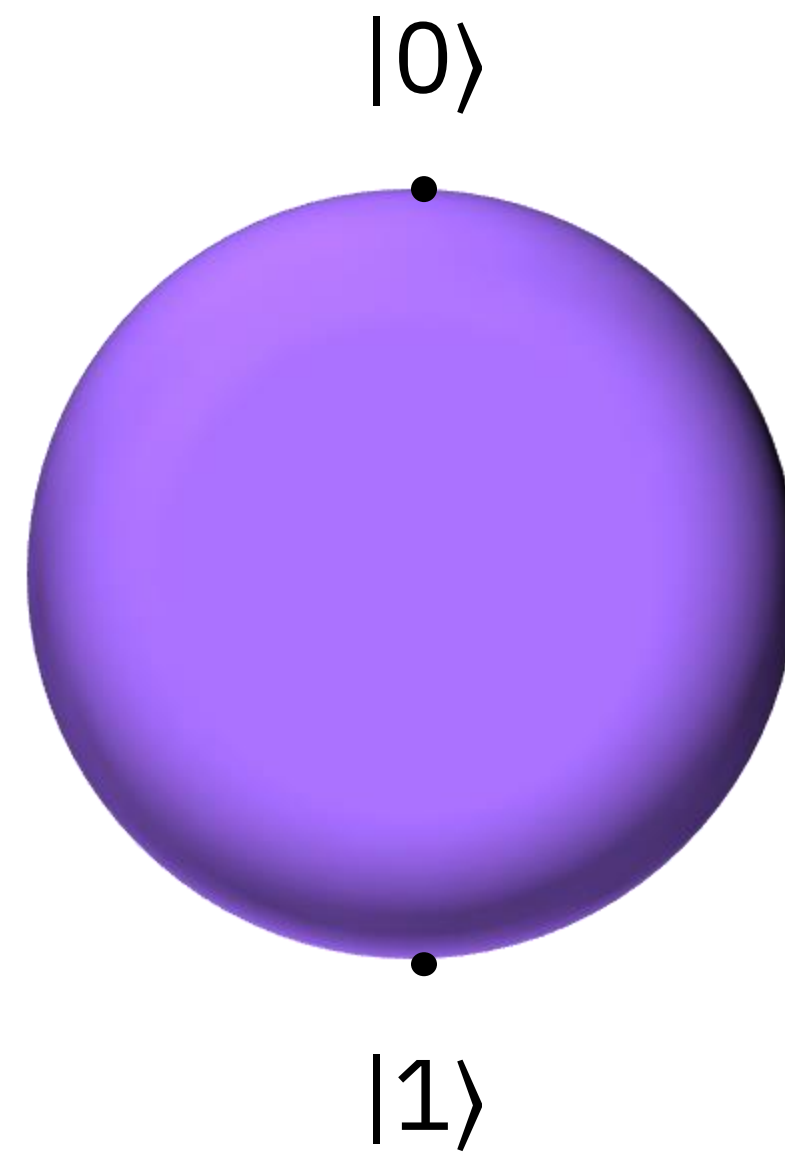


A **quantum bit** or qubit is a controllable quantum object that is the unit of information



A **quantum** circuit is a set of quantum gate operations on qubits and is the unit of computation

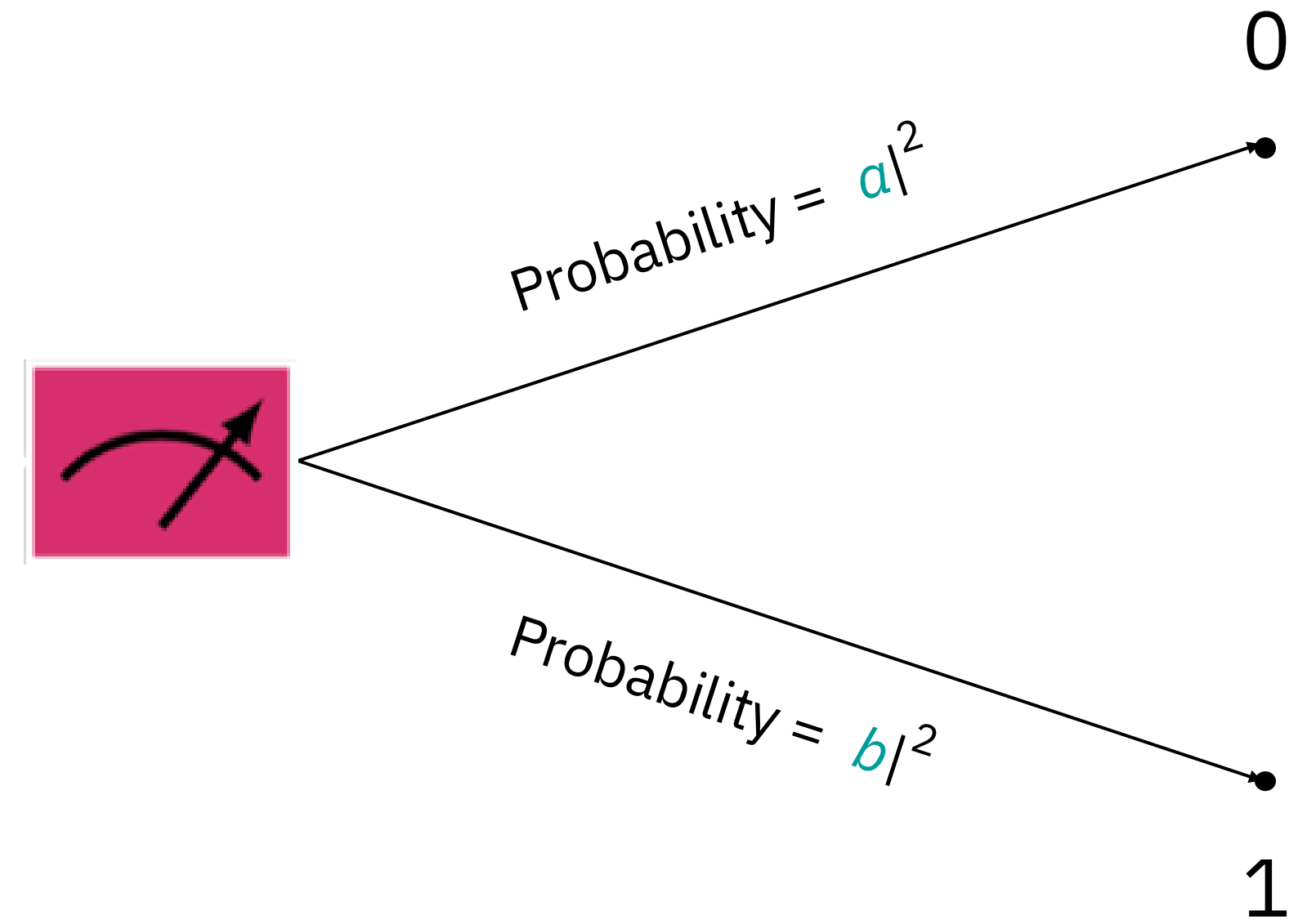
Bits and qubits



A qubit's **state** is a combination of $|0\rangle$ and $|1\rangle$:

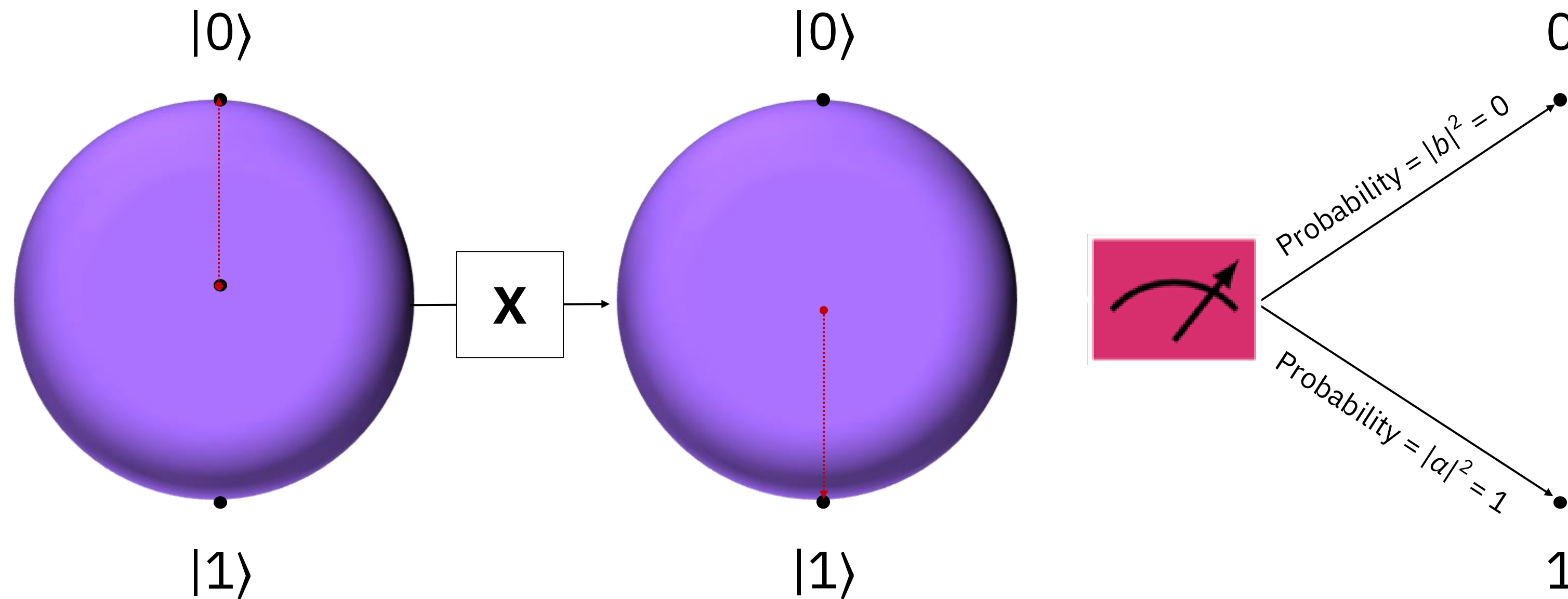
$$a |0\rangle + b |1\rangle$$

This means that a single qubit contains **two** pieces of information.



When we measure a qubit, it becomes **0** or **1** based on probability.

Bits and qubits: the effect of the **X** gate on $|0\rangle$



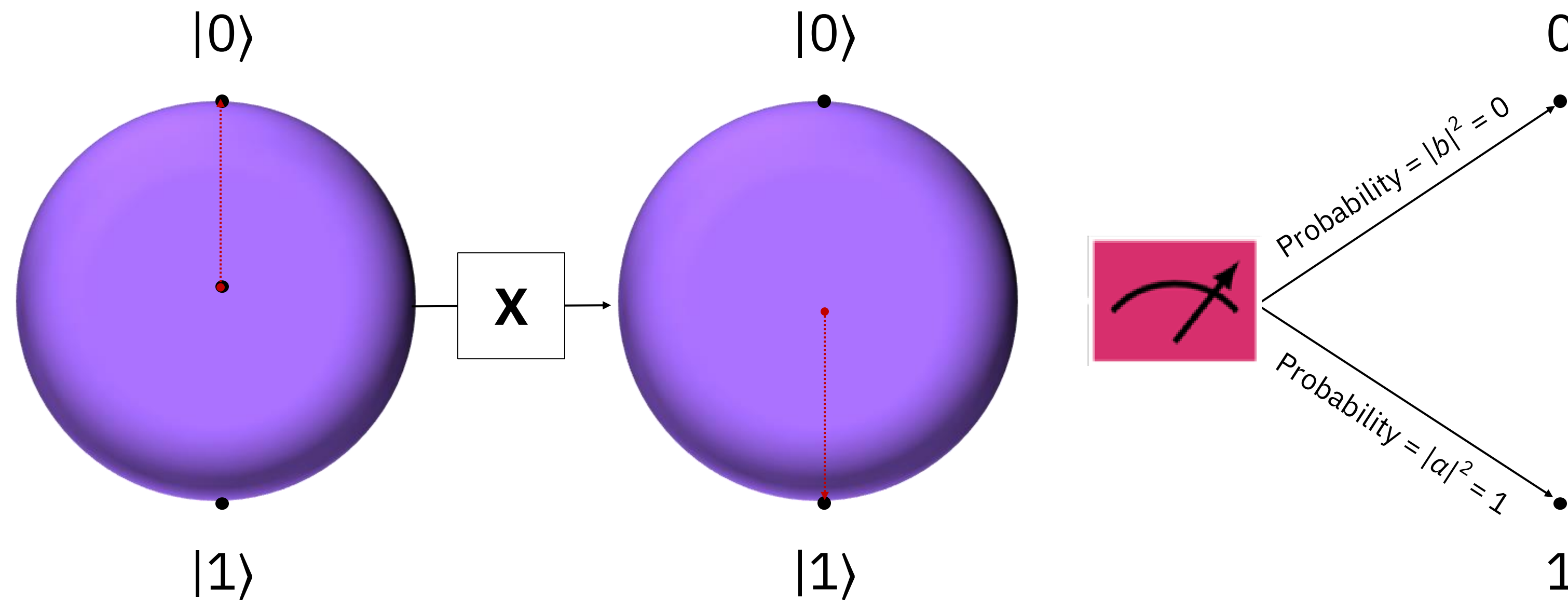
The **X** gate reverses $|0\rangle$ and $|1\rangle$:

$$a |0\rangle + b |1\rangle \mapsto b |0\rangle + a |1\rangle$$

$a = 1$ and $b = 0$, so $|0\rangle$ is mapped to $|1\rangle$.

When measured, the result is **1**
with 100% probability.

Bits and qubits: the effect of the **X** gate on $|0\rangle$



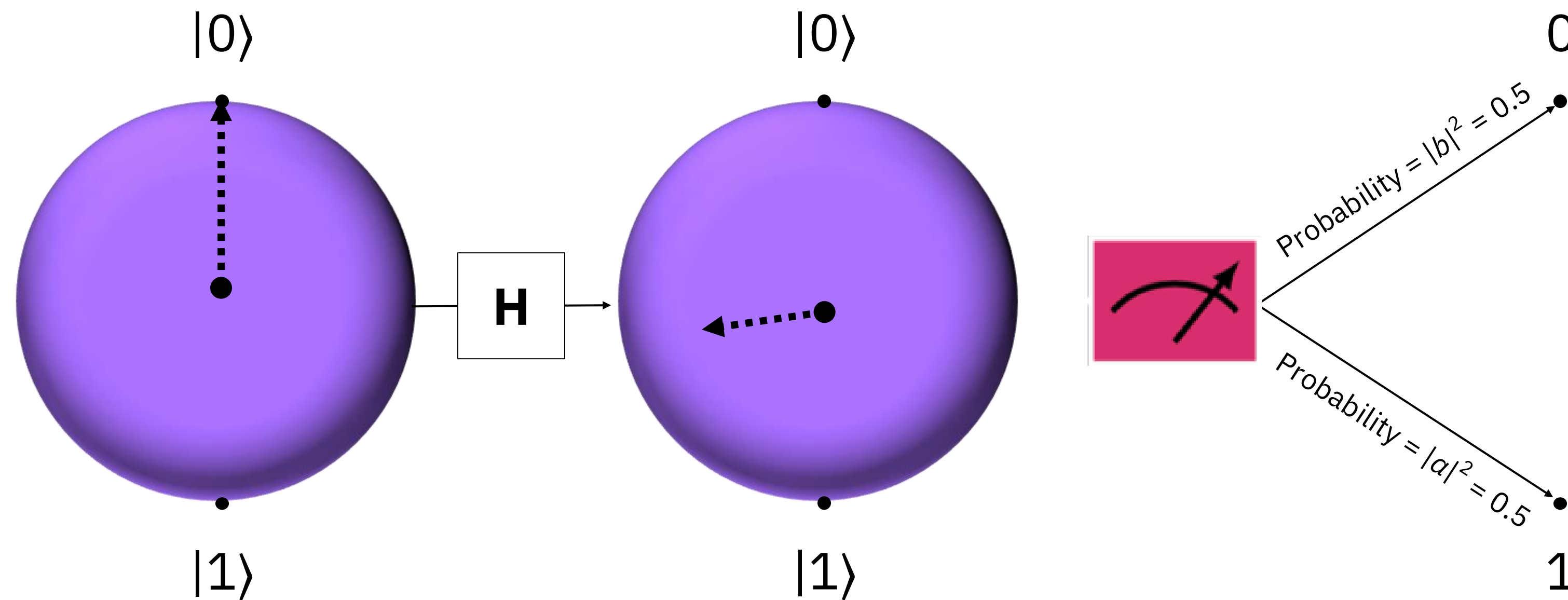
The **X** gate reverses $|0\rangle$ and $|1\rangle$:

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When measured, the result is **1**
with 100% probability.

Bits and qubits: the effect of the **H** gate on $|0\rangle$



The **H** gate maps $|0\rangle$ via

$$|0\rangle \mapsto \frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle = a |0\rangle + b |1\rangle$$

Since $a = b = 1/\sqrt{2}$, $|a|^2 = |b|^2 = 1/2$.

When measured, the probability of getting **0** or **1** is the same, 0.5.
Quantum randomness!

Quantum computing
uses essential ideas from
quantum mechanics

Superposition

$|0\rangle$ and $|1\rangle$ are vectors in the two-dimensional complex vector space $\mathbb{C}^2$:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{and} \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

So we can write any vector in $\mathbb{C}^2$ as

$$a |0\rangle + b |1\rangle$$

We pronounce $|0\rangle$ and $|1\rangle$ as “ket zero” and “ket one.” These are called the *computational basis*.

Quantum computing
uses essential ideas from
quantum mechanics

Superposition

Superposition is creating a quantum state
that is a combination of $|0\rangle$ and $|1\rangle$

$$a |0\rangle + b |1\rangle$$

where

a and b are complex numbers

$$|a|^2 + |b|^2 = 1$$

Two quantum states are equivalent if they
differ only by a constant multiple u
where $|u| = 1$.

This is because

$$|a|^2 + |b|^2 = |au|^2 + |bu|^2 = 1$$

Quantum computing
uses essential ideas from
quantum mechanics

Superposition

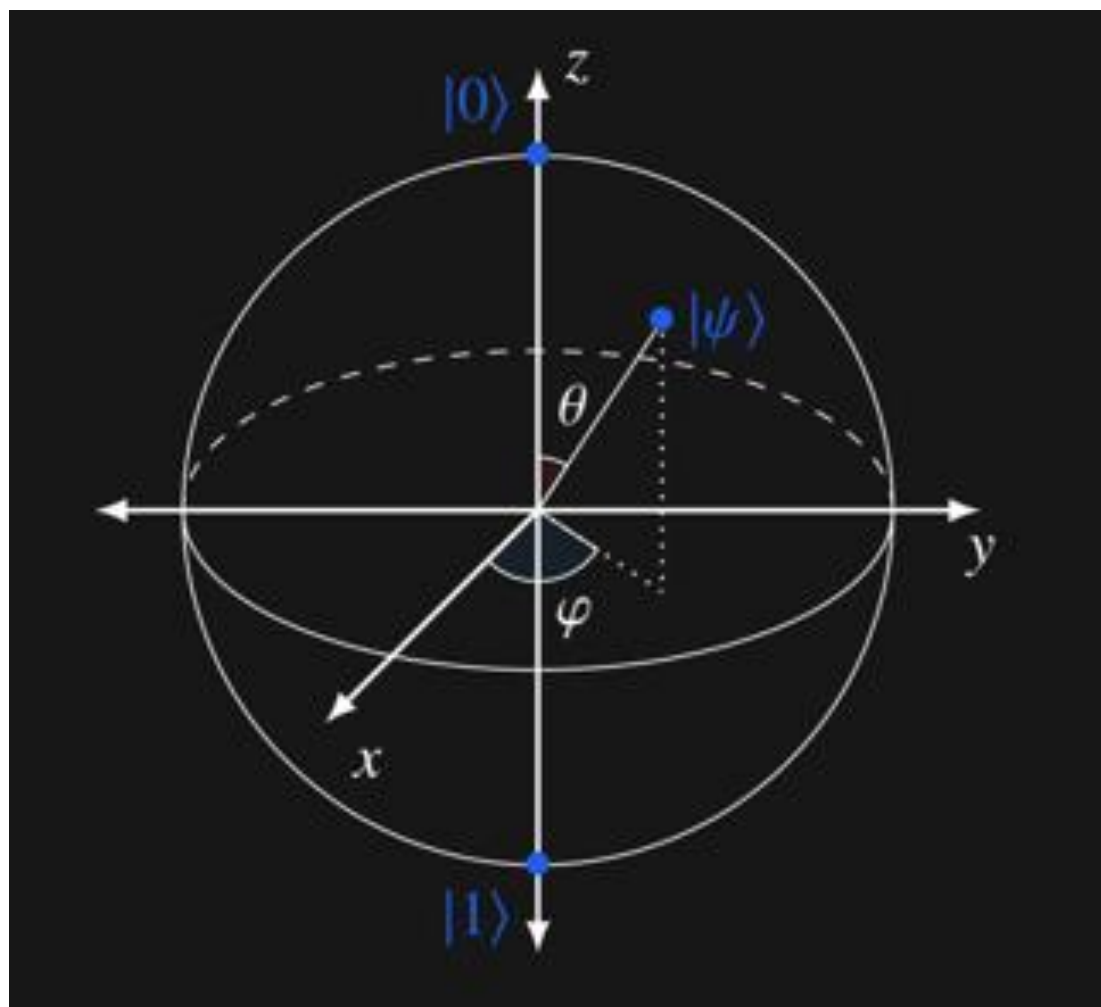
Superposition is creating a quantum state
that is a combination of $|0\rangle$ and $|1\rangle$

$$a |0\rangle + b |1\rangle$$

These conditions allow us to map the qubit
onto the *Bloch Sphere*.

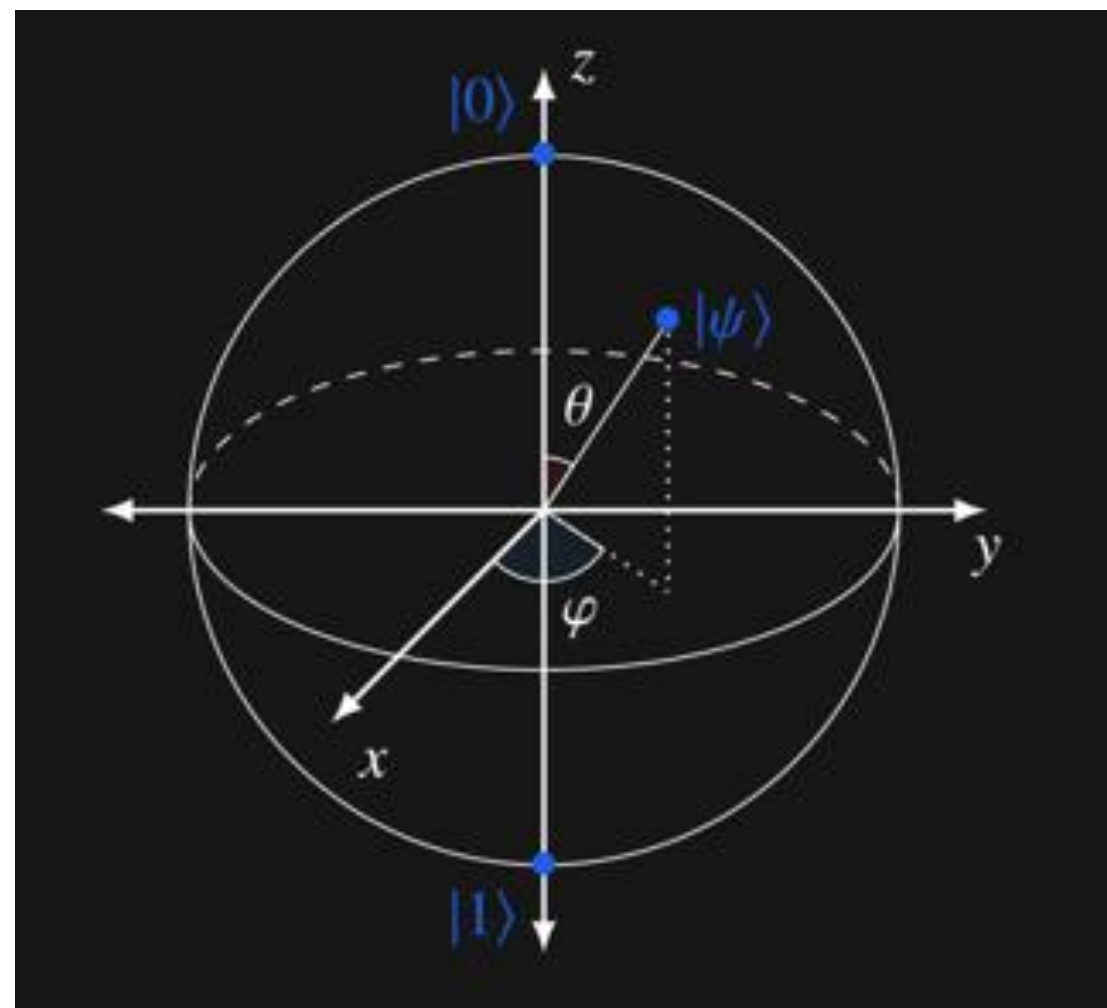
Note that if a and b are non-zero, then the
qubit's state contains both $|0\rangle$ and $|1\rangle$.

This is what people mean when they say that
a qubit can be “0 and 1 at the same time.”



Quantum computing
uses essential ideas from
quantum mechanics

Measurement



Measurement is forcing the qubit's state

$$a |0\rangle + b |1\rangle$$

to $|0\rangle$ or $|1\rangle$ by observing it, where

$|a|^2$ is the probability we will get $|0\rangle$ when we measure

$|b|^2$ is the probability we will get $|1\rangle$ when we measure

For example,

$$\frac{\sqrt{2}}{2} |0\rangle + \frac{\sqrt{2}}{2} |1\rangle$$

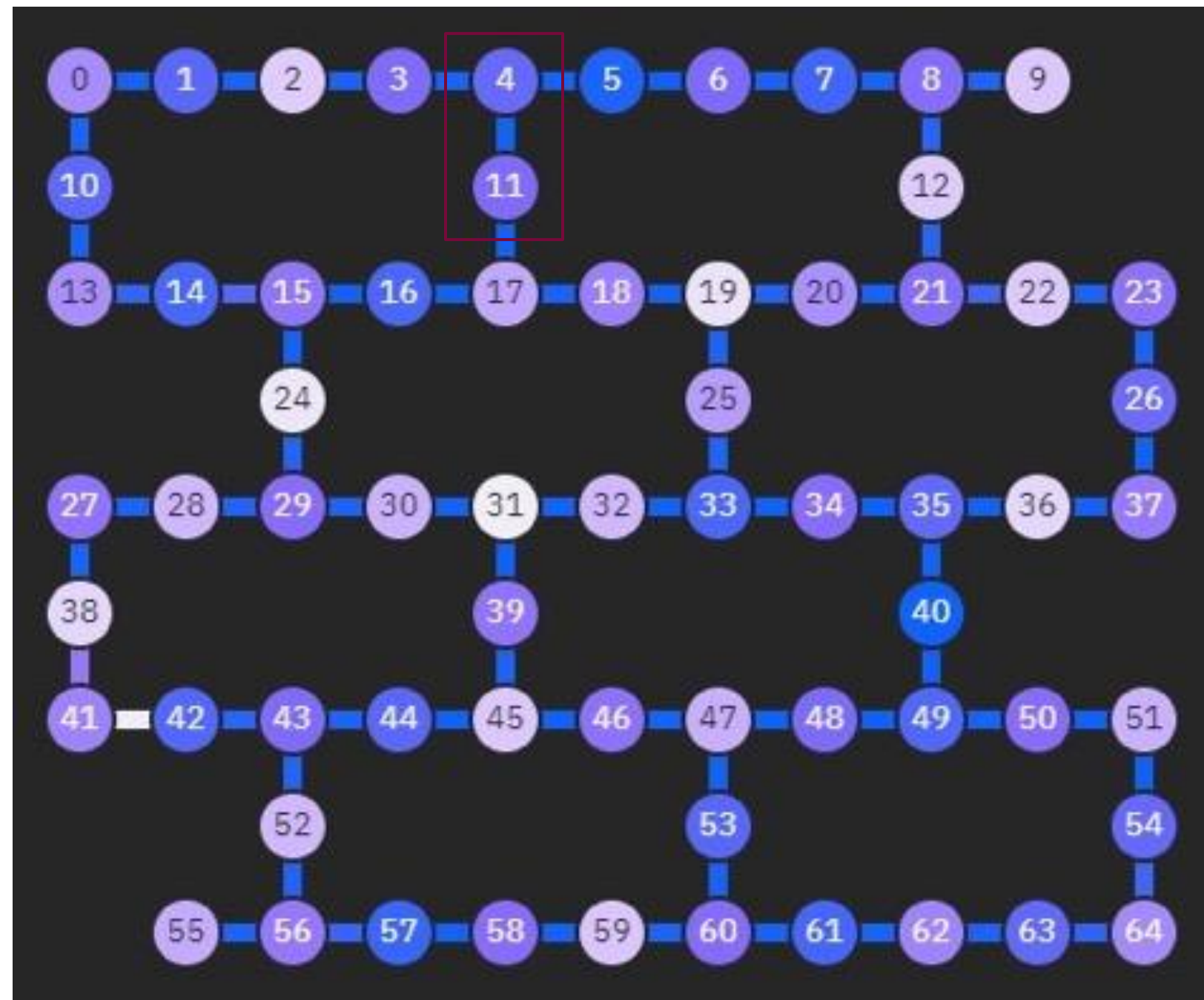
has an equal probability of becoming $|0\rangle$ or $|1\rangle$, and

$$\frac{\sqrt{3}}{2} |0\rangle - \frac{1}{2} i |1\rangle$$

has a 75% chance of becoming $|0\rangle$.

Quantum computing uses essential ideas from quantum mechanics

Entanglement



With two qubits we get combinations like

$$a |00\rangle + b |01\rangle + c |10\rangle + d |11\rangle$$

where

$|01\rangle$ means the first qubit is $|0\rangle$ and
the second is $|1\rangle$

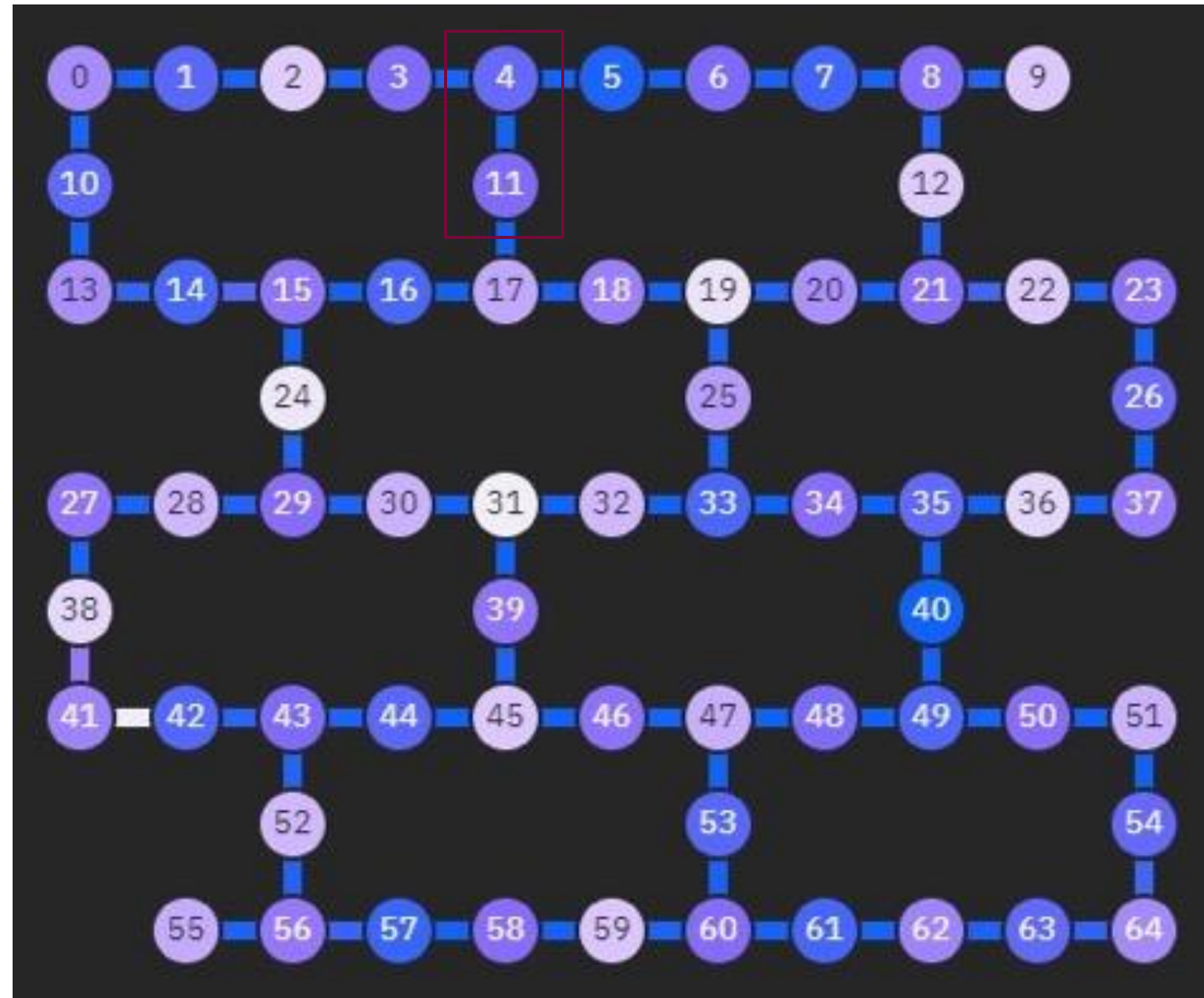
a , b , c and d are complex numbers and

$$|a|^2 + |b|^2 + |c|^2 + |d|^2 = 1$$

If two or more of the a , b , c , and d are non-zero, and we cannot separate the qubits, they are entangled with perfect correlation and are no longer independent.

Quantum computing
uses essential ideas from
quantum mechanics

Entanglement



For example,

$$\frac{\sqrt{2}}{2} |00\rangle + \frac{\sqrt{2}}{2} |01\rangle$$

not entangled

$$\frac{\sqrt{2}}{2} |01\rangle - \frac{\sqrt{2}}{2} |10\rangle$$

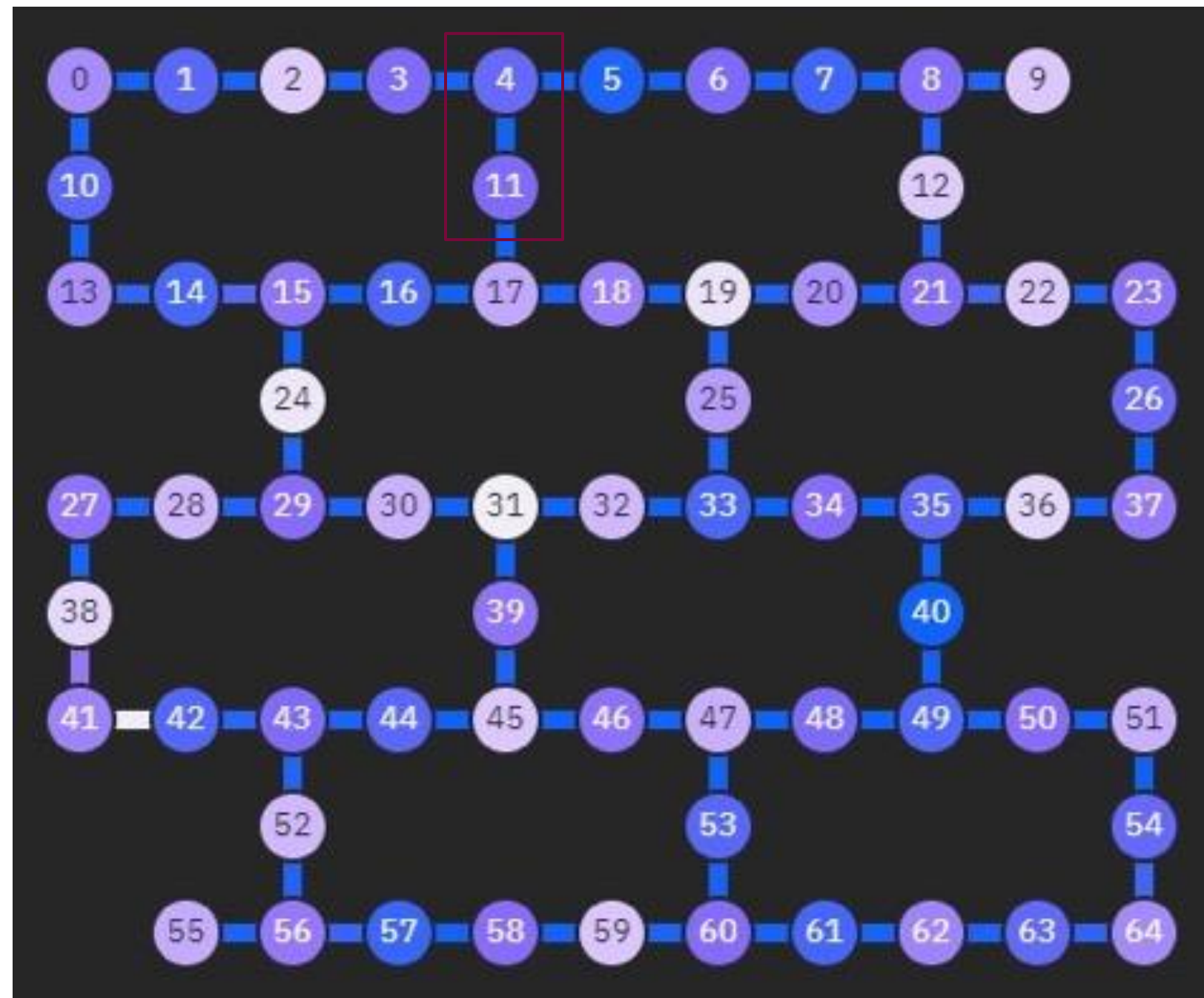
entangled

$$\frac{\sqrt{2}}{2} |00\rangle + \frac{\sqrt{2}}{2} |11\rangle$$

entangled

Quantum computing uses essential ideas from quantum mechanics

Entanglement



We can write

$$\frac{\sqrt{2}}{2} |00\rangle + \frac{\sqrt{2}}{2} |01\rangle$$

as

$$|0\rangle \left(\frac{\sqrt{2}}{2} |0\rangle + \frac{\sqrt{2}}{2} |1\rangle \right)$$

but we cannot write

$$\frac{\sqrt{2}}{2} |00\rangle + \frac{\sqrt{2}}{2} |11\rangle$$

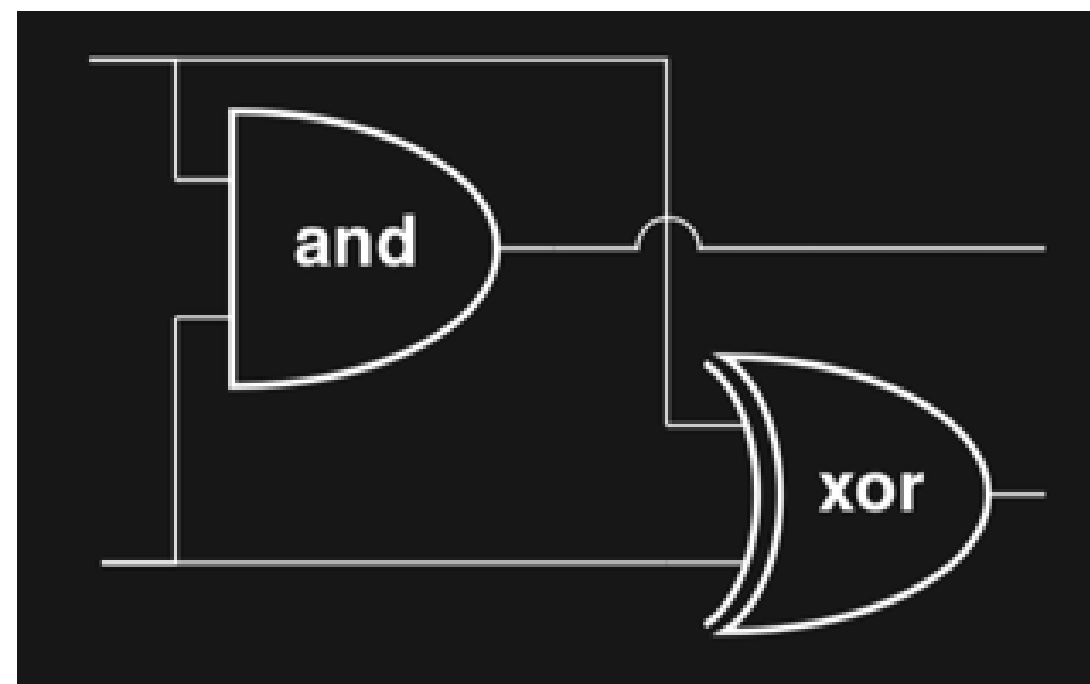
as the “product” of two single
qubit states.

They are **entangled**!

Once you measure the first qubit,
the second is uniquely determined.

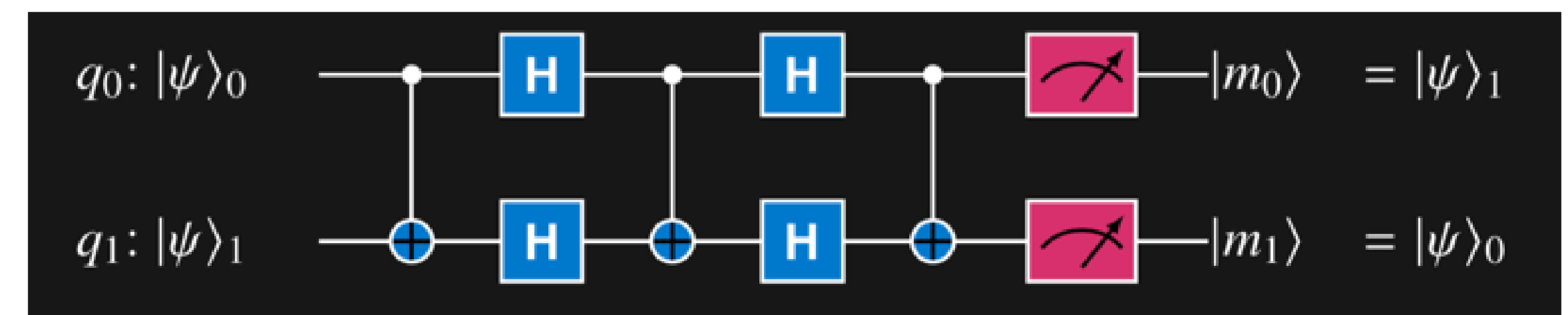
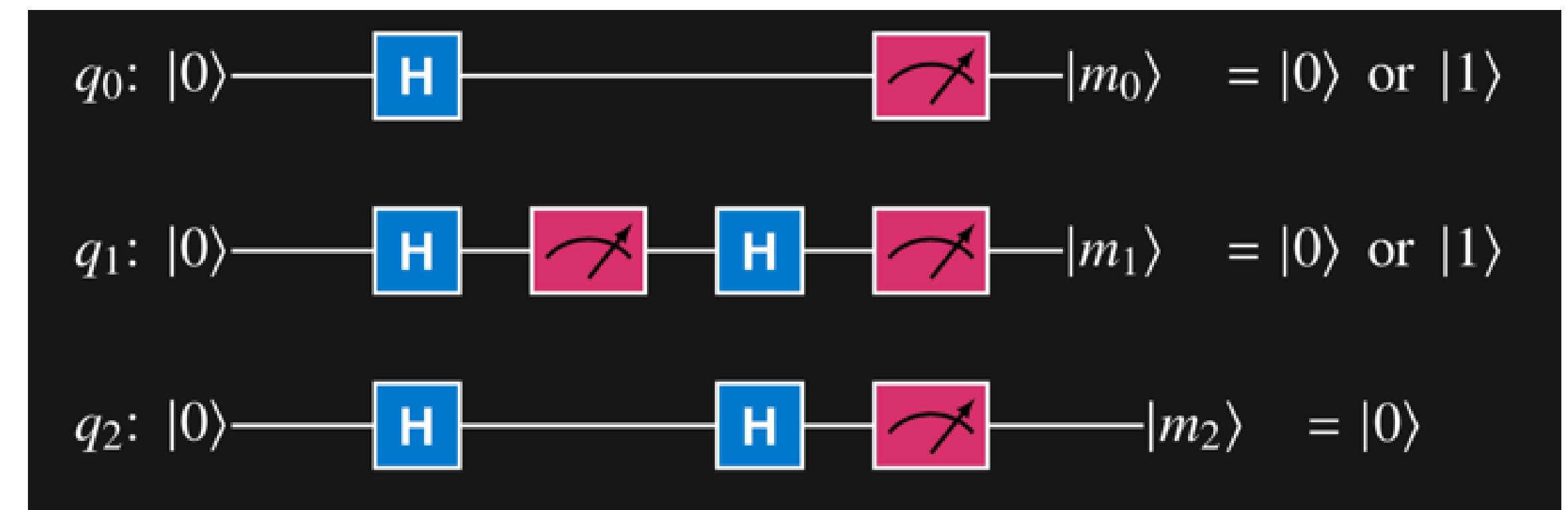
Quantum computing uses essential ideas from quantum mechanics

Gates / operations



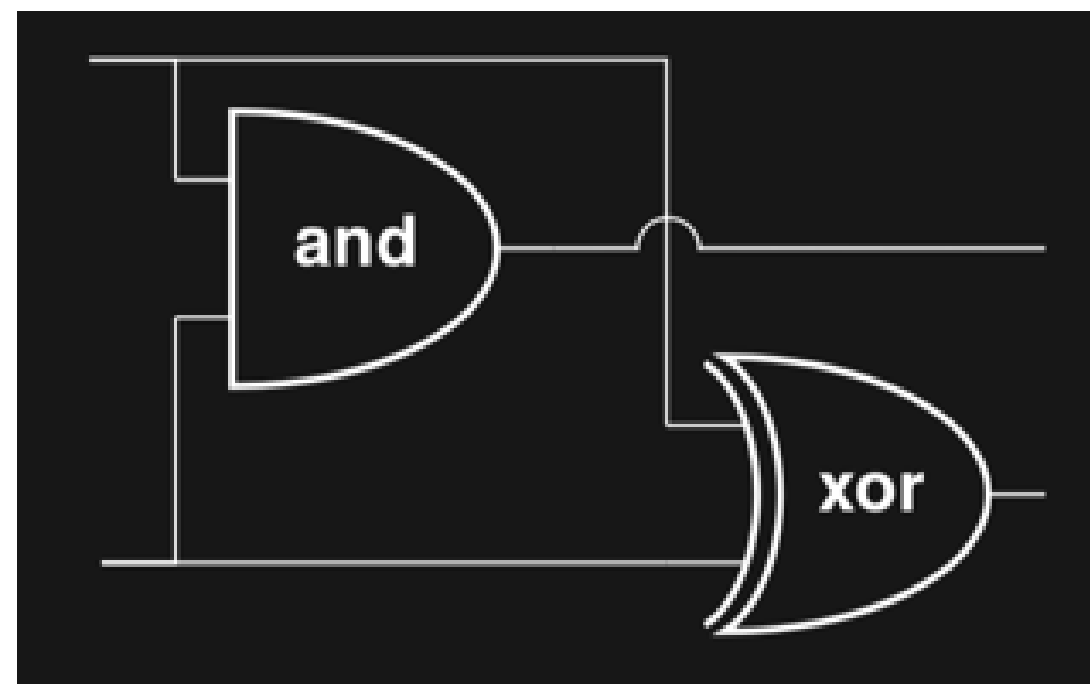
Classical logical circuits use operations like **and**, **or**, **not**, **nand**, and **xor**. We also call these gates.

Quantum circuits use reversible gates that change the quantum states of one, two , or more qubits.



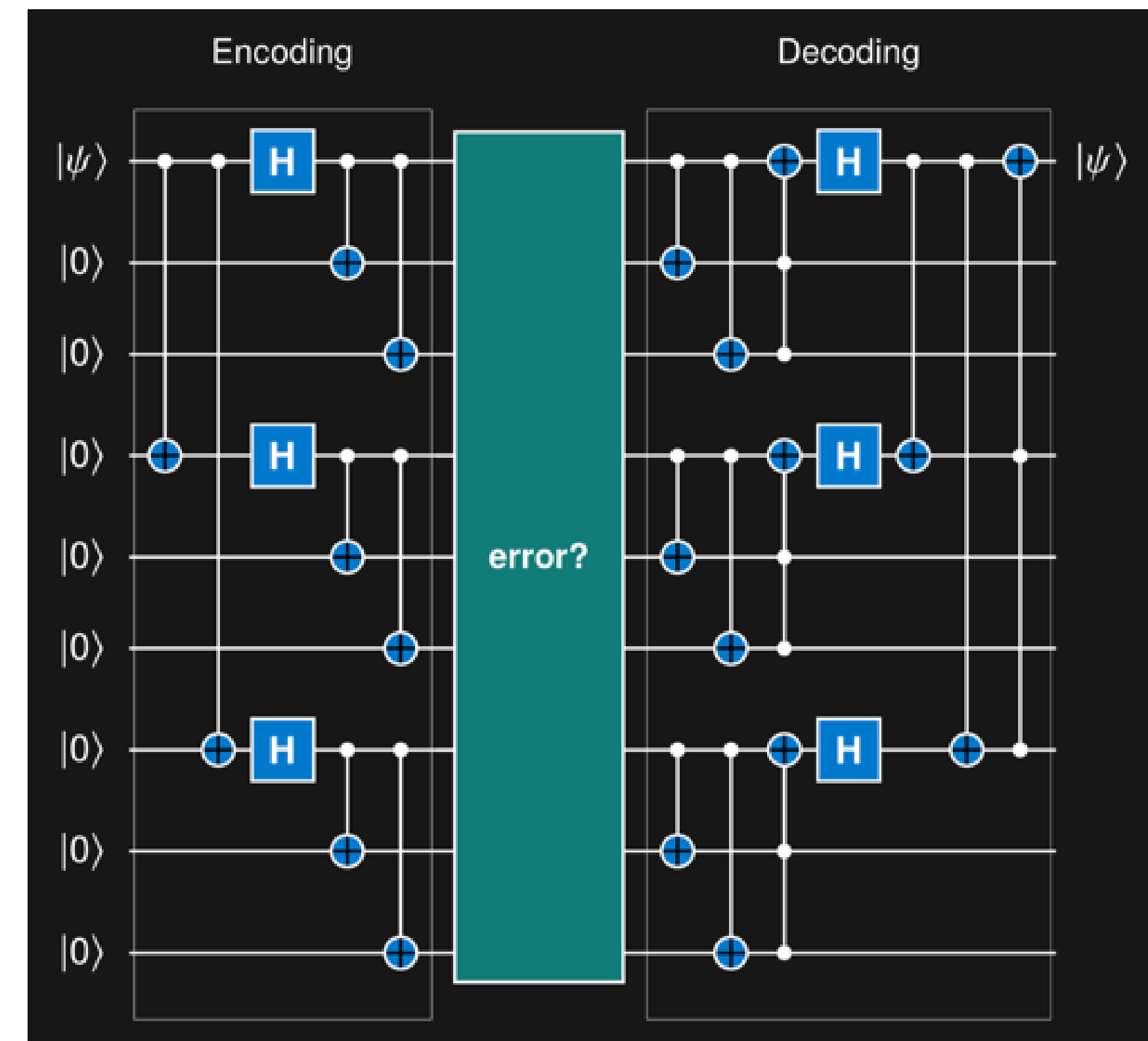
Quantum computing uses essential ideas from quantum mechanics

Gates / operations



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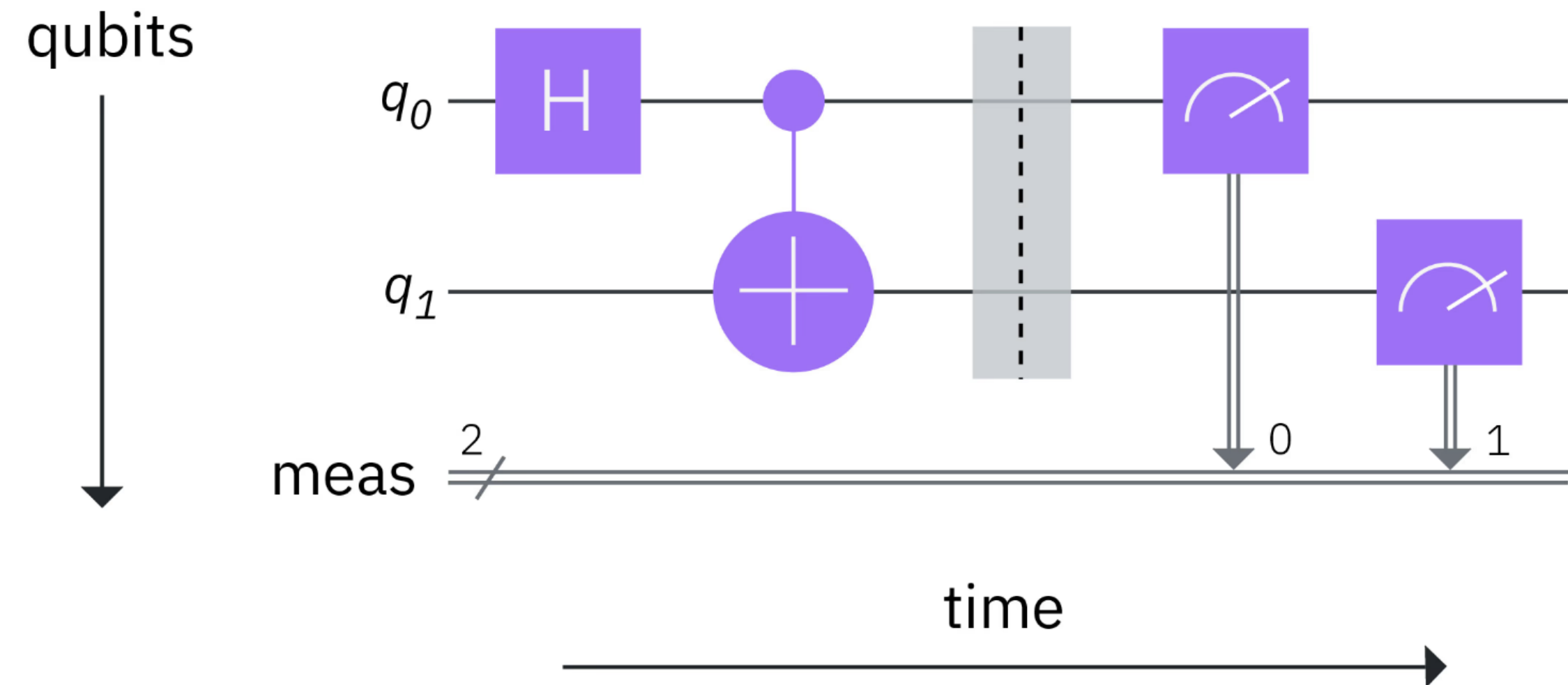
Quantum gates and circuits

Quantum computers process information by applying quantum gates to qubits.

Examples of quantum gates:

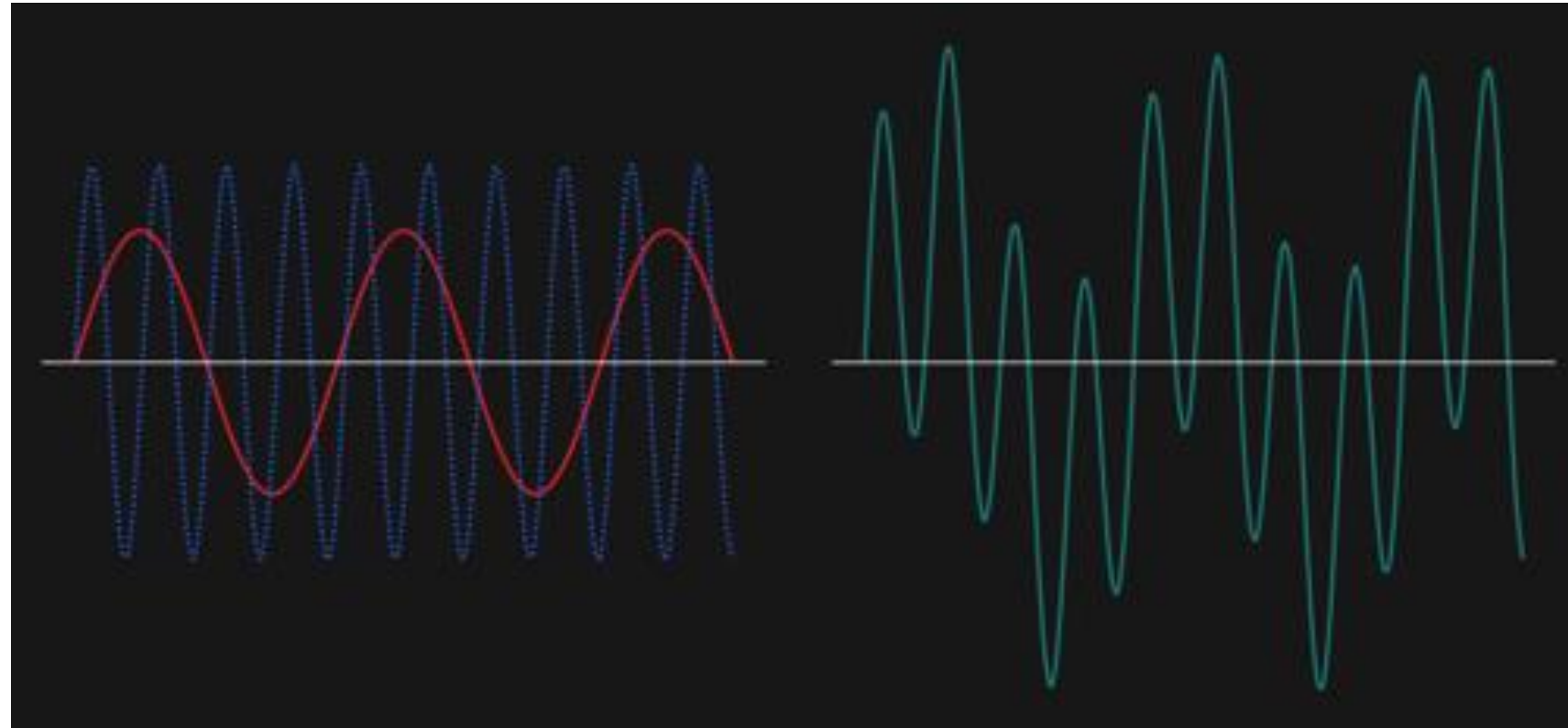
- Hadamard (H)
- X (not), Y, Z
- S, T, P
- $S^\dagger$, $T^\dagger$
- CX
- Swap

Example of a quantum circuit:

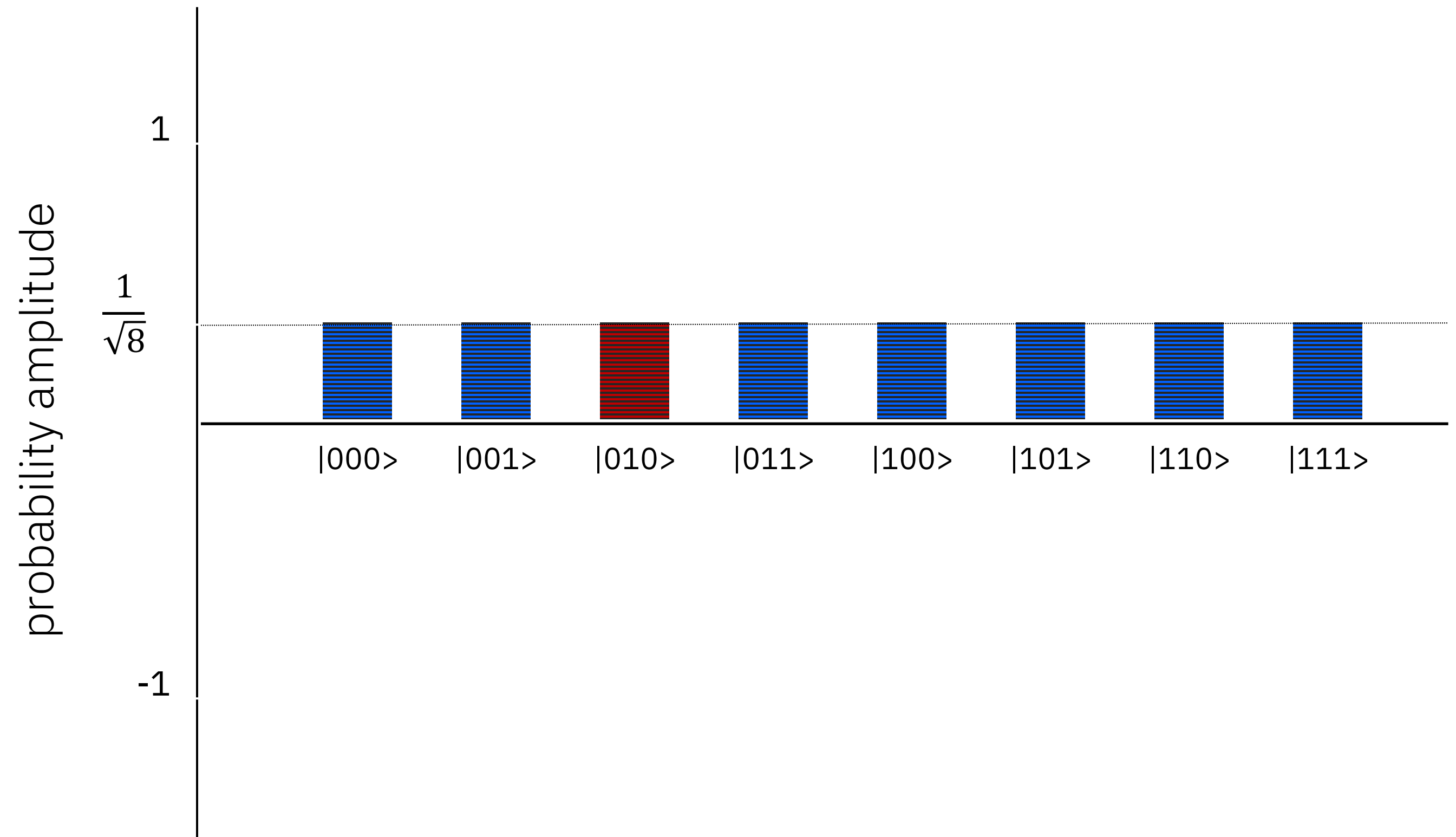


Quantum computing uses essential ideas from quantum mechanics

Interference

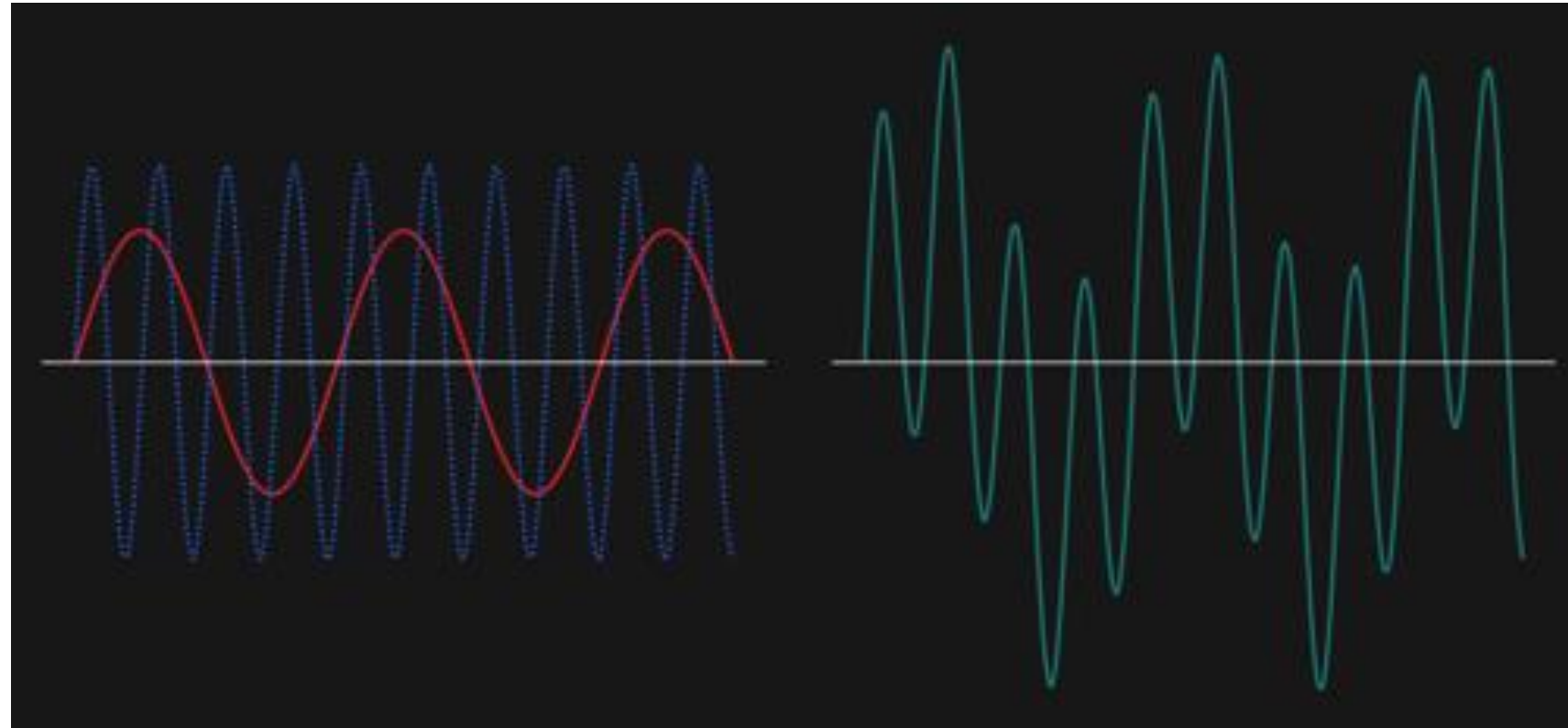


Interference allows us to increase the probability of getting the right answer and decrease the chance of getting the wrong one.

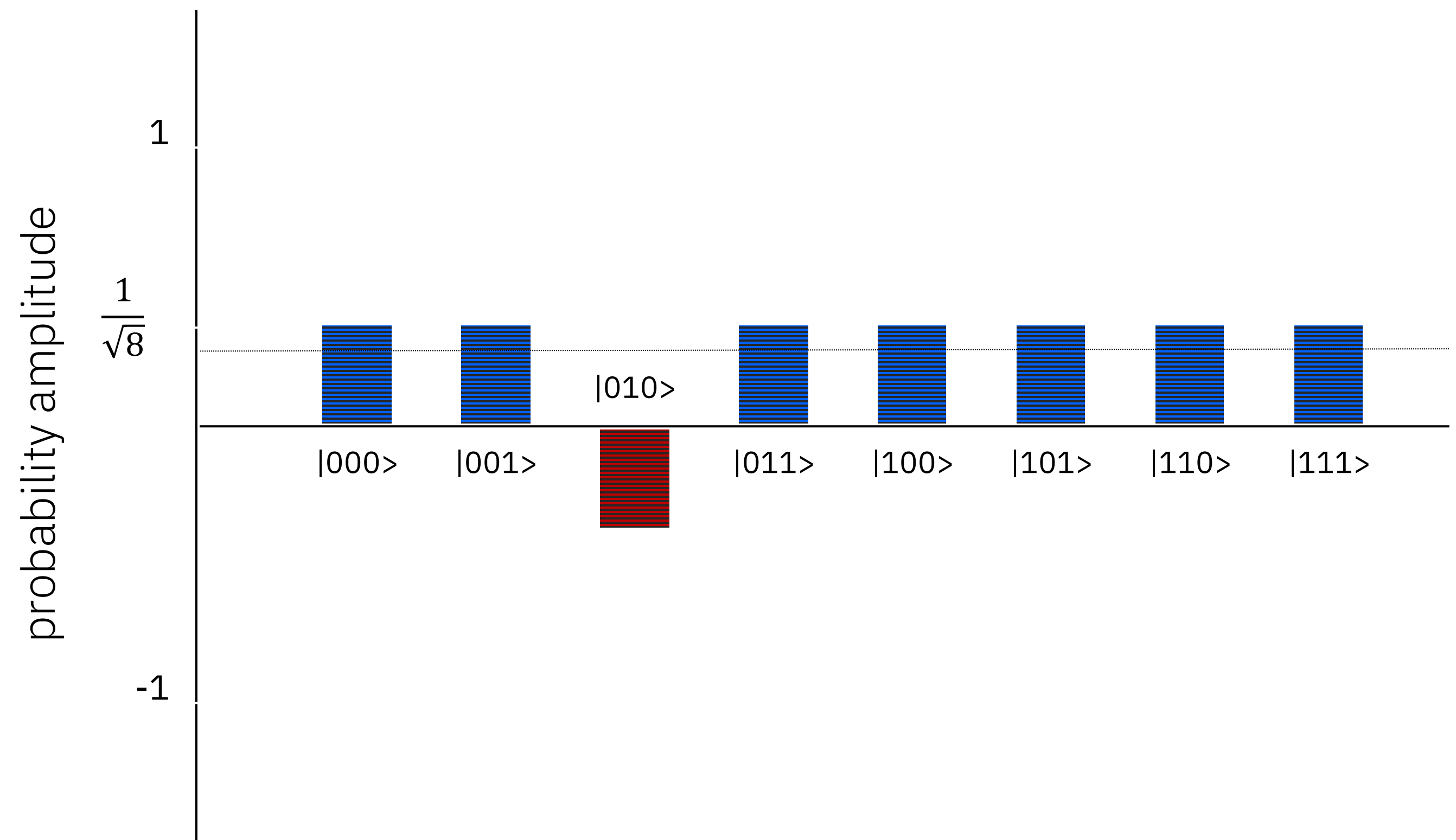


Quantum computing
uses essential ideas from
quantum mechanics

Interference

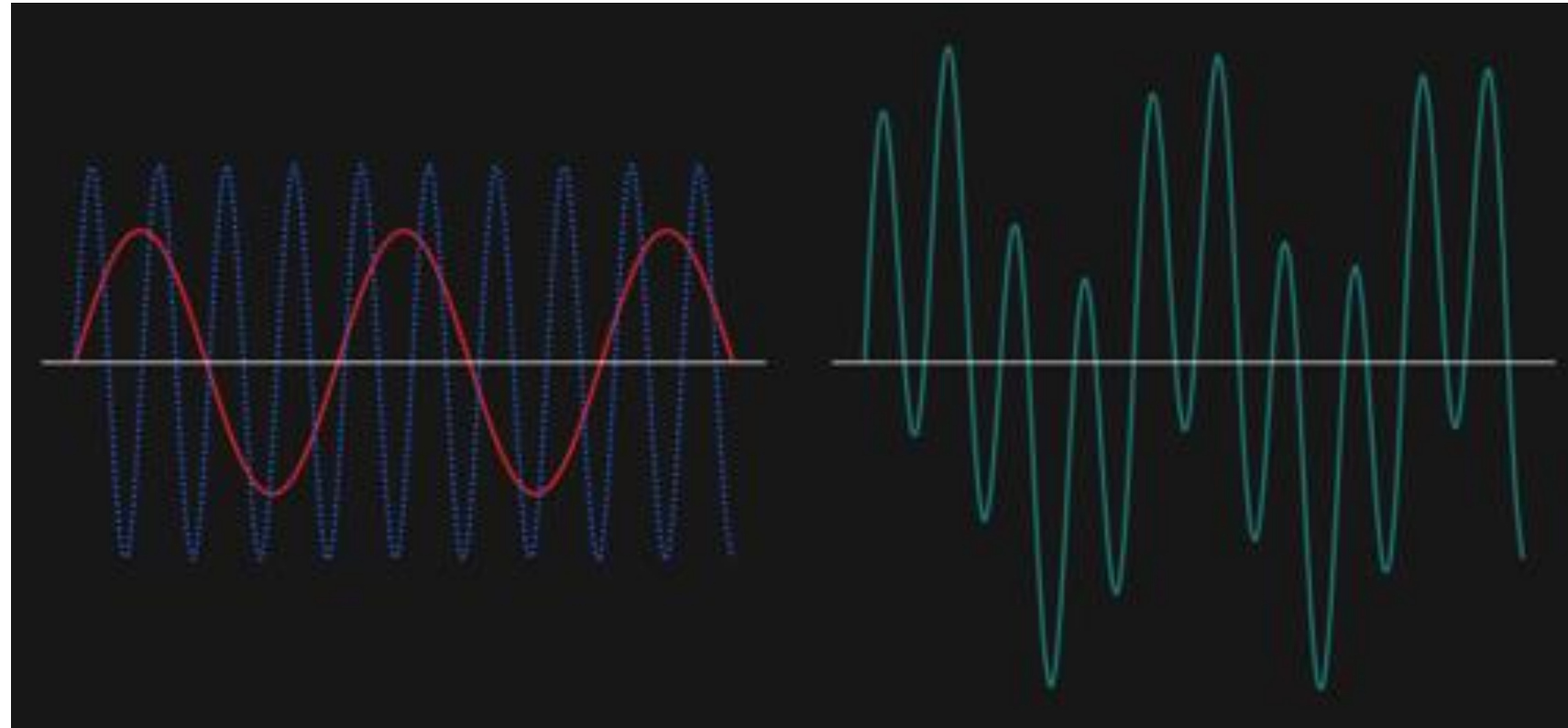


Interference allows us to increase the probability of getting the right answer and decrease the chance of getting the wrong one.

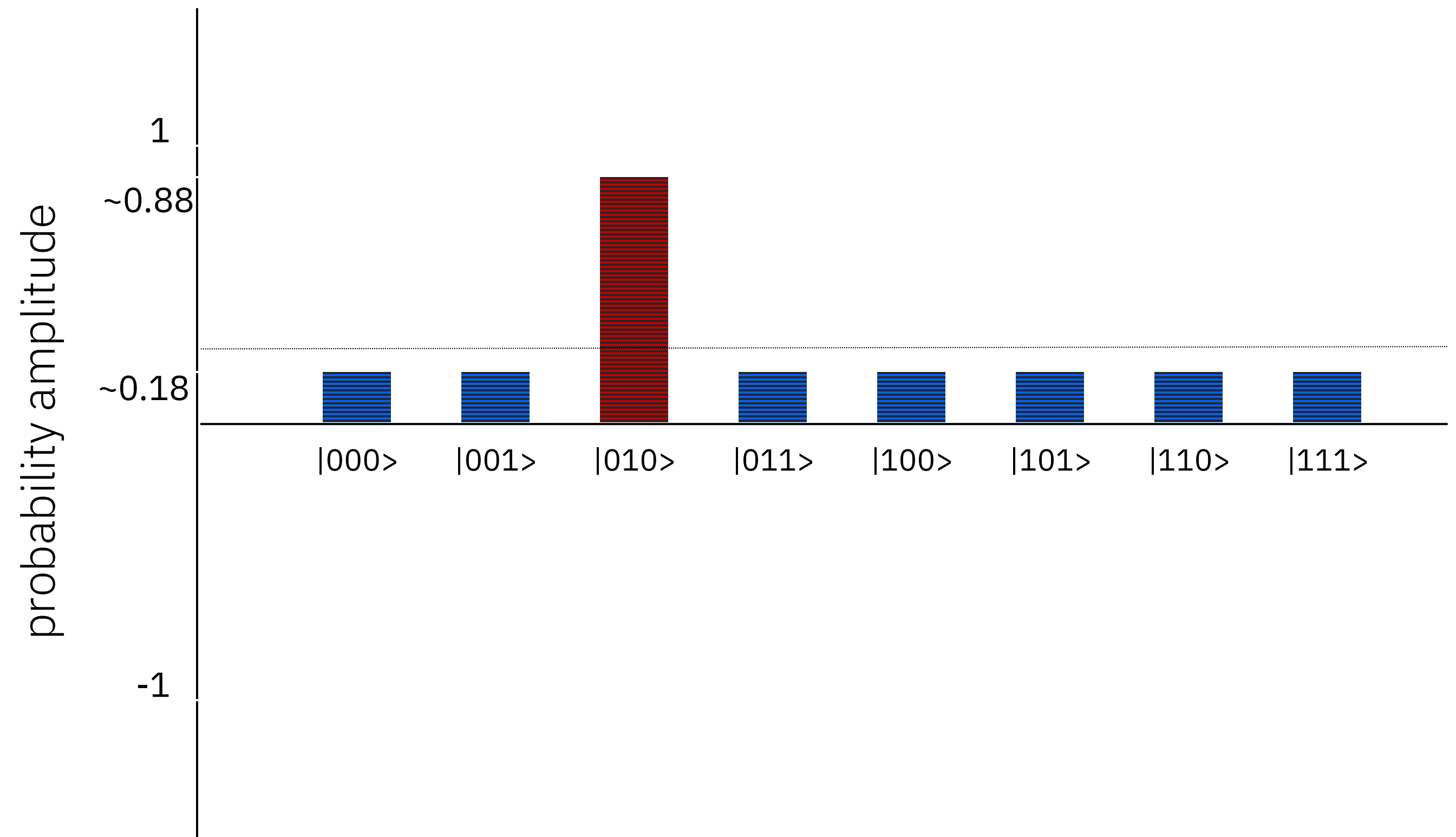


Quantum computing uses essential ideas from quantum mechanics

Interference



Interference allows us to increase the probability of getting the right answer and decrease the chance of getting the wrong one.



The **new** wave of computing



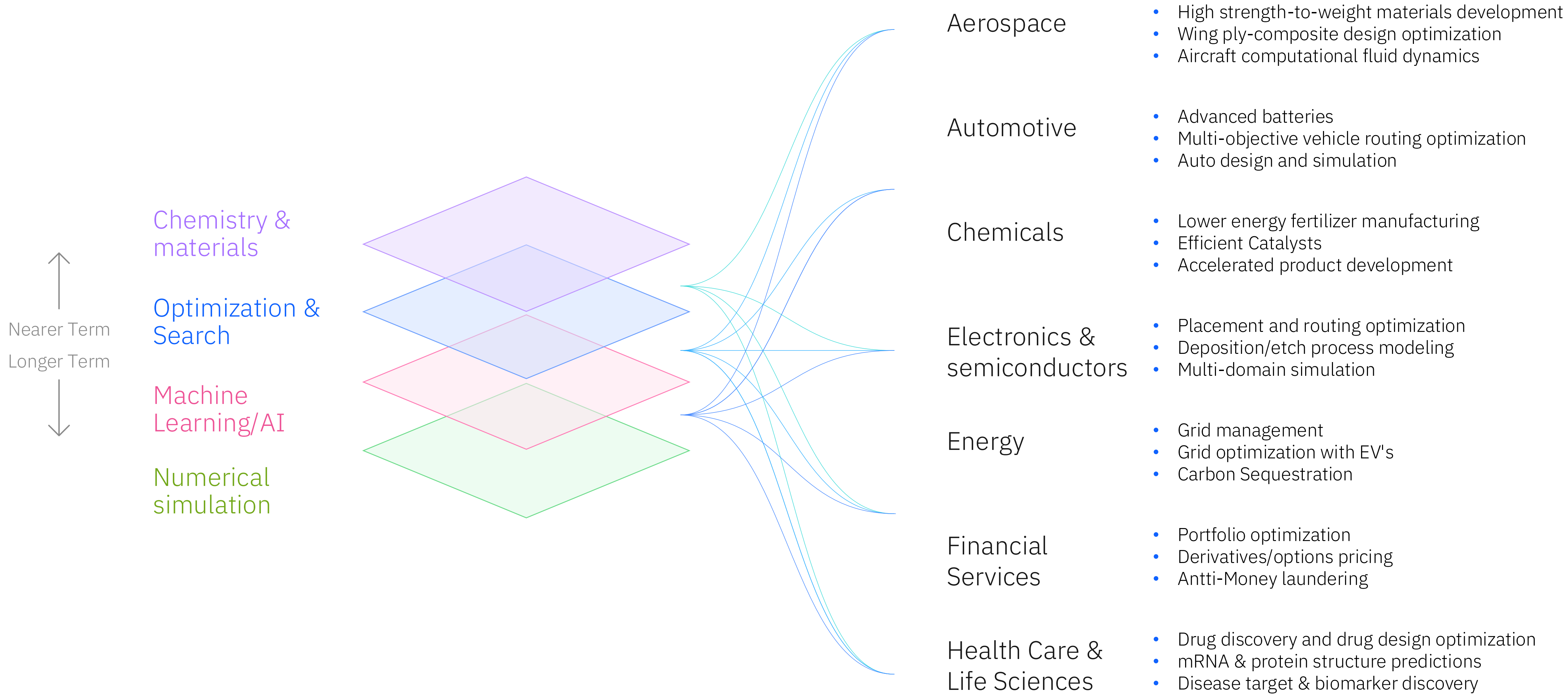
Classical computer
Well suited for many problems



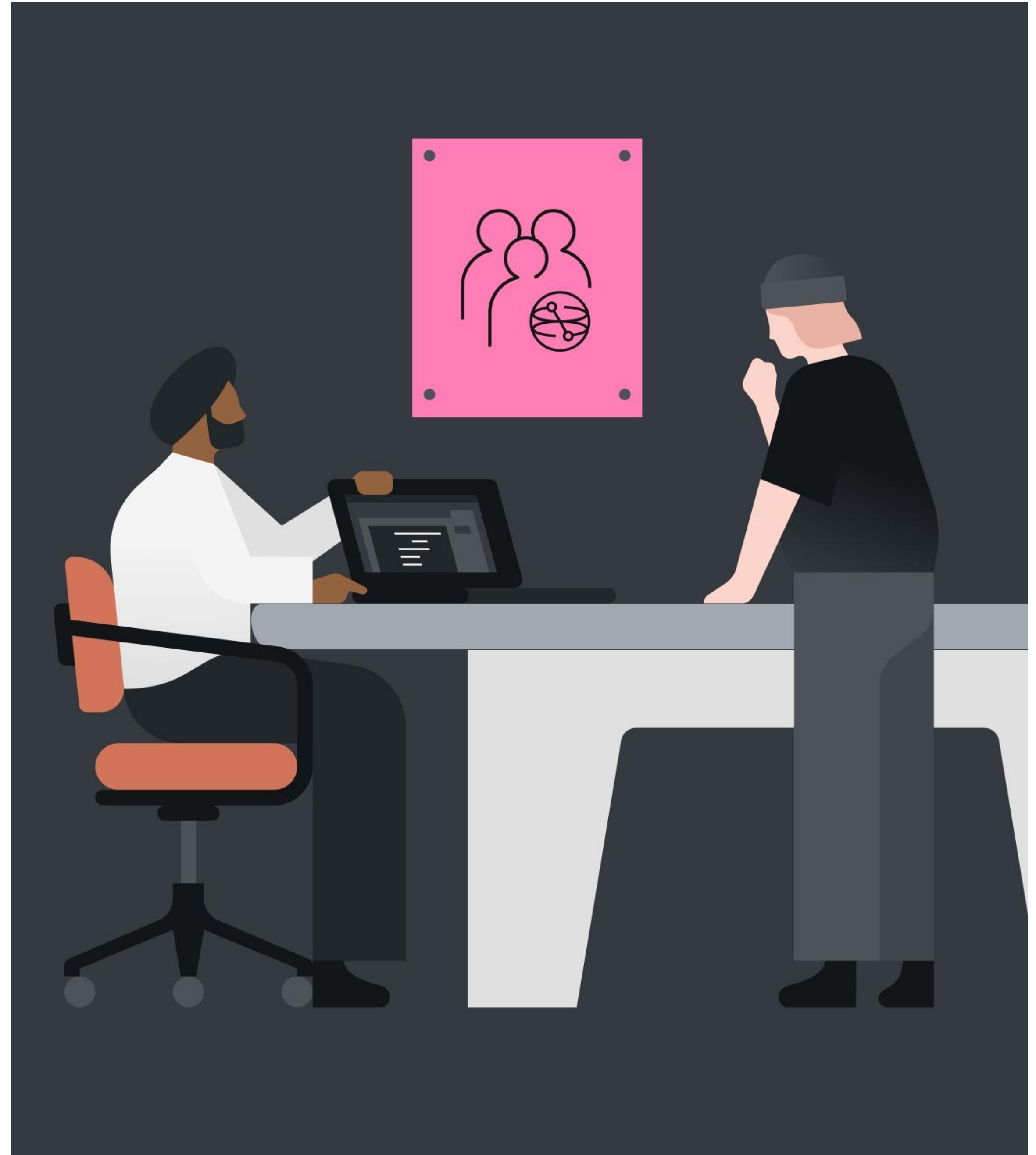
Quantum computer
Unlock classically intractable problems

Quantum computing is expected to have impact across industries

Example use cases



Challenges ahead



Quantum noise and error

Quantum decoherence: the process in which quantum particles and systems can decay, collapse or change, converting into single states measurable by classical physics.

- Intentionally triggered by measuring qubits to send results back via bits.
- Unintentionally triggered due to environmental factors – undesirable!

Noise: anything that affects our quantum system in an undesirable way

Source of noise	What is it?
Coherent noise	Imperfect quantum gates causing over- or under-rotations, resulting in outcomes different to the desired
Incoherent noise	Loss of quantum properties in a qubit due to interaction with its environment
State preparation and measurement	Imperfections in preparing initial quantum states and measuring quantum states
Projection noise	The non-deterministic nature of quantum computers resulting in fluctuations in measurement outcomes

Quantum error mitigation

Quantum error mitigation: tools and methods that allow us to evaluate accurate expectation values from noisy, shallow depth quantum circuits, even before the introduction of fault tolerance.

Examples:

- Probabilistic error cancellation (PEC)
- Zero noise extrapolation (ZNE)
- Twirled-readout error extinction (TRES)

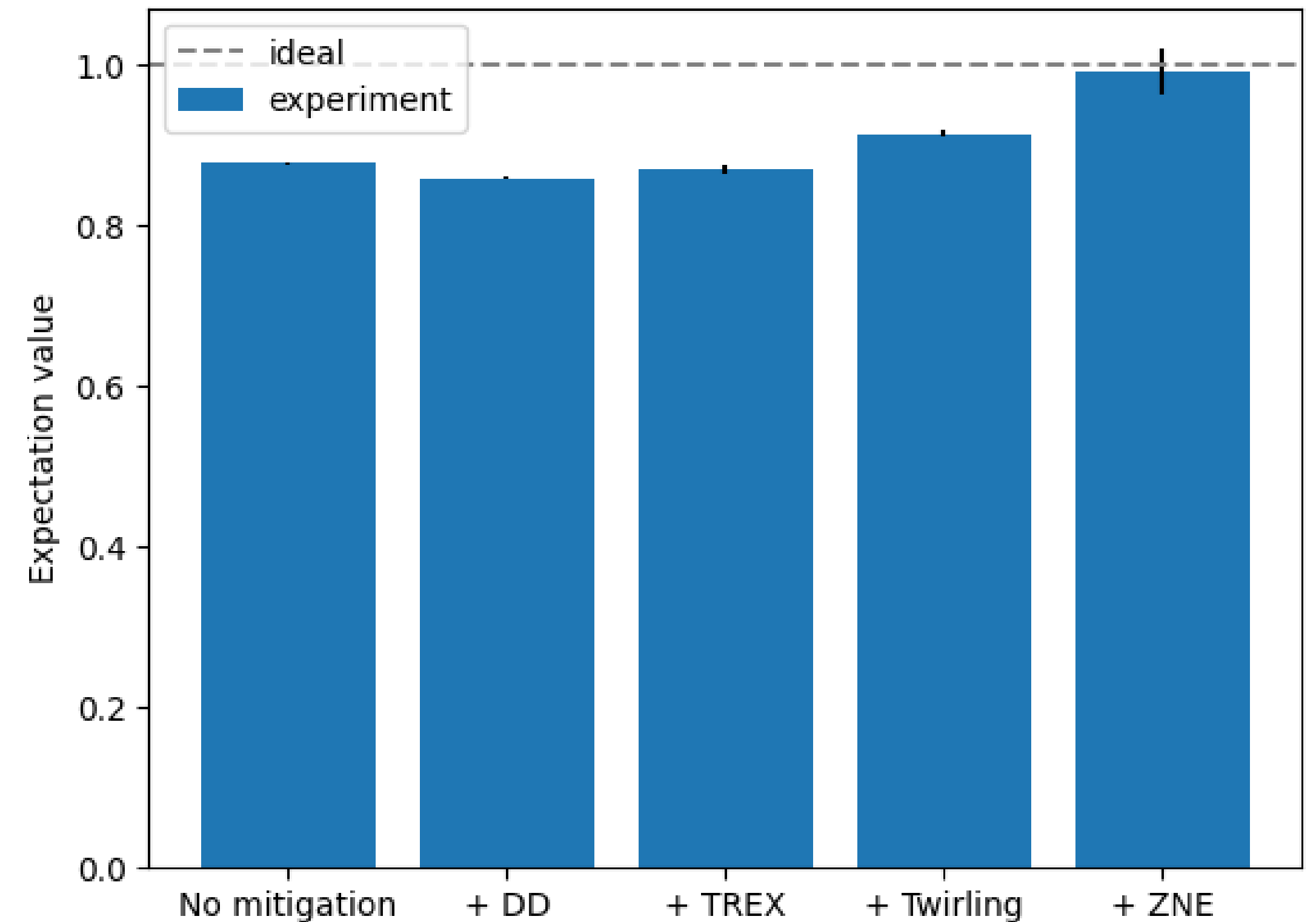
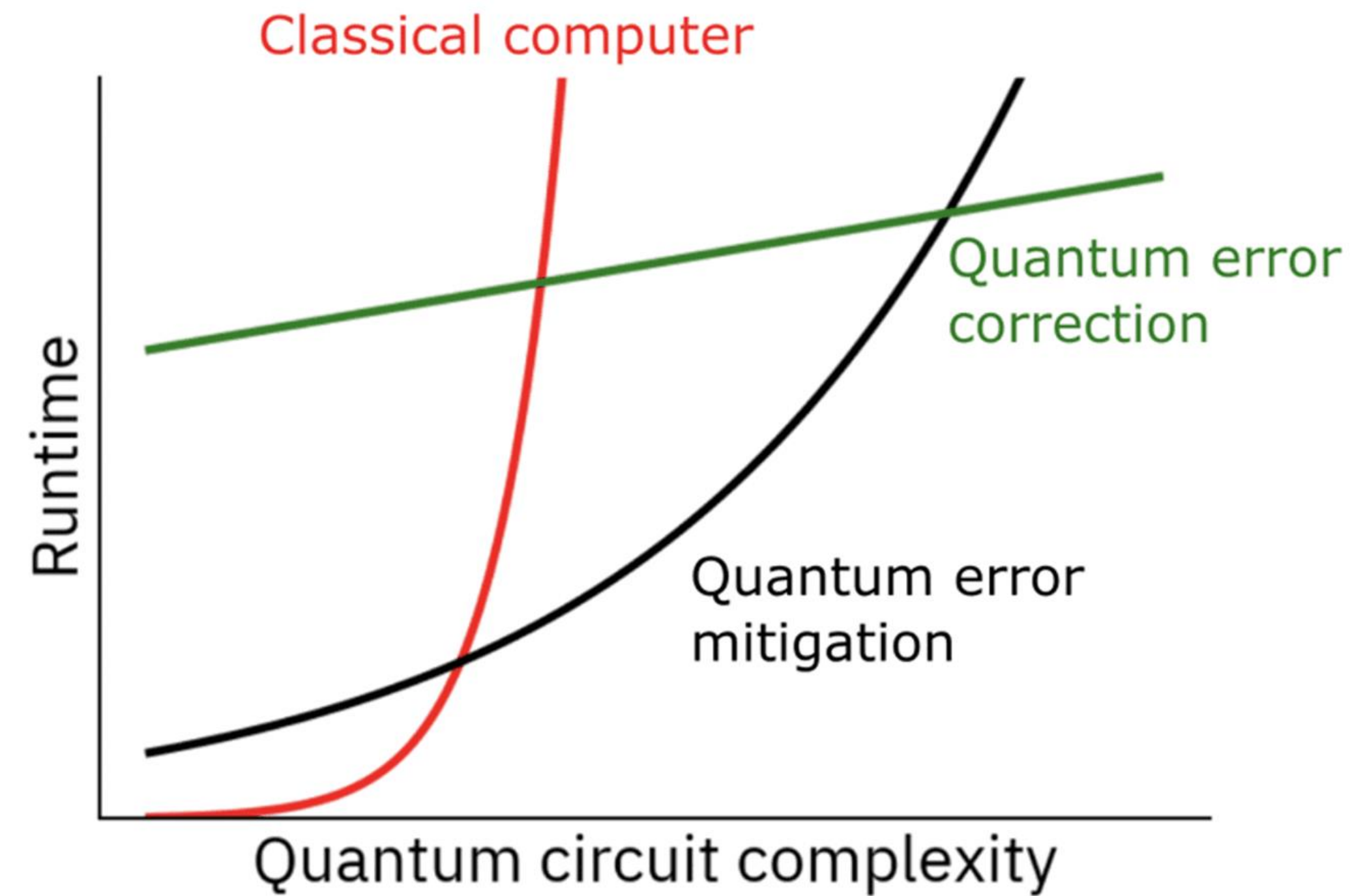


Fig: 50-qubit EfficientSU2 circuit experiment using quantum error mitigation and suppression techniques vs no mitigation (source: IBM, 2025)

Quantum error correction

Quantum error correction: the techniques that identify and correct intrinsic errors in quantum computers.

Lots of error correction already happens in classical computers!



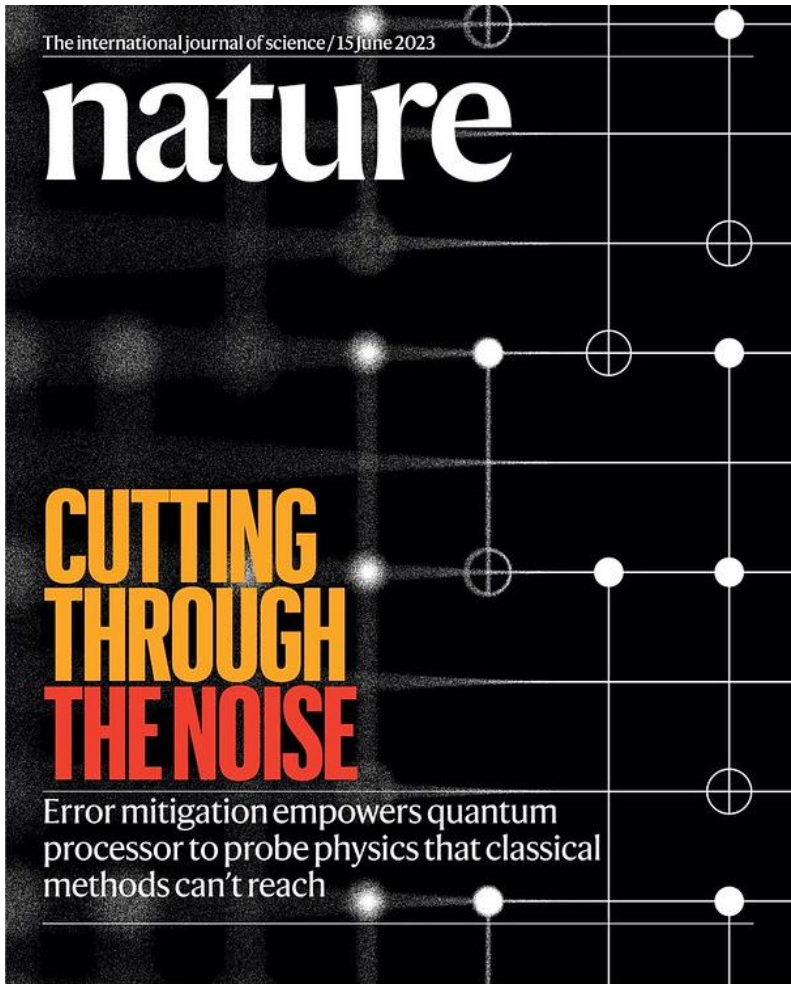
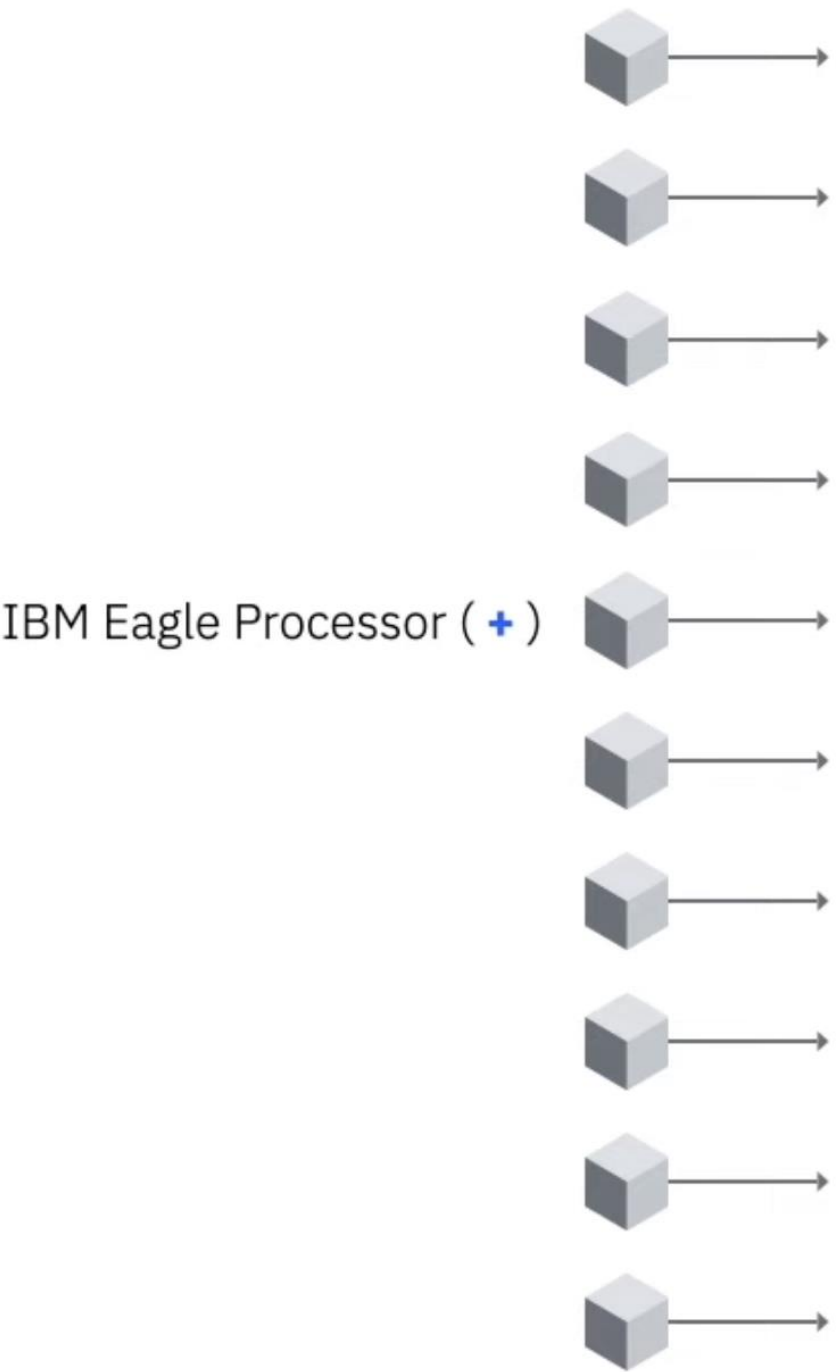
Source: IBM, 2022

Quantum utility

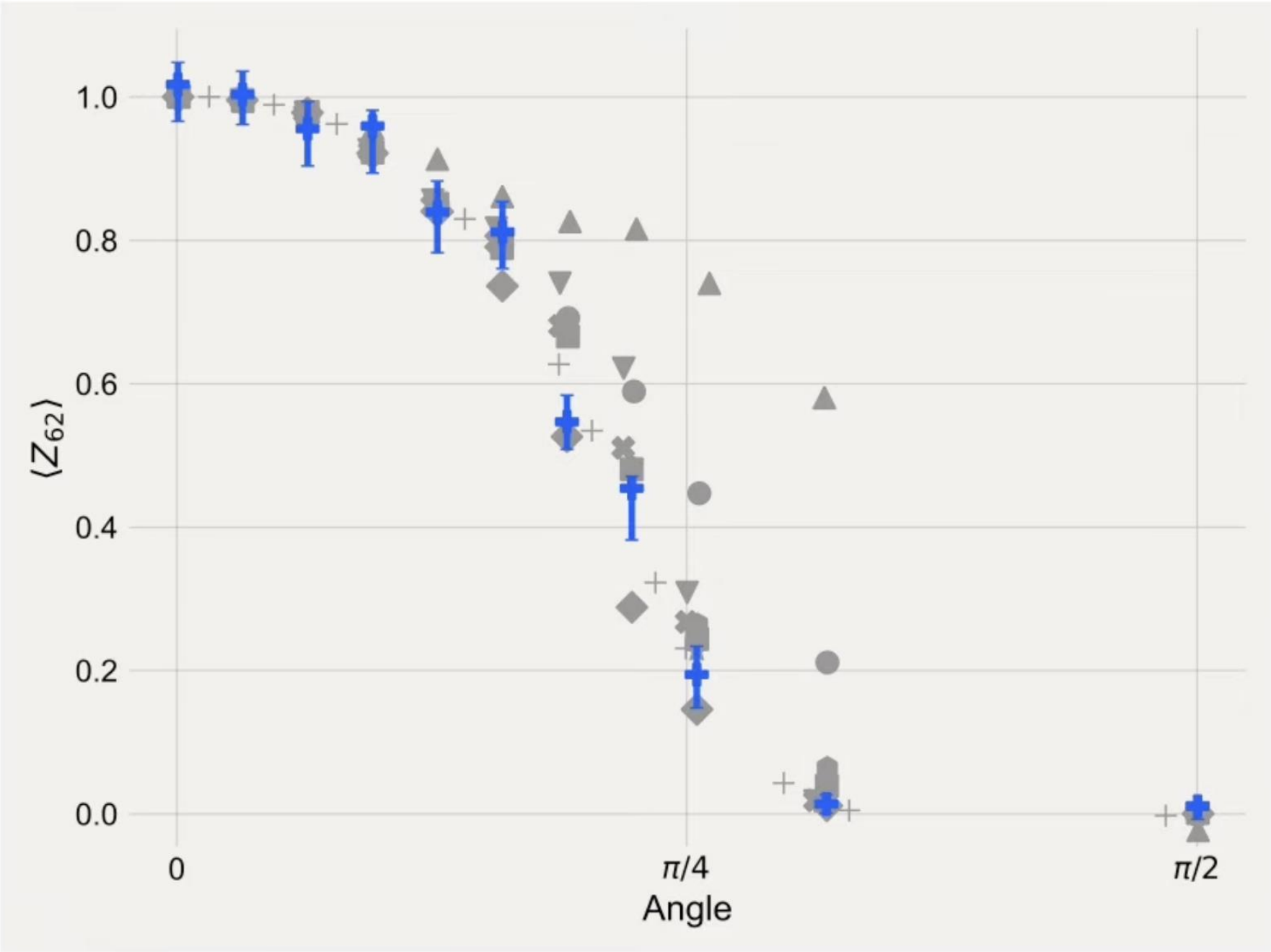
Quantum utility: refers to any quantum computation that provides reliable, accurate solutions to problems that are beyond the reach of brute force classical computing quantum-machine simulators.

“In their experiment, the IBM team demonstrates that it is possible for a quantum computer to outperform leading classical simulations by learning and mitigating errors in the system. The team used the IBM Quantum ‘Eagle’ quantum processor composed of 127 superconducting qubits on a chip to generate large, entangled states that simulate the dynamics of spins in a model of material and accurately predict properties such as its magnetization.”

- IBM Press Newsroom, Jun 14, 2023



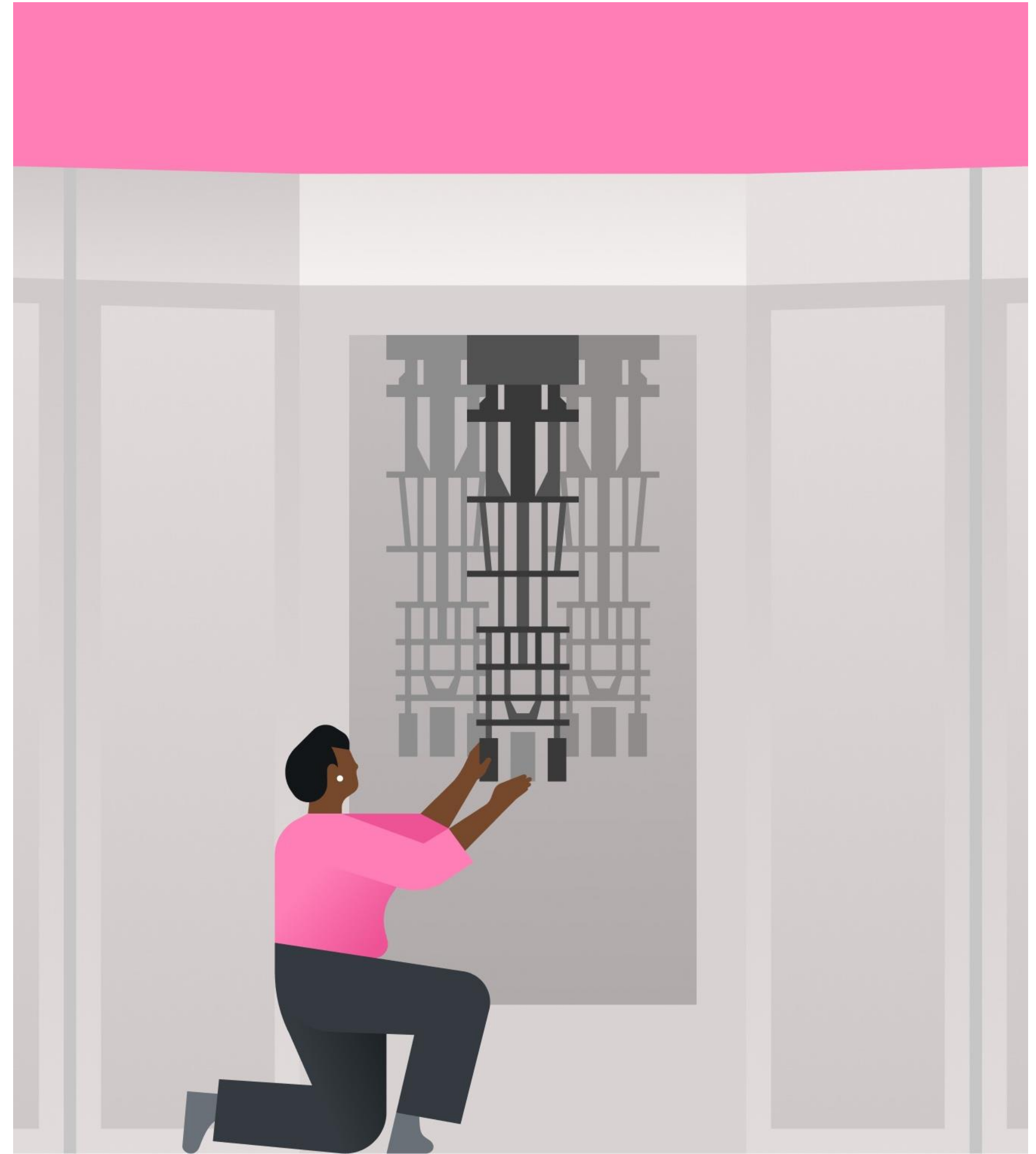
Source: IBM, 2023



Results of IBM “utility paper”: IBM’s Eagle QPU (blue) vs quantum simulators on CPU/GPU (grey).

Quantum advantage

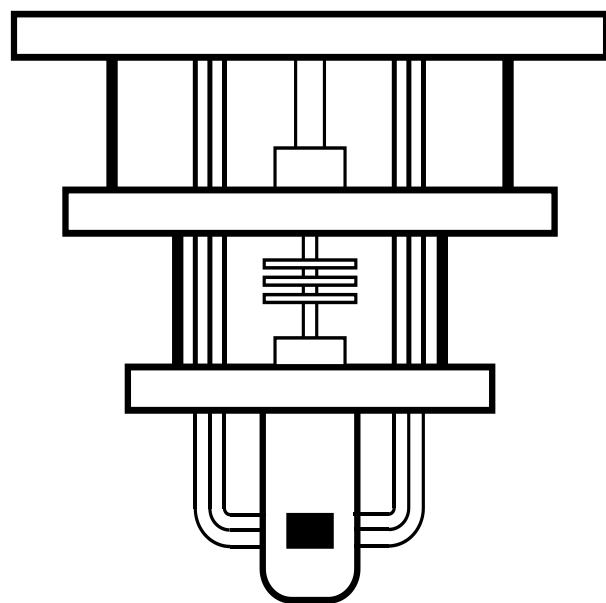
Quantum advantage: said to be achieved when a quantum computer can deliver solutions more efficiently, cost effectively, or more accurately over classical computers.



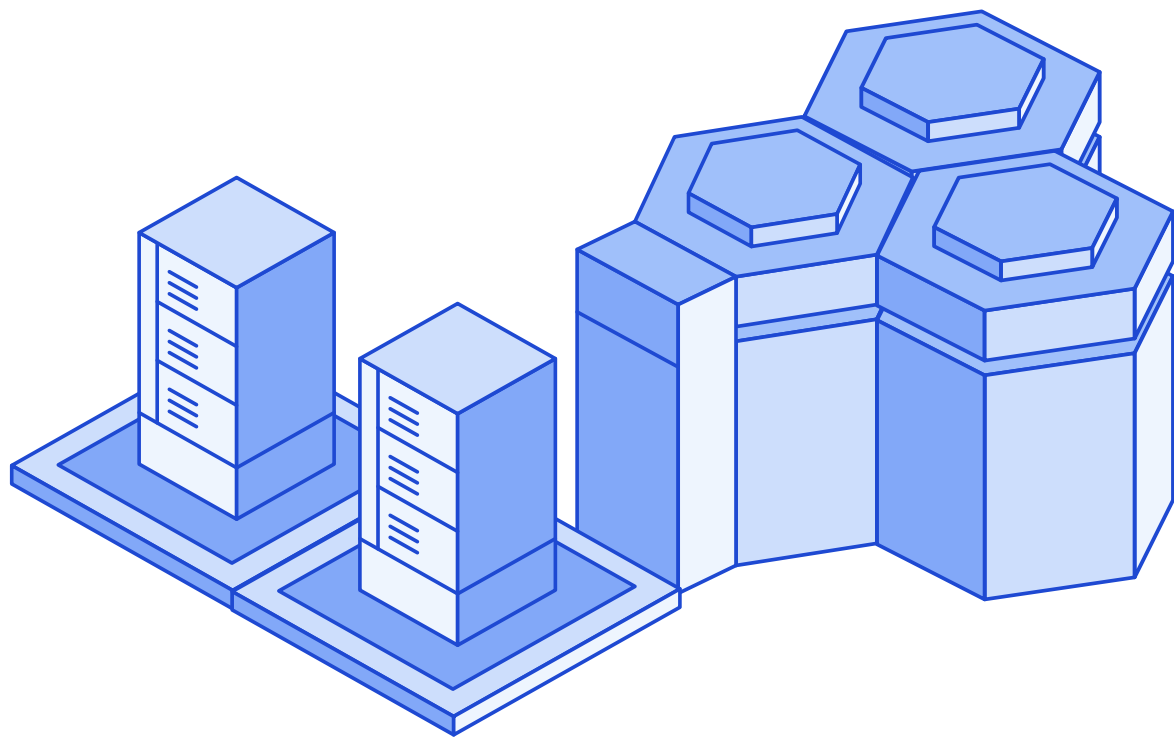
Future of quantum computing



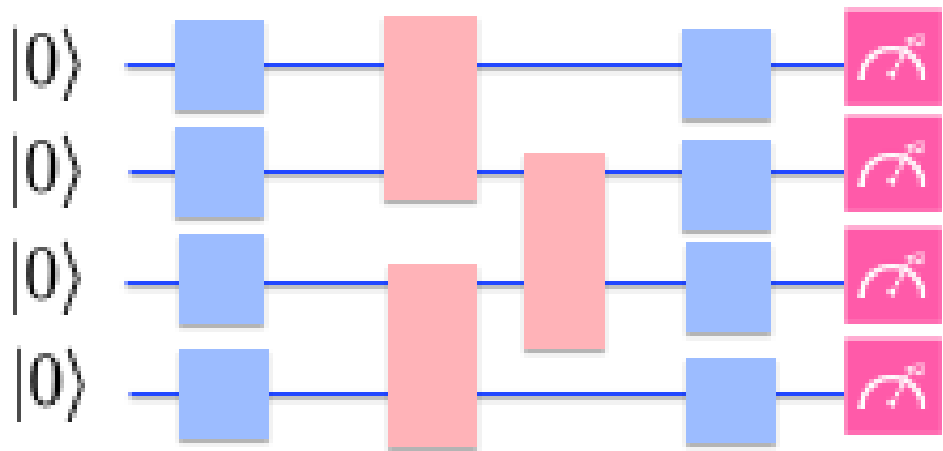
There are many components required for useful quantum computing...



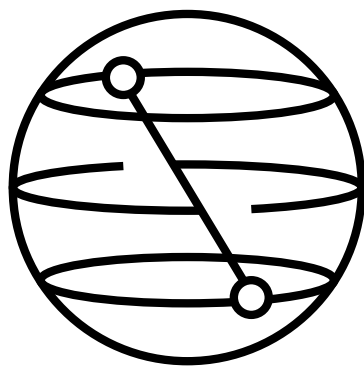
Large-scale quantum processors and hardware



Quantum-centric supercomputing



Quantum algorithms



Quantum software and middleware, like Qiskit

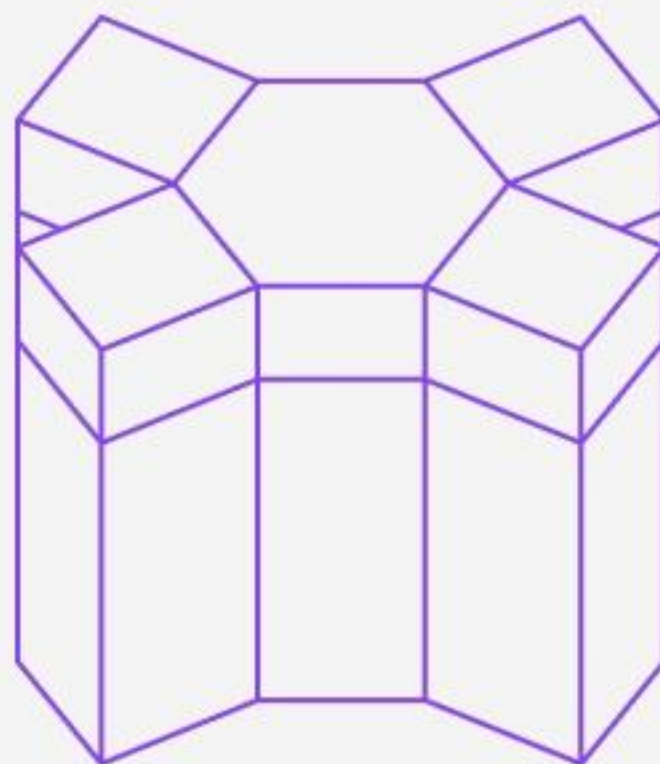


Quantum error correction



Qiskit

+



IBM QPUs

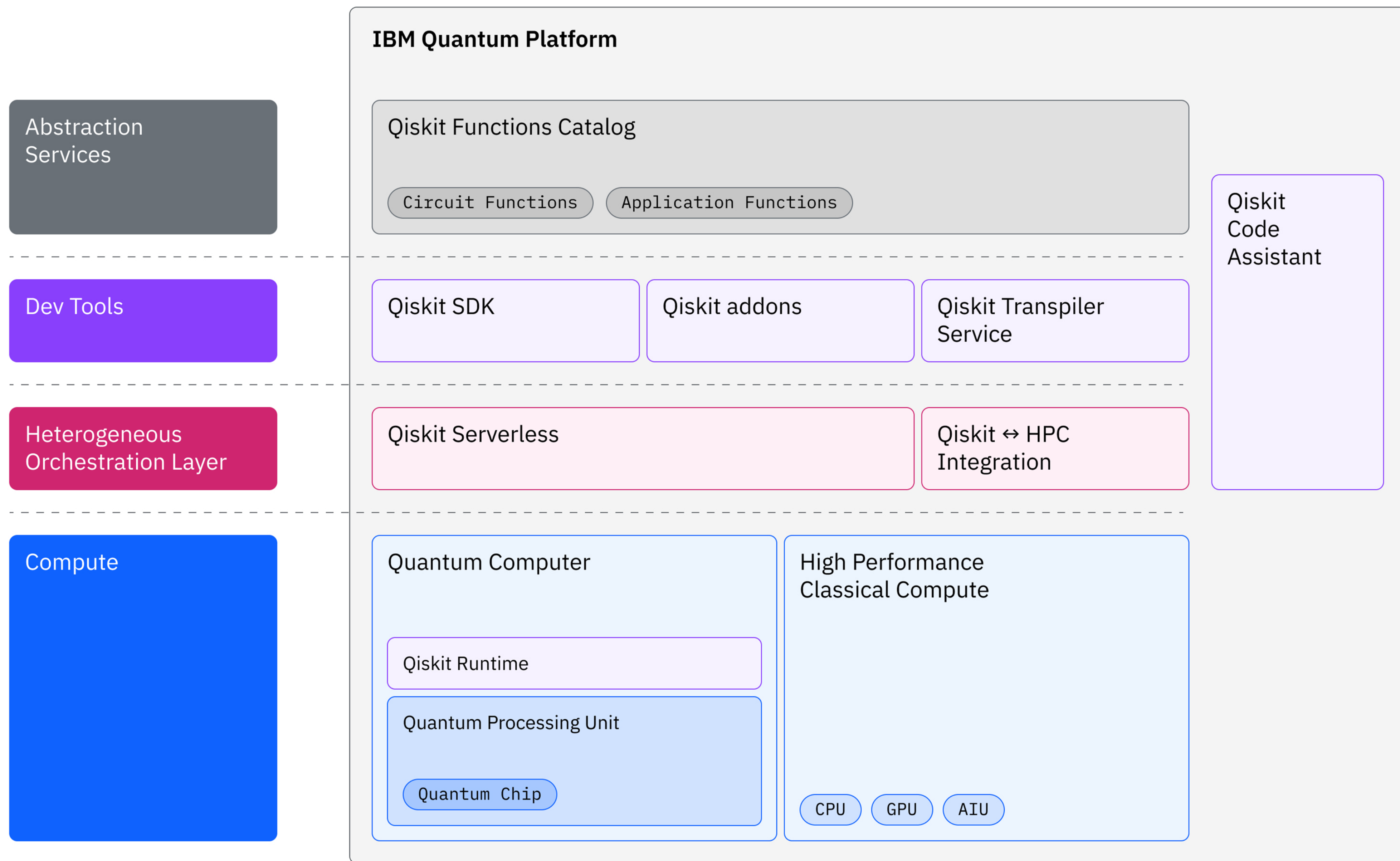
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Utility Scale Work

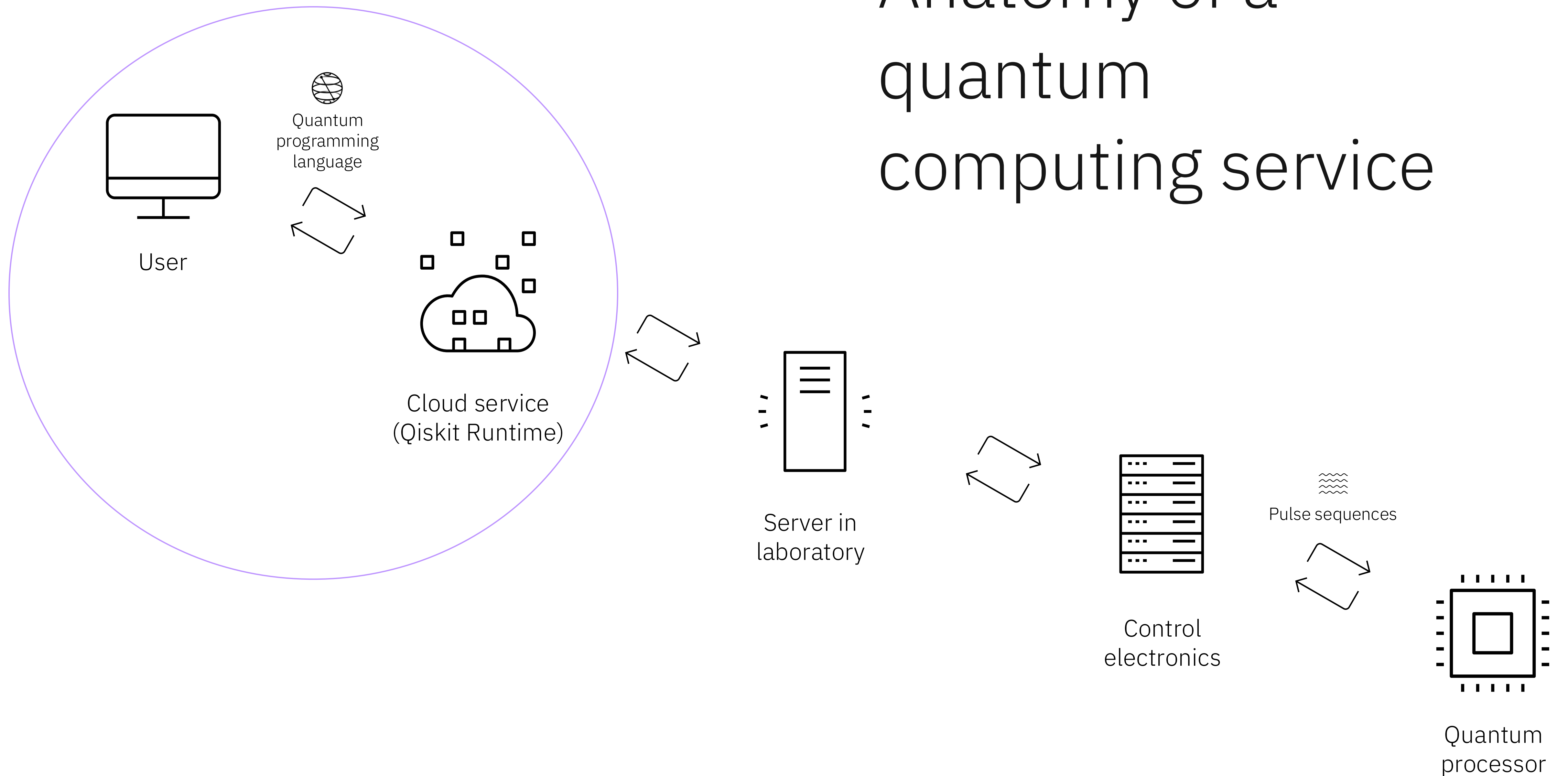
Quantum must be
performant

Quantum must be
easy-to-use



Qiskit SDK

Anatomy of a quantum computing service



What is Qiskit?

The name "Qiskit" is a general term referring to a collection of software for executing programs on quantum computers.

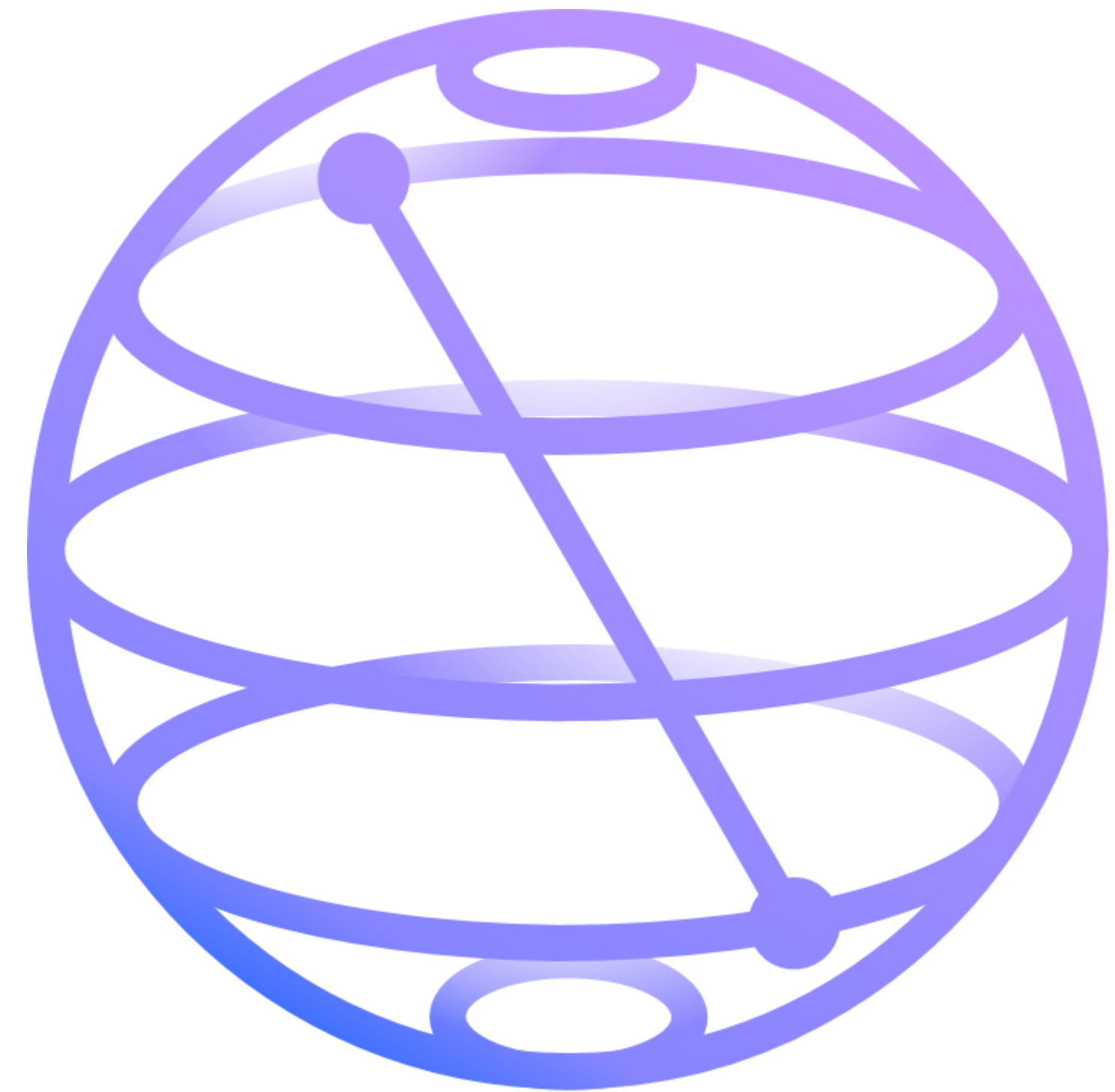
Qiskit SDK is an open-source SDK for working with quantum computers at the level of extended quantum circuits, operators, and primitives.

github.com/Qiskit/qiskit
quantum.cloud.ibm.com/docs

Qiskit Runtime is a cloud service to execute workloads on IBM quantum processing units (QPUs)

The **Qiskit Ecosystem** is a collection of software and projects that build on or extend Qiskit.

qiskit.github.io/ecosystem



Qiskit 2.0

Released April 2025

Qiskit 2.0 is the first release to introduce a C API beyond the usual Python interface, marking the beginning of multi-language support. This sets the foundation for accessing Qiskit from C directly and building bindings for other programming languages on top of C interface.

Two modes of operation:

- Standalone C usage (no Python runtime dependency) - [Installation Guide](#)
- Embedded extension, for calling the C API from a Python context Python - [Extension Guide](#)

Currently, the C interface is minimal but growing: [C API Reference](#)

Other Qiskit 2.0 highlights:

- Rust-powered performance boosts: Up to 2× faster circuit construction and ~20% faster transpilation.
- Support for abstract/deferred timing in dynamic circuits: Enables use of stretches for complex timing logic.
- New 'Box' support: Atomic blocks for the transpiler; useful for twirling, noise learning, and more.

Qiskit SDK

[arXiv:2405.08810](https://arxiv.org/abs/2405.08810)

The lingua franca of quantum computing;
write once and execute quantum circuits on
10+ different hardware providers

IBM Quantum
AQT
IQM
Azure Quantum
Alice & Bob
IonQ
Amazon Braket
Superstaq
Quantinuum
Rigetti

Quantum SDK Preferred
(2024 Unitary Fund Survey)

74%

Qiskit contributors that
are external to IBM

75%

Dependent Qiskit Projects

5185

Qiskit Runtime

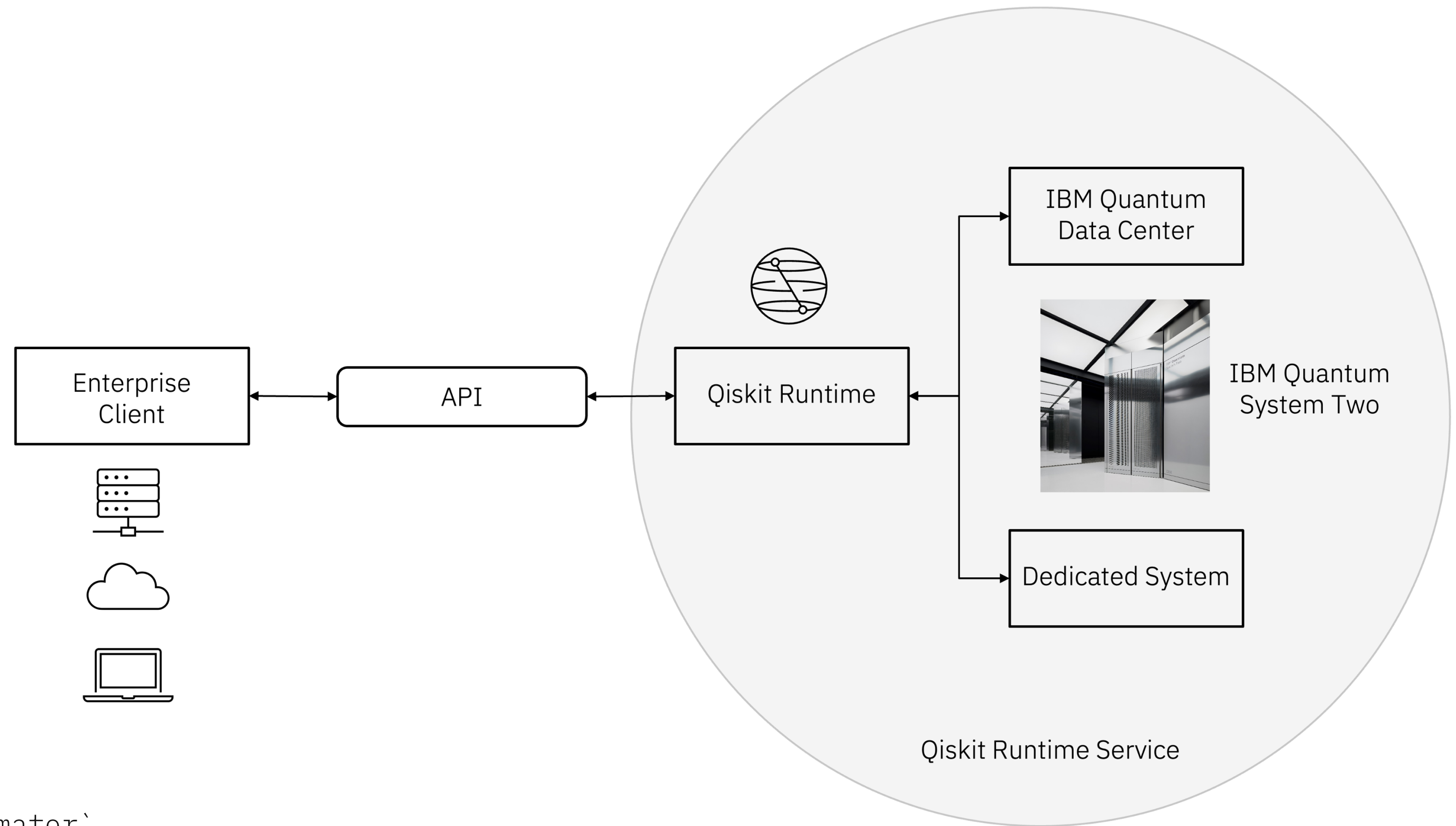
Qiskit Runtime is a cloud-based service for executing quantum computations on IBM Quantum hardware that streamlines quantum computations and provides optimal implementations of Qiskit primitives.

- Runs quantum programs on IBM Quantum hardware through IBM Quantum Platform or IBM Cloud
- Uses additional classical and quantum compute resources including error suppression and error mitigation techniques
- Offers three execution modes: Job, Session, and Batch for different use cases
- Simple installation via pip:

```
`pip install qiskit-ibm-runtime`
```

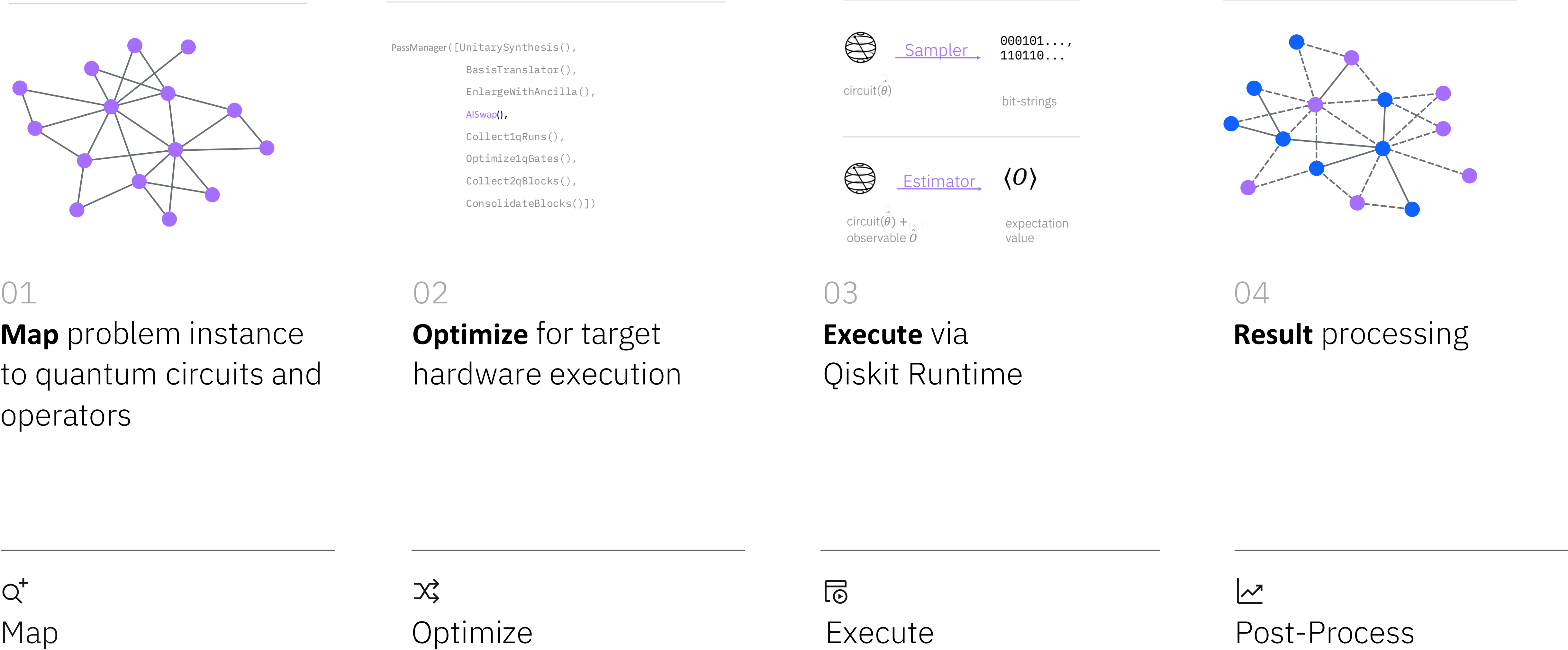
How to import runtime primitives:

```
`from qiskit_ibm_runtime import EstimatorV2 as Estimator`  
`from qiskit_ibm_runtime import SamplerV2 as Sampler`
```



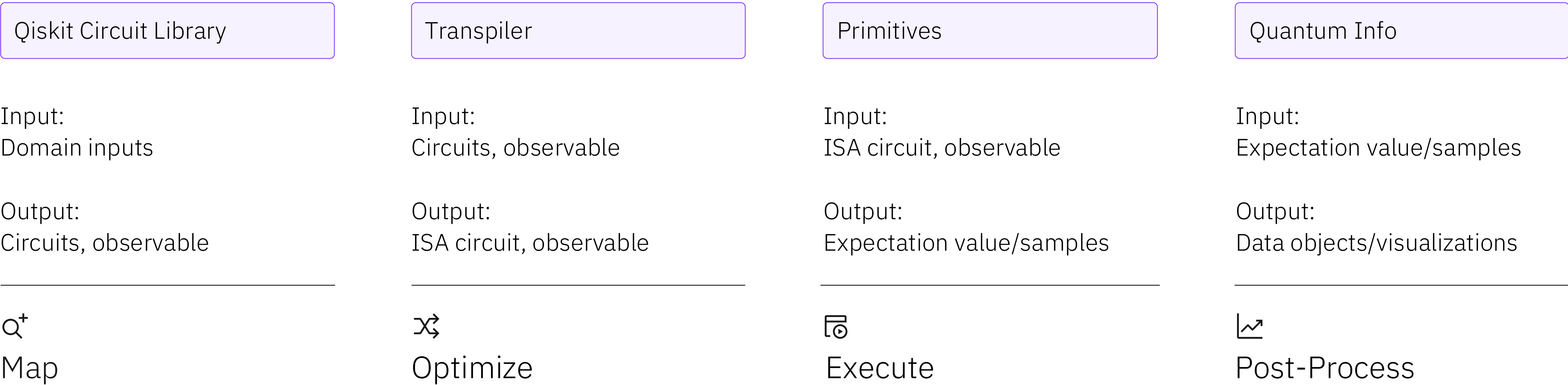
Qiskit Pattern:

The anatomy of a quantum algorithm



Qiskit SDK sets the foundation

Qiskit SDK gives us a base layer of building blocks for building and running quantum algorithms



Qiskit addons build on the Qiskit SDK

A collection of research capabilities developed as modular tools that can plug into a workflow to design new algorithms at the utility scale

AQC-Tensor

MPF

Qiskit Circuit Library

Input:
Domain inputs

Output:
Circuits, observable

Q⁺
Map

OBP

Circuit cutting

Transpiler

Input:
Circuits, observable

Output:
ISA circuit, observable

↗
Optimize

Starting with multi-product formulas (MPF), approximate quantum compilation (AQC-Tensor), operator backpropagation (OBP), and sample-based quantum diagonalization (SQD).

Primitives

Input:
ISA circuit, observable

Output:
Expectation value/samples

📄
Execute

SQD

M3

Quantum Info

Input:
Expectation value/samples

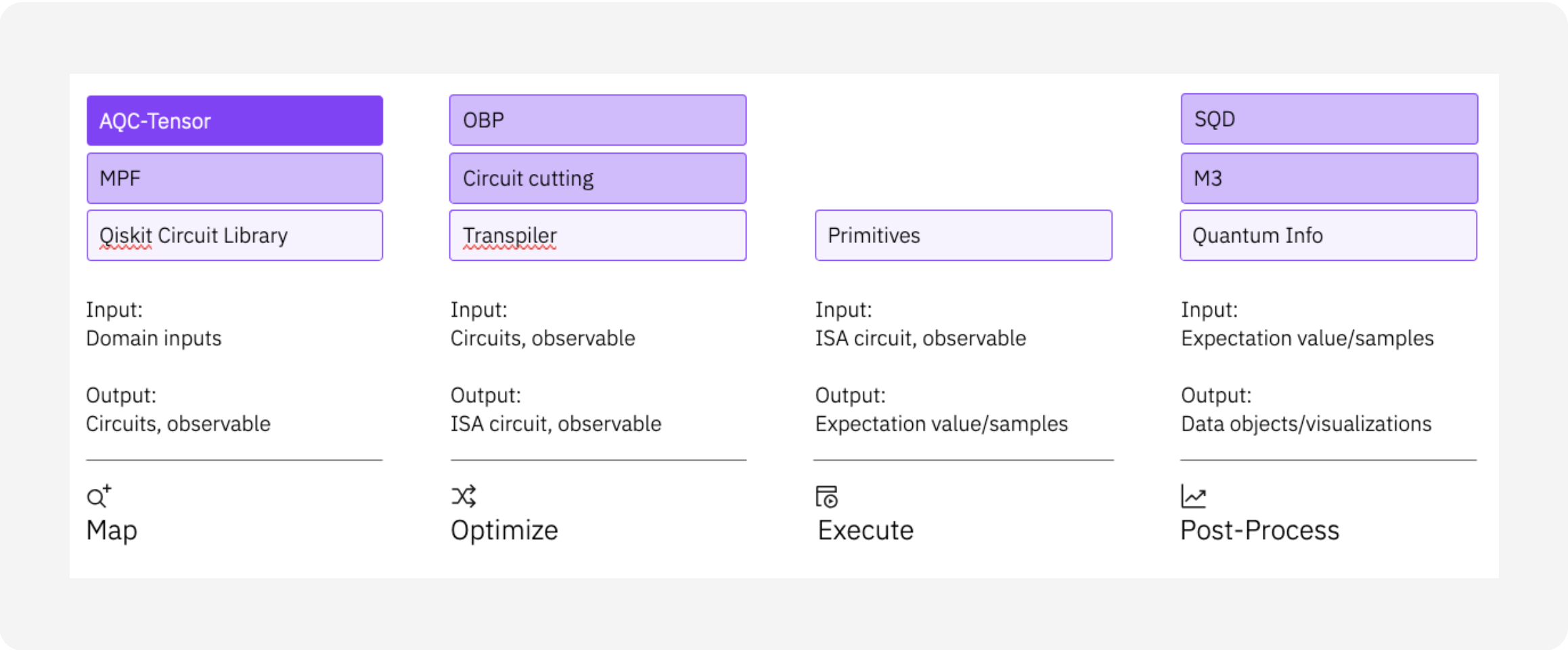
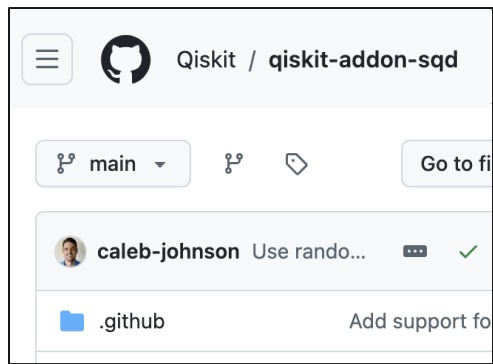
Output:
Data objects/visualizations

📈
Post-Process

Accelerating discovery with Qiskit addons

Qiskit addons - a collection of research capabilities developed as modular tools that can plug into a workflow to design new algorithms at the utility scale, starting with:

- Multi-product formulas (MPF)
- Approximate quantum compilation (AQC-Tensor)
- Operator backpropagation (OBP)
- Sample-based quantum diagonalization (SQD)



Faster research
Up to:

↑ 5x

1 year of research

“Chemistry Beyond Exact Solutions on a Quantum-Centric Supercomputer”

—
arXiv:2405:05068

RIKEN

2.5 months of research

“Accurate quantum-centric simulations of supramolecular interactions”

—
arXiv:2410:09209

Cleveland Clinic

2.5 months of research

“Quantum-Centric Study of Methylene Singlet and Triplet States”

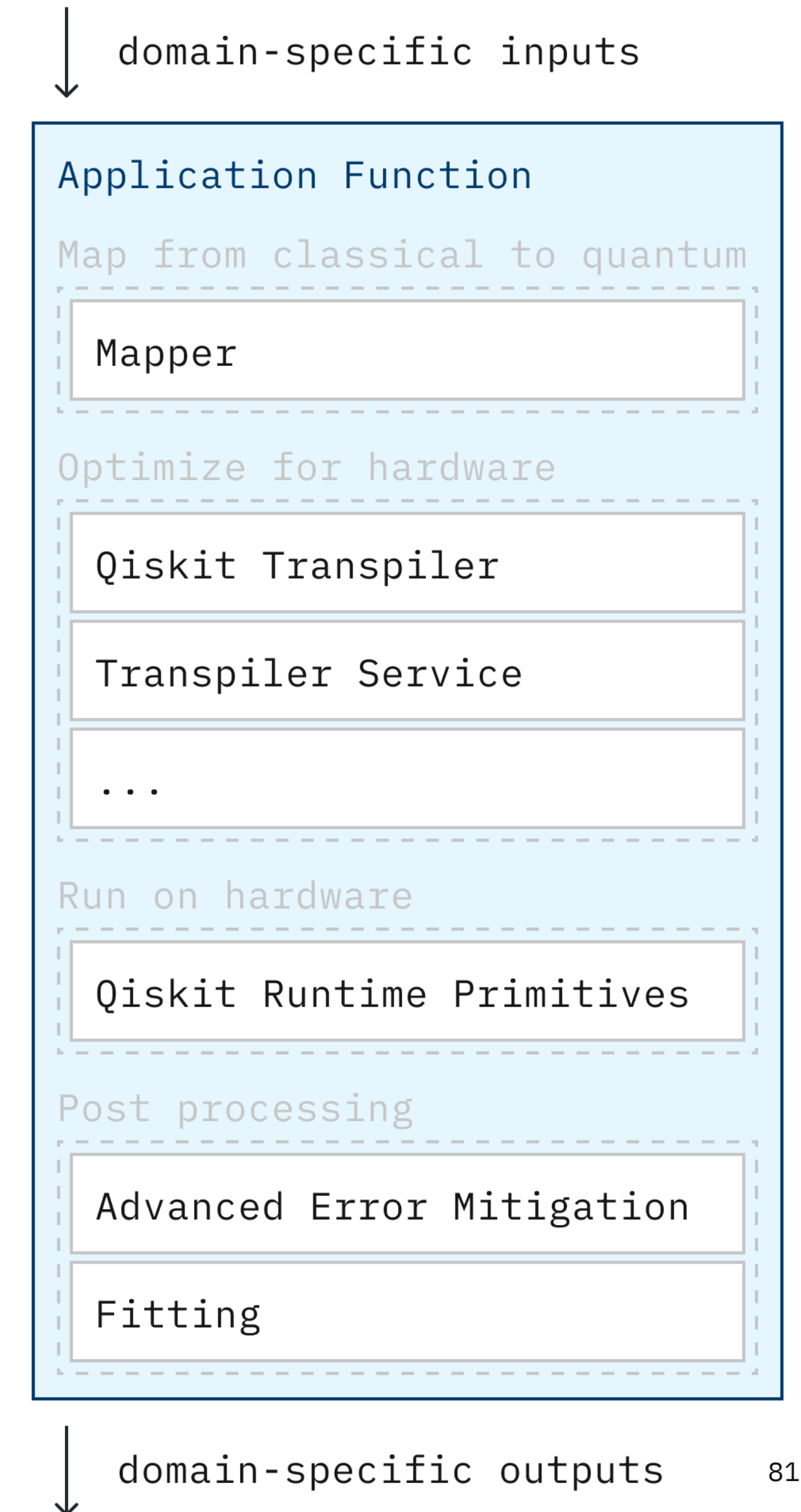
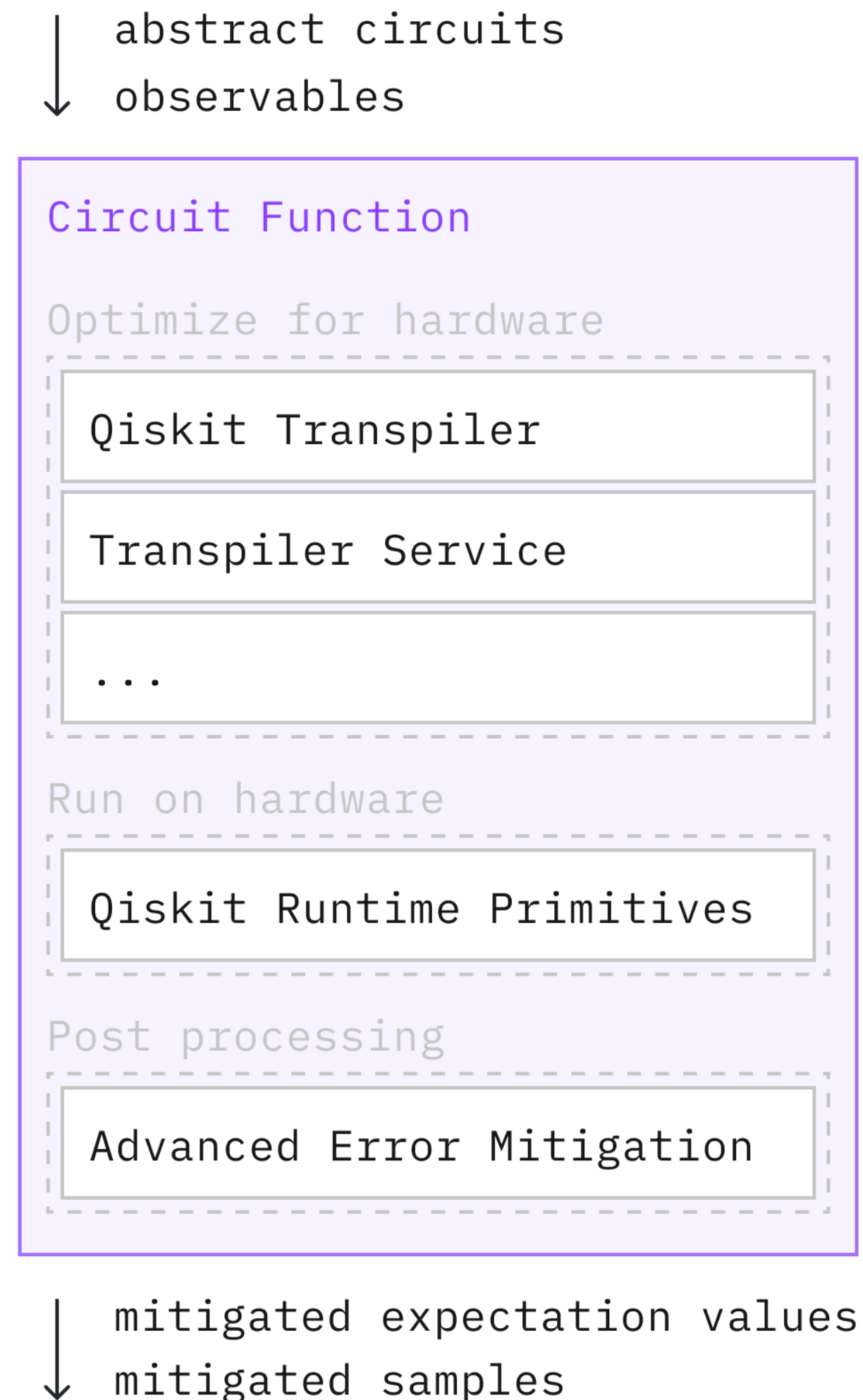
—
arxiv.org/abs/2411.04827

LOCKHEED MARTIN

Qiskit Functions are abstracted services, designed to accelerate development

IBM is [partnering](#) with the broader quantum computing [ecosystem](#) to offer a catalog of [managed, utility-scale services](#) to unlock new possibilities:

- **Circuit Functions:** Enabling quantum computational scientists to discover new algorithms and applications, without needing to manage transpilation, error suppression, or error mitigation
- **Application Functions:** Enabling data scientists and enterprise developers to integrate quantum into industry workflows, while leveraging familiar domain abstractions



Qiskit Ecosystem

Beyond Qiskit there are many open-source projects that use the "Qiskit" name but interface with Qiskit to provide additional functionality.

The Qiskit SDK is designed in a modular, extensible way to make it easy for developers to create projects like these that extend its capabilities.

Notable Projects:

- Qiskit Aer (qiskit-aer): Quantum computing simulators with realistic noise models (IBM Quantum)
- qBraid SDK (qbraid): Platform-agnostic quantum runtime framework for lifecycle management of quantum jobs (qBraid)
- mthree (mthree): Package implementing Matrix-free Measurement Mitigation technique (IBM Quantum)

For more projects, visit the [Qiskit ecosystem page](#).



Featured

Circuit Knitting Toolbox Repo Documentation Cut large circuits into smaller, hardware-executable circuits, then use their results to reconstruct the original outcome. ☆ 87 IBM-maintained Algorithms Circuit building tool	Qiskit SAT Synthesis Repo Synthesis plugins for Cliffords, linear functions, permutations, and more. ☆ 7 Community-maintained Transpiler plugin	qiskit-qubit-reuse Repo A Qiskit transpiler stage plugin that reuses qubits through mid-circuit measurement and reset. ☆ 20 Community-maintained Paper implementation Transpiler plugin	mthree Repo Documentation Matrix-free Measurement Mitigation (M3). Reduce the effects of readout errors from noisy quantum hardware. ☆ 41 IBM-maintained Error mitigation
---	---	---	---

All projects

Filter

Clear

Types

Application package

Compute provider

Circuit building tool

Circuit simulator

Converter

Cloud service

Game

Paper implementation

Software development kit

Transpiler plugin

Subjects

Algorithms

Chemistry

Error correction

Error mitigation

Finance

Machine learning

Optimization

Physics

Search projects

Stars (High to low)

TorchQuantum

[Repo](#)

A PyTorch-centric hybrid classical-quantum dynamic neural networks framework.

☆ 1435 Community-maintained

Machine learning Application package

Mitiq

[Repo](#)

Mitiq is a Python toolkit for implementing error mitigation techniques on quantum computers.

☆ 381 Community-maintained

Error mitigation

Qiskit Machine Learning

[Repo](#) [Documentation](#)

Use flexible building blocks, such as quantum kernels and neural networks, to create quantum machine learning algorithms.

☆ 744 Community-maintained

Algorithms Machine learning

Application package

Aer

[Repo](#) [Documentation](#)

High-performance quantum computing simulators with realistic noise models.

☆ 537 IBM-maintained

Circuit simulator

Qiskit Nature

[Repo](#) [Documentation](#)

Model and solve problems in physics, chemistry, material science, and biology by using quantum simulations.

☆ 326 Community-maintained

Algorithms Chemistry Physics

Application package

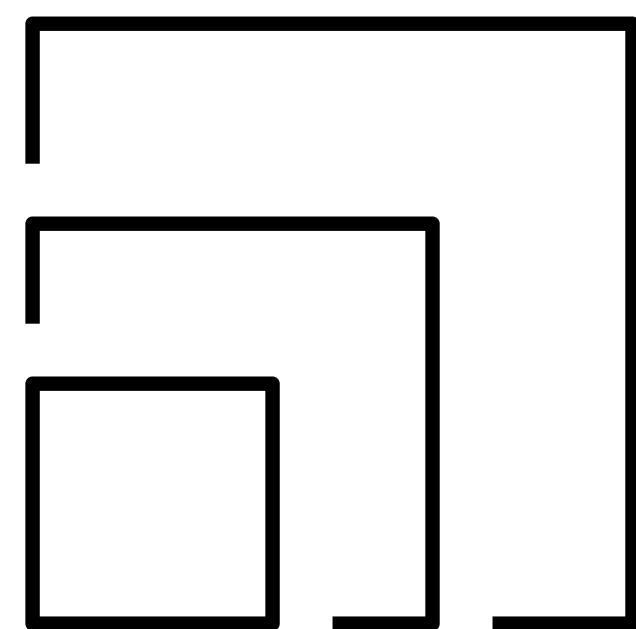
Qiskit Finance

[Repo](#) [Documentation](#)

Includes uncertainty components for stock/ securities problems, Ising translators for portfolio optimizations, and data providers.

☆ 258 Community-maintained

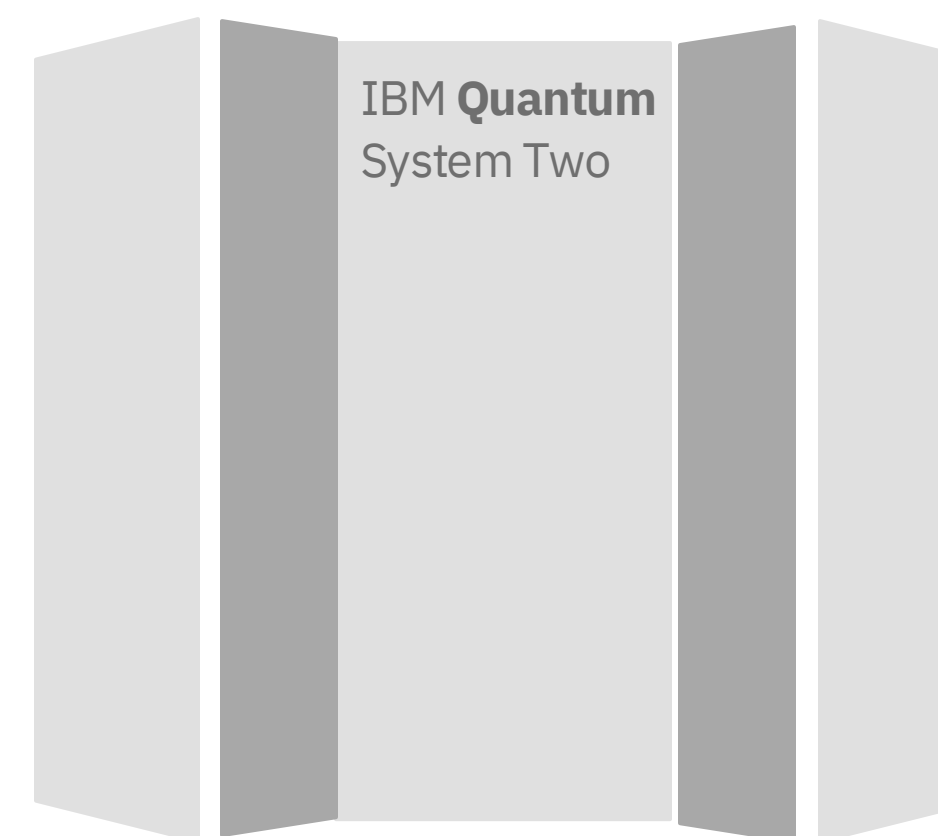
Algorithms Finance Application package



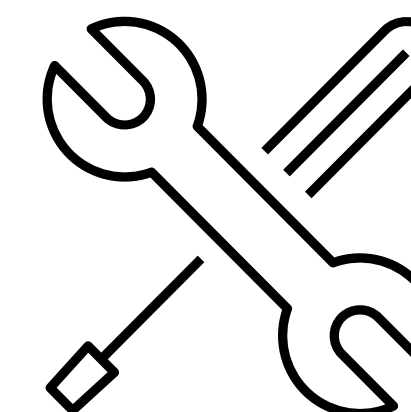
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Quantum
Software

(powered by Qiskit Addons
and Qiskit Functions)

Qiskit

QPUs

Useful Work

The IBM Quantum platform unlocks **research with quantum**

IBM Quantum

Check in!



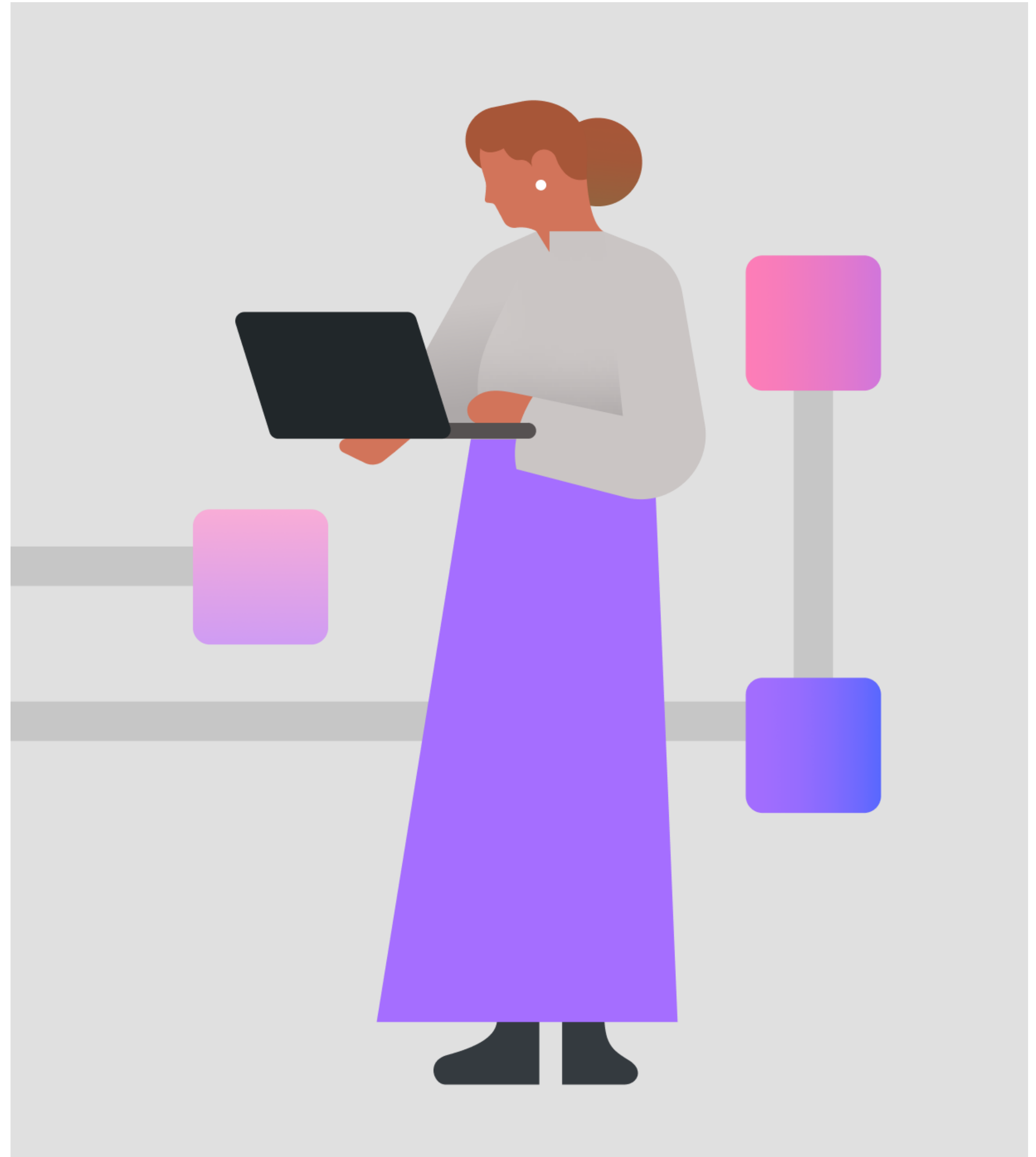
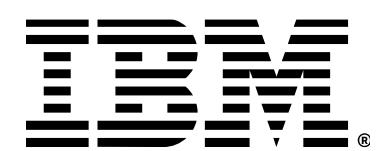
Next session in 5 minutes:

github.com/JackGBray/Qiskit-Spring-Fest-Monash



IBM Quantum

Qiskit Fundamentals Workshop



Agenda

Set up in qBraid

Fundamentals Lab notebook

Hello World notebook

Advanced notebook

Check in!



Lunch

provided by

**Quantum
Australia**

Agenda

10:00 – 10:10

Welcome / Opening Remarks

10:10 – 10:20

Introduction to Quantum Australia — Olivia Dennis

10:20 – 11:20

Quantum & Qiskit 101

Intro to qubits, gates, measurement, entanglement, and how Qiskit is used in practice

11:20 – 12:15

Qiskit Fundamentals Lab

Learn the basics of coding for a quantum computer

Hands-on notebook session: *run your first circuits on a real quantum computer via IBM Quantum*

12:15 – 1:00

Lunch Break

1:00 – 1:45

Guest Seminar — Dr Anna Phan

Asia Pacific Lead for IBM Quantum Innovation Centres

1:50 – 2:30

Guest Lecture — Nikai Jagganath

QKD and PQC protocols

2:30 – 3:00

Showcase: Quantum at Monash

Short presentations and demos of current research highlights

3:00 – 3:30

Mini Competition — *Test your understanding and **win a prize!***

Slido for Q&A





Quantum Cryptography & Post-Quantum Cryptography

NIKAI JAGGANATH

NIKAI.JAGGANATH@MONASH.EDU

FACULTY OF INFORMATION TECHNOLOGY, SOFTWARE SYSTEMS AND
CYBERSECURITY
MONASH UNIVERSITY

What is cryptography?

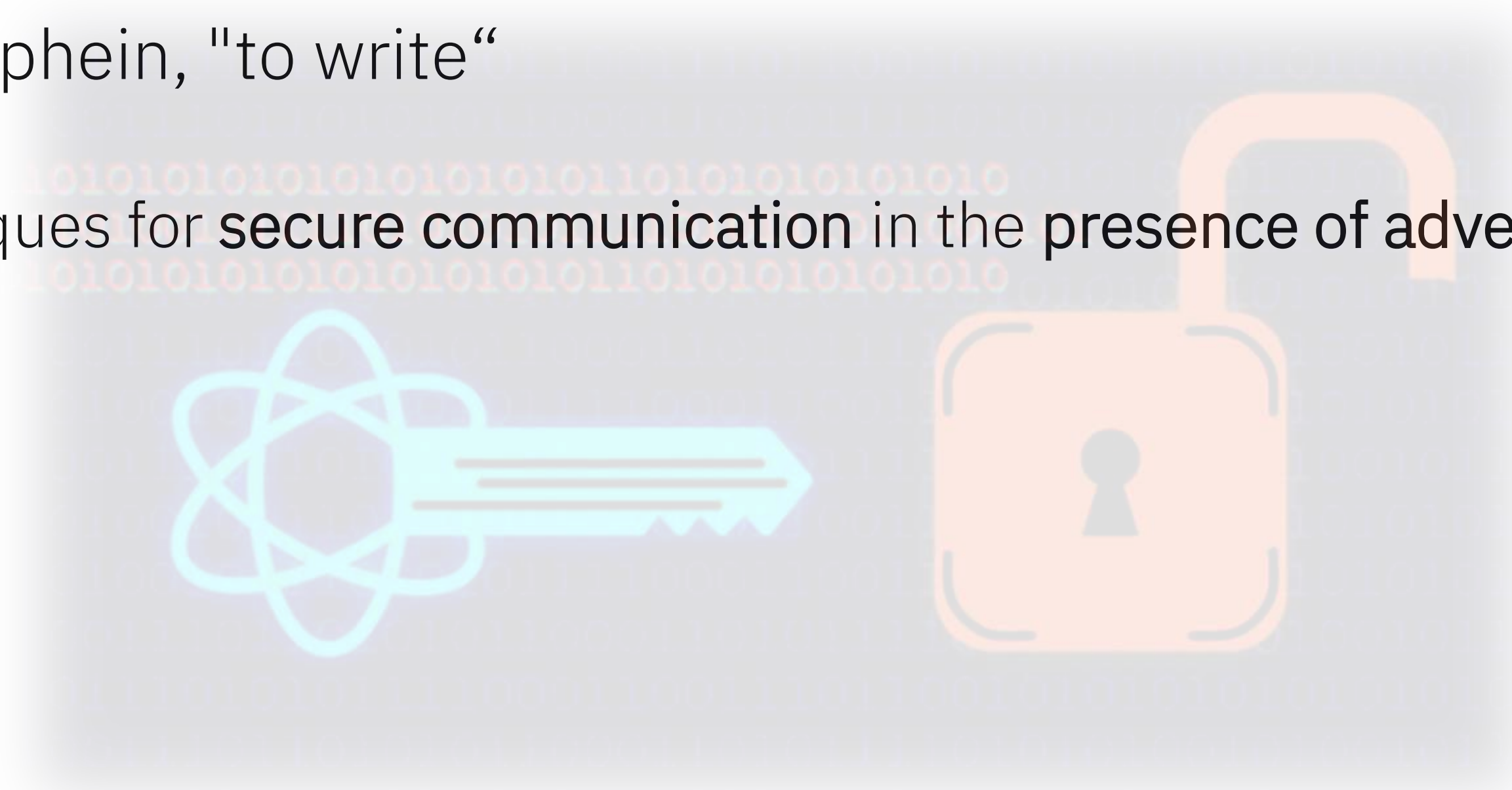
From Ancient Greek:

- κρυπτός, romanized: kryptós "hidden, secret";
- and γράφειν graphein, "to write"

The study of techniques for **secure communication** in the presence of adversarial behavior.

Goals:

- Confidentiality
- Integrity
- Authentication



Caesar Cipher

~50 BCE: Caesar cipher – letter shifting.

Plain	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Cipher	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W

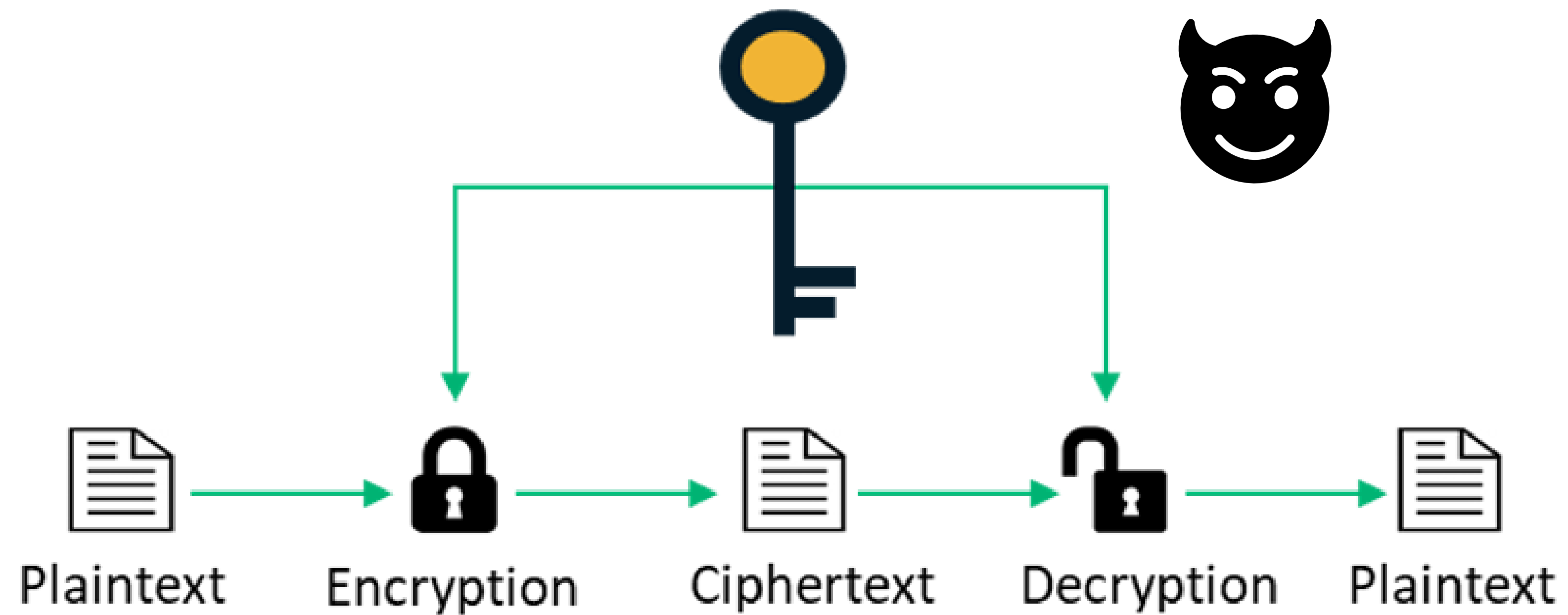
Plaintext: THE QUICK BROWN FOX JUMPS OVER THE LAZY DOG
Ciphertext: QEB NRFZH YOLTK CLU GRJMP LSBO QEB IXWV ALD

Timeline

Year	Cryptographic algorithm	
~50 BCE	Caesar cipher	Letter shifting
16th century	Vigenère cipher	Polyalphabetic substitution
1940s	Enigma machine	Mechanical encryption in WWII
1970s	DES	First modern symmetric cipher
1977	RSA	Public-key cryptography based on integer factorisation
1990s – Present	AES, ECC	Efficient modern standards.
Present – Future	QKD, PQC	Lattice-based Cryptography

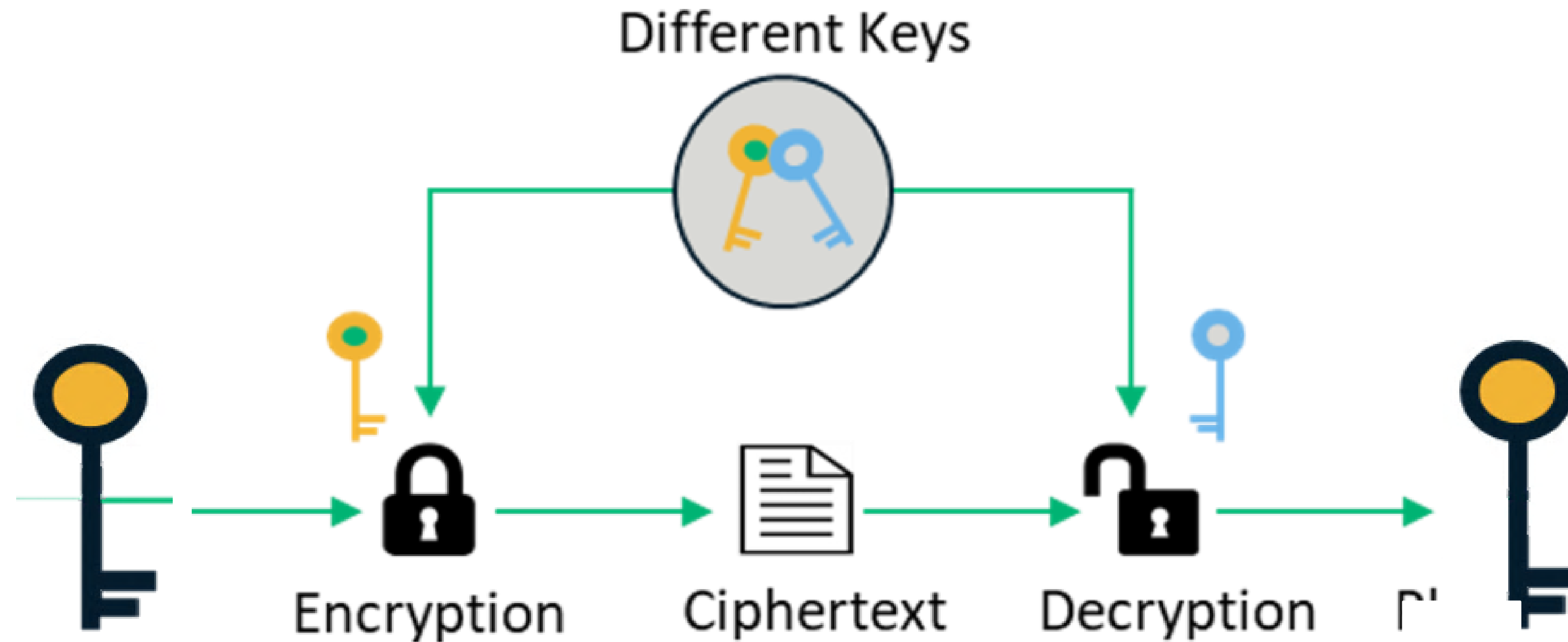
“The Key Distribution Problem”

How to share symmetric key over public channel?



“The Key Distribution Problem” (cont.)

Use asymmetric/public key cryptography



(Classical) Public-Key Algorithms

RSA

– Based on the difficulty of factoring large composite numbers.

Algorithm 1 The structure of RSA algorithm as follows.

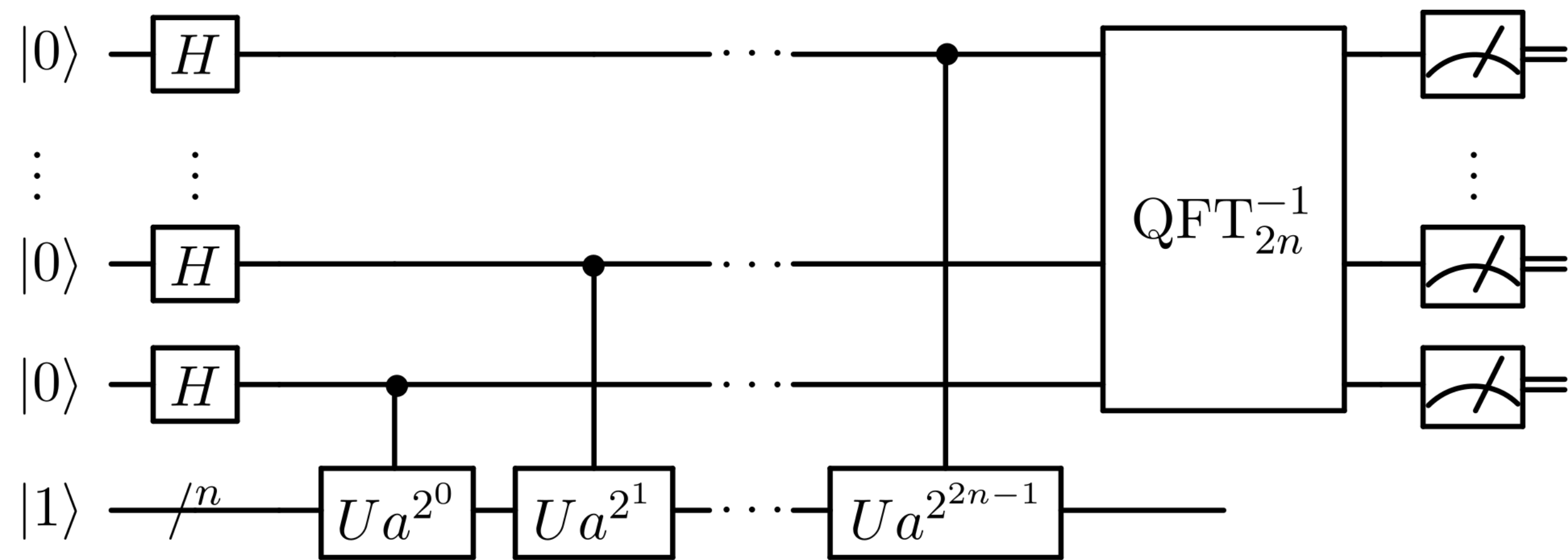
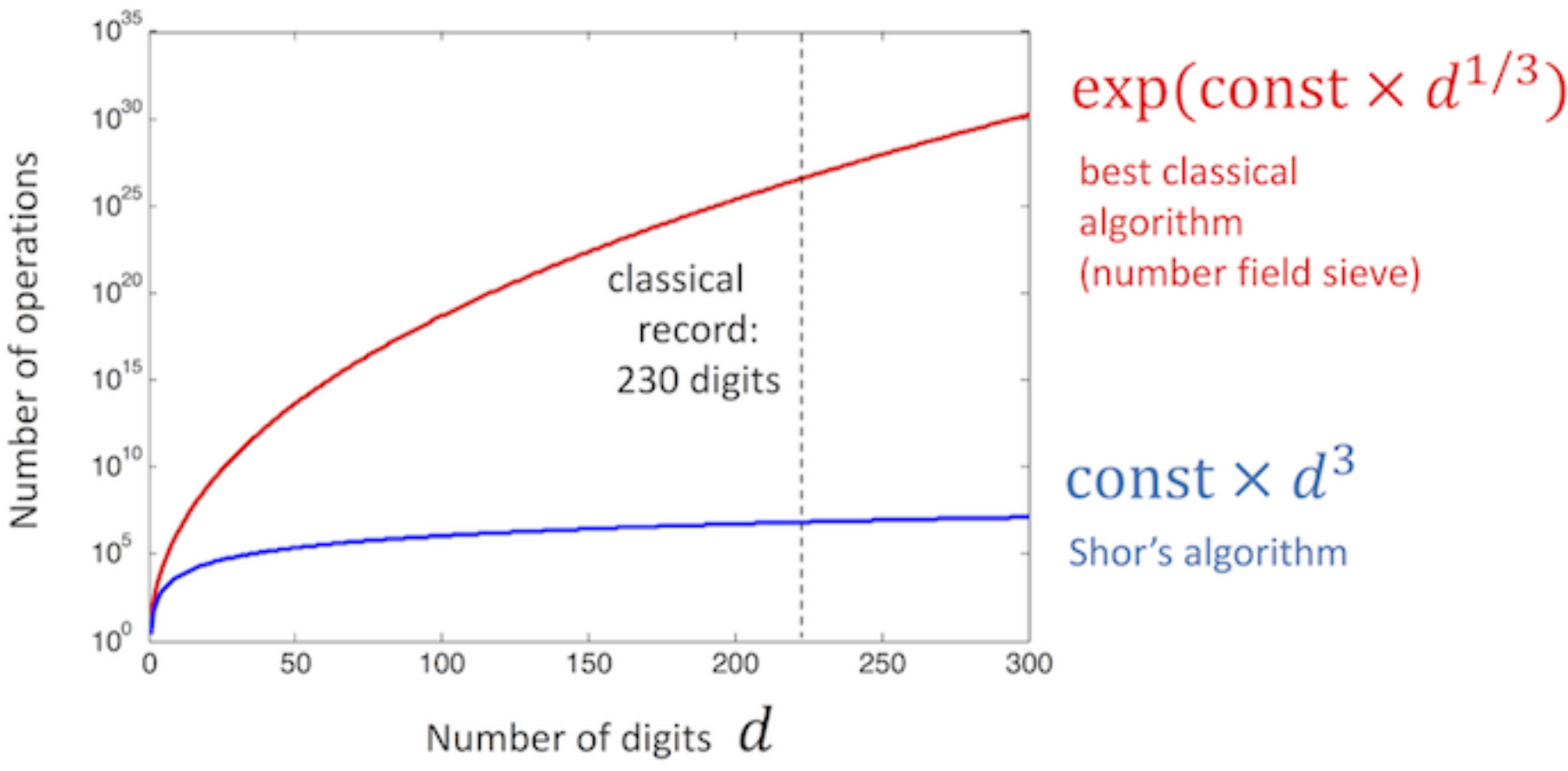
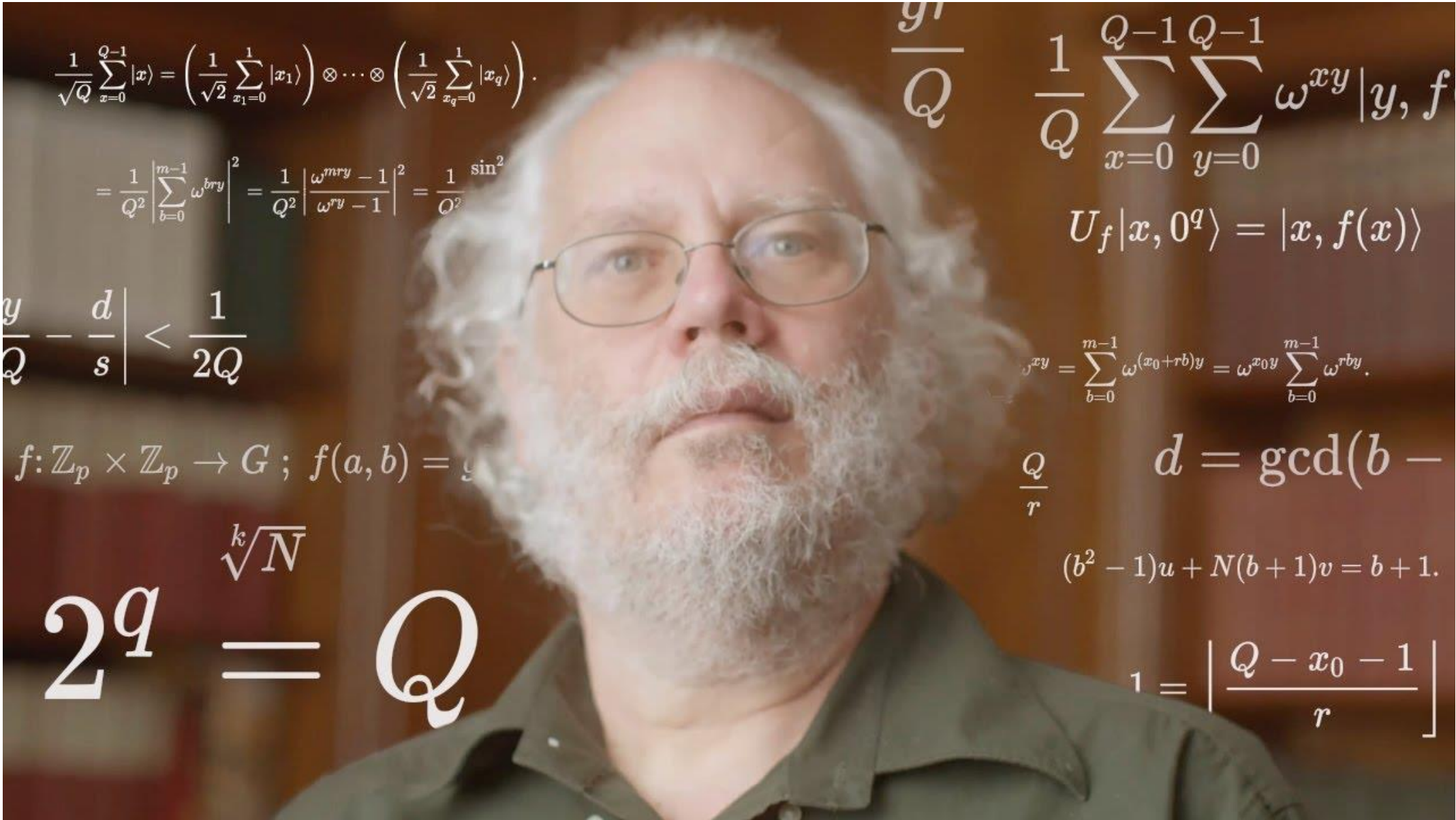
- 1: **Input Values:** p and q
 - 2: **Compute:**
 - 3: $n = p \times q$
 - 4: $\phi(n) = (p-1)(q-1)$
 - 5: **Select Integer values:** e [$(\gcd(\phi(n), e) = 1; 1 < e < \phi(n))$]
 - 6: **Compute:** d $de \bmod \phi(n) = 1$
 - 7: $C = M^e \bmod n$
 - 8: **Encryption:** $M < n$ $C = M^e \bmod n$
 - 9: **Decryption:** $M = C^d \bmod n$
-

ECC / Diffie–Hellman

– Based on the Discrete Logarithm Problem

Shor's Algorithm

- Developed by Peter Shor (1994).
- Efficiently factors large integers and solves DLP using quantum mechanics in polynomial time.

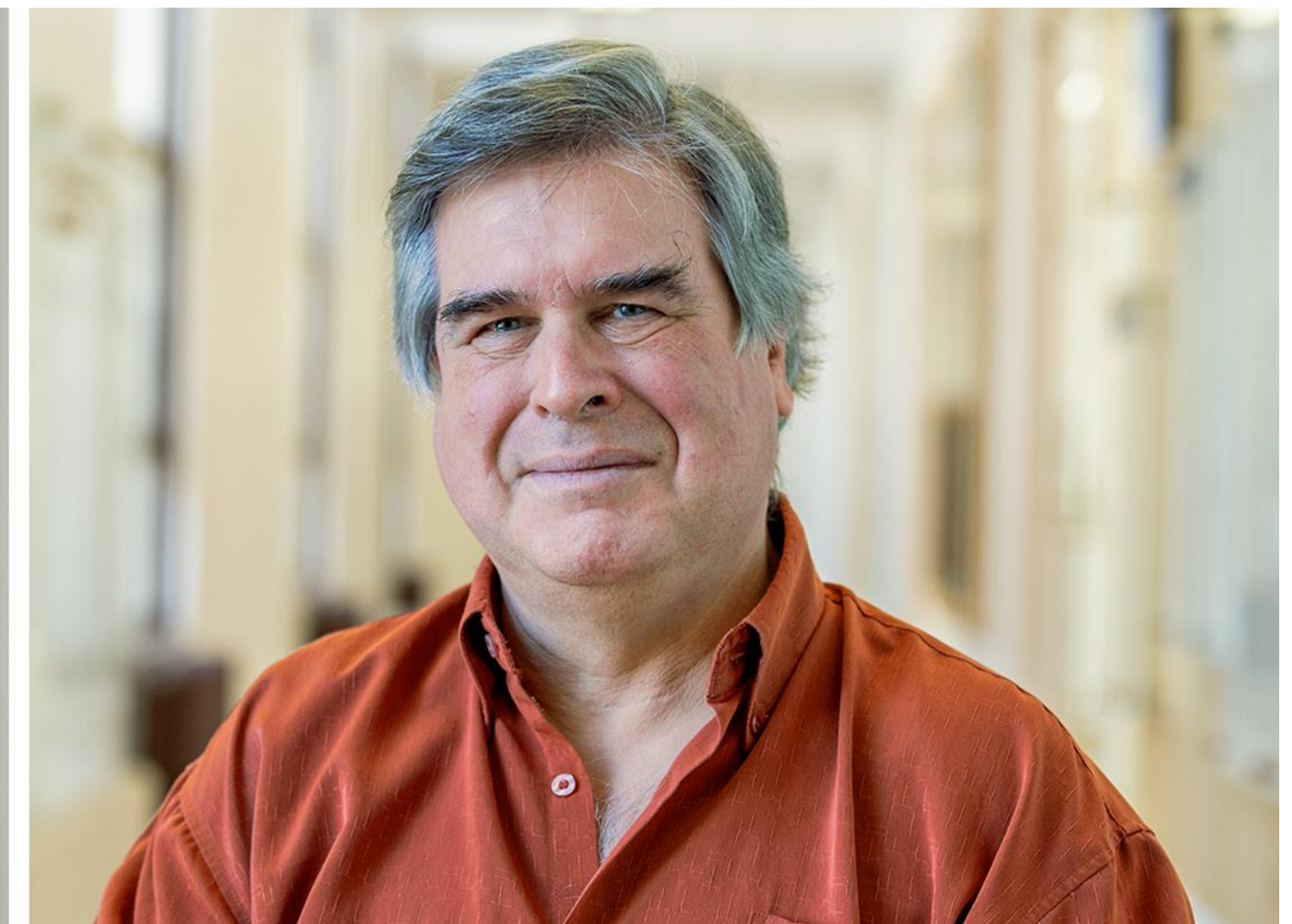
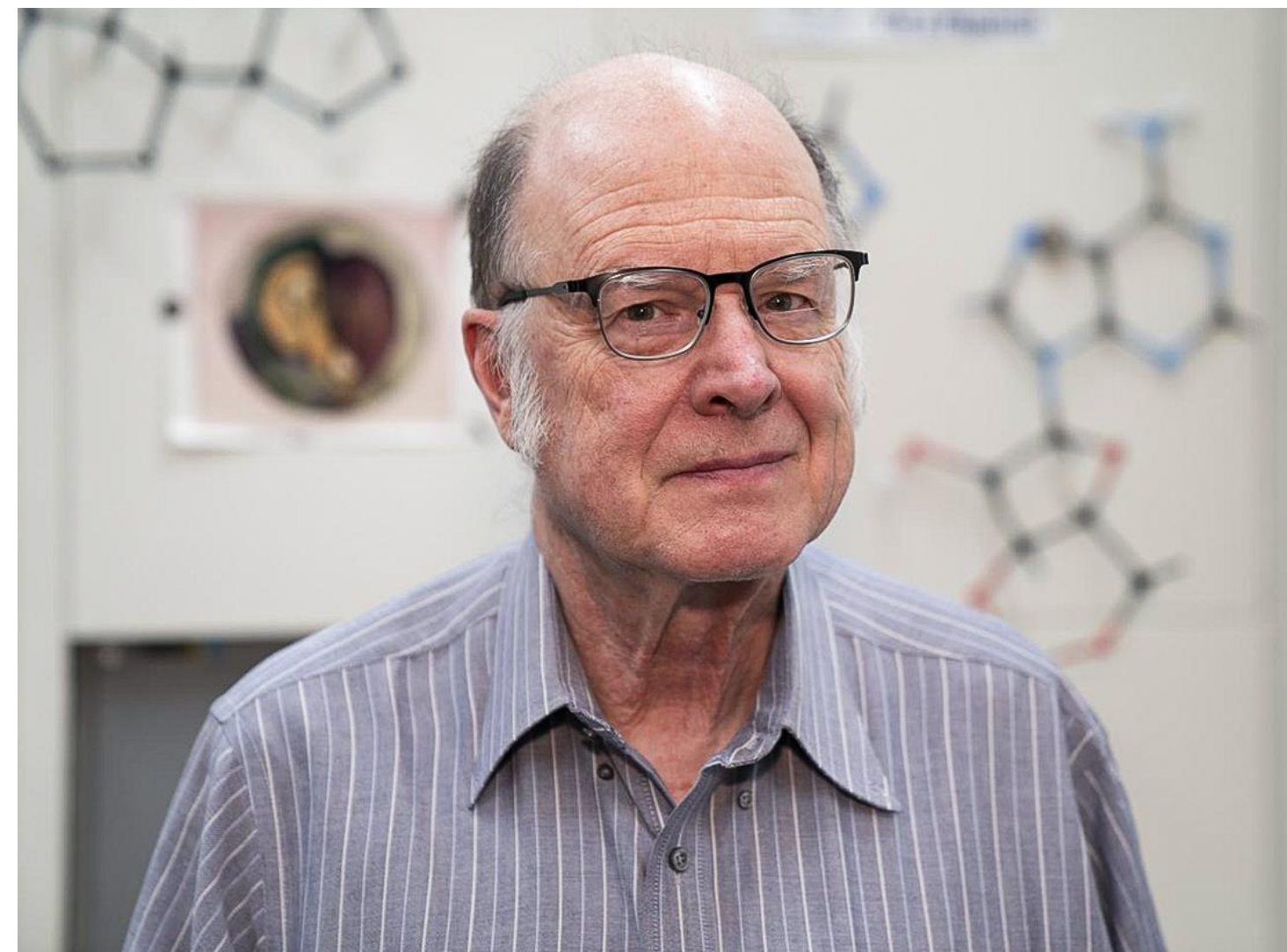


Can we instead
distribute symmetric key
another way?



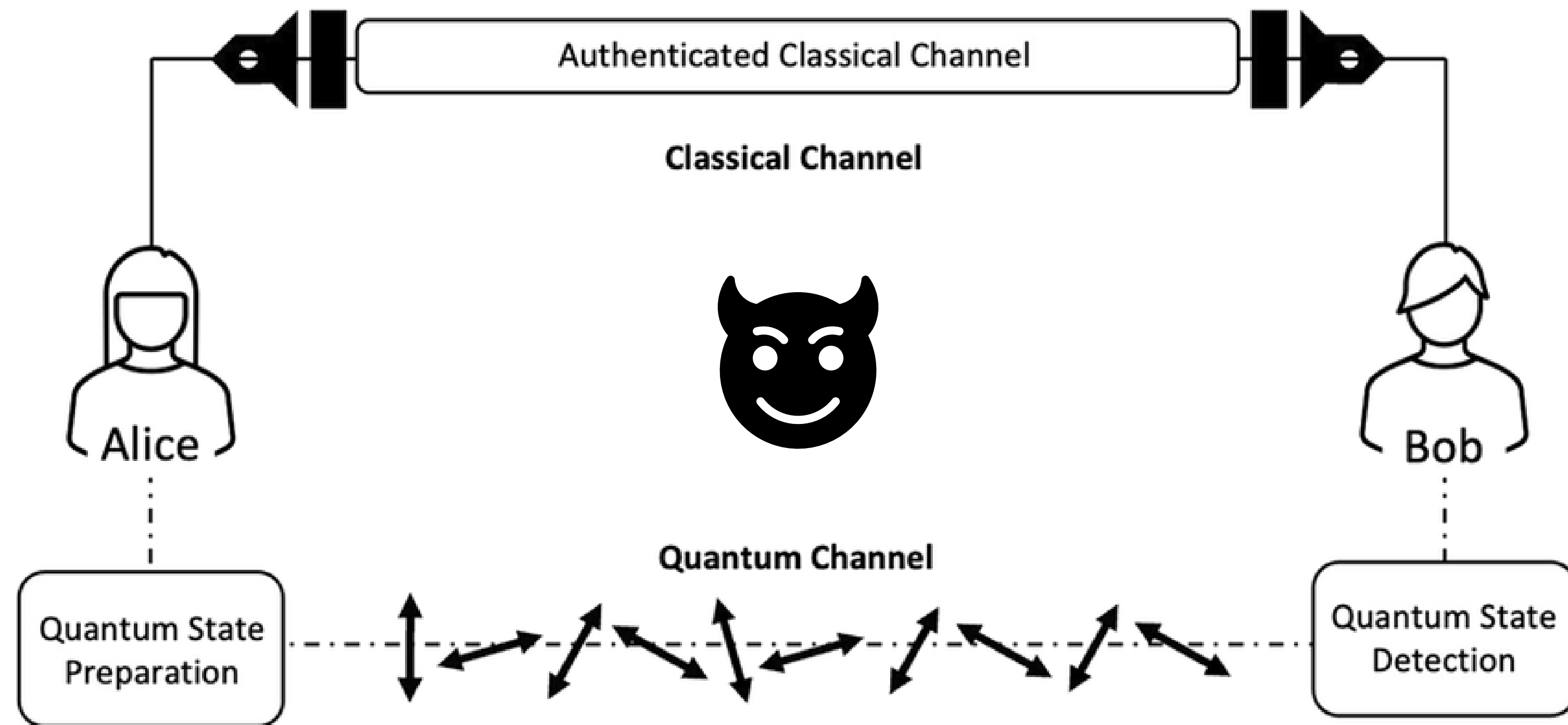
Quantum Key Distribution (QKD)

- BB84: Proposed by **Bennett** and **Brassard** in **1984**.
- **First practical Quantum Key Distribution (QKD) protocol**. Foundation of nearly all later QKD protocols (e.g. E91, decoy-state QKD).
- **Goal**: Generate a **shared secret key** between two distant parties with security guaranteed by quantum physics.



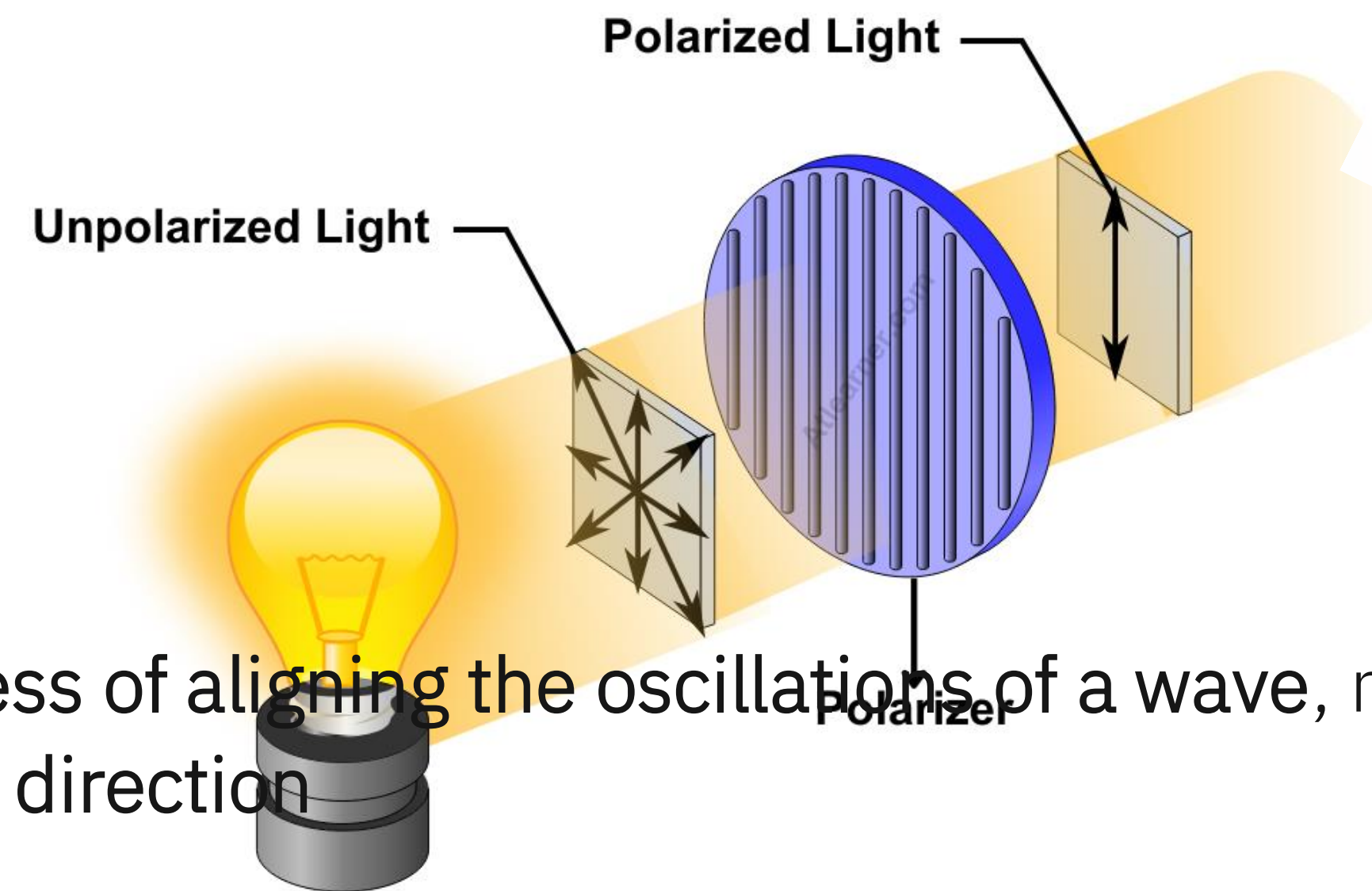
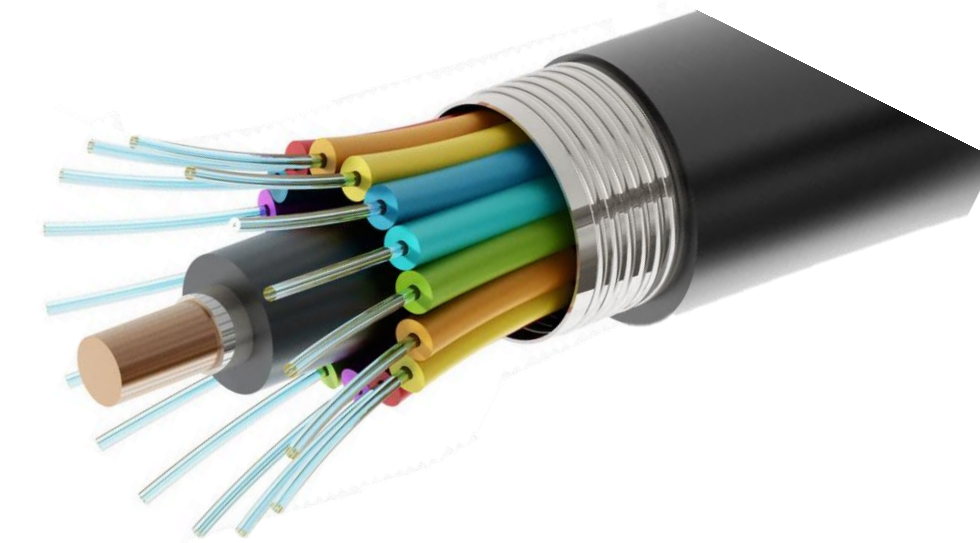
BB84 Protocol

Core principle: Measuring a quantum system **disturbs** it, so eavesdropping can be detected.



Photons as Qubits

- Can transmit through fibre optics.



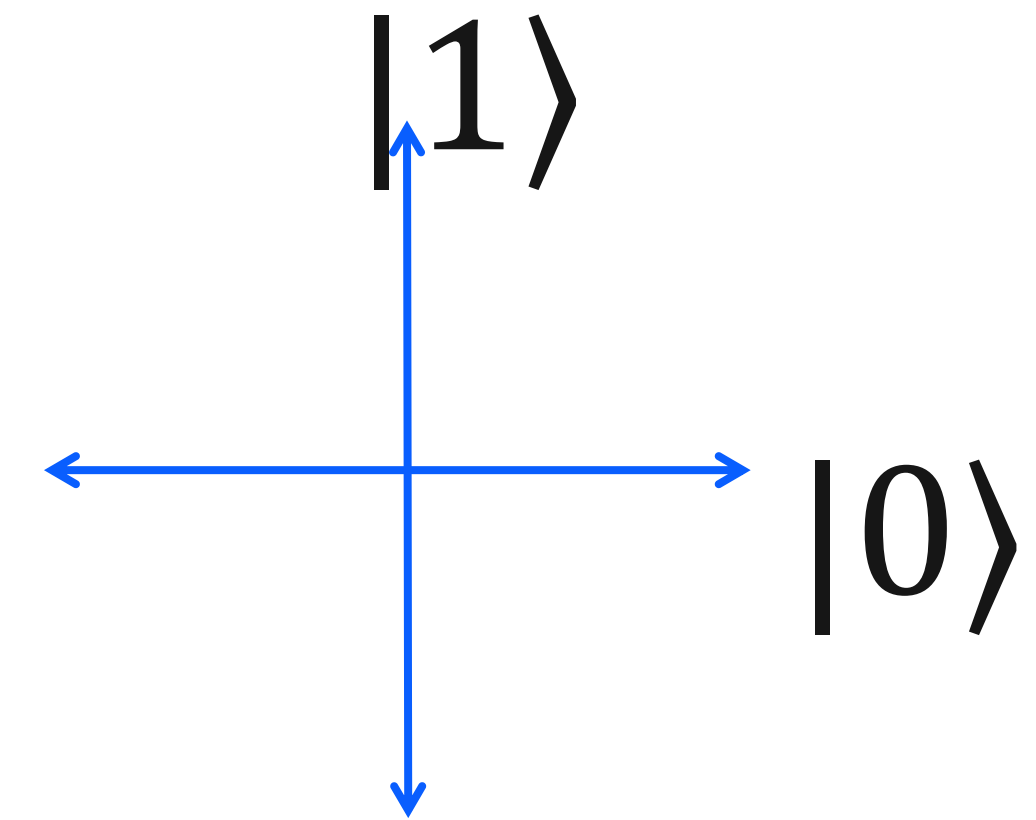
- Polarization is the process of aligning the oscillations of a wave, most commonly an electromagnetic wave like light, to a single direction

Linear Polarization

Rectilinear polarization (Computational basis $+$):

- Horizontal polarization (H) $|0\rangle$
- Vertical polarization (V) $|1\rangle$

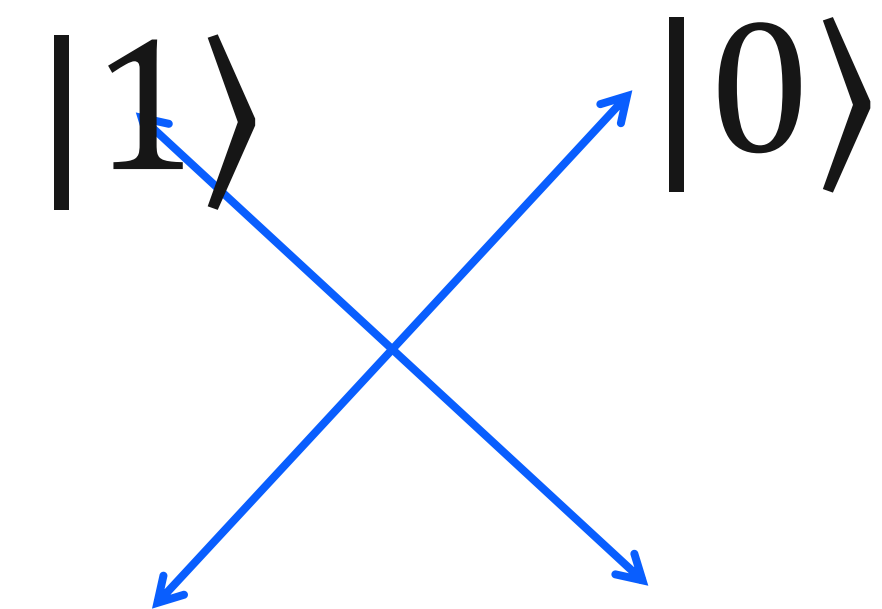
“Binary 0”
“Binary 1”



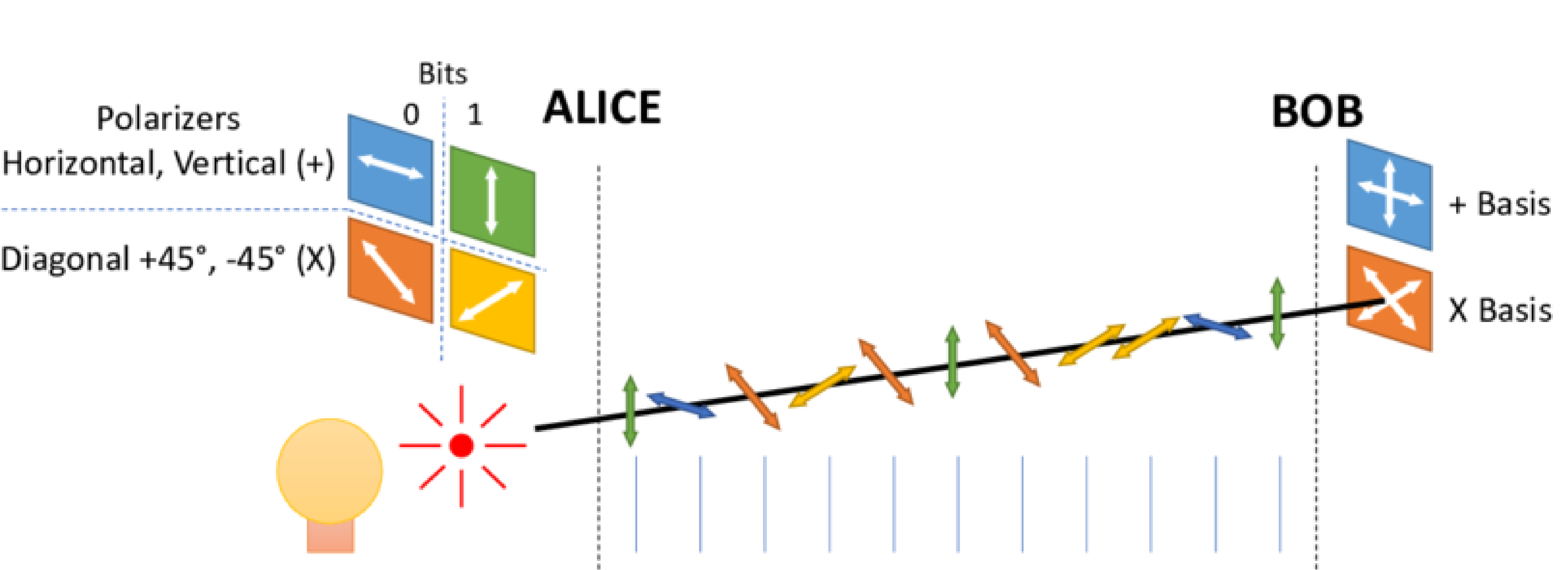
Diagonal polarization (Hadamard basis $\times$):

- Diagonal polarization (D) $|+\rangle$
- Anti-diagonal polarization (A) $|-\rangle$

“Binary 0”
“Binary 1”



BB84 Protocol



BB84 (cont.)

Alice's bit string	1	0	0	1	1	0	1	0	0	1
Alice's basis	+	×	×	+	×	+	+	+	+	+
Alice's polarization	V	D	D	V	A	H	V	H	H	V
Bob's basis	+	+	×	+	+	×	+	+	+	×
Bob's polarization	V	V	D	V	H	A	V	H	H	D
Bob's bit string	1	1	0	1	1	1	1	0	0	0
Sifting										
Key	?	?	?	?	?	?	?	?	?	?

BB84 (cont.)

Alice's bit string	1	0	0	1	1	0	1	0	0	1
Alice's basis	+	×	×	+	×	+	+	+	+	+
Alice's polarization	V	D	D	V	A	H	V	H	H	V
Bob's basis	+	+	×	+	+	×	+	+	+	×
Bob's polarization	V	V	D	V	H	A	V	H	H	D
Bob's bit string	1	1	0	1	1	1	1	0	0	0
Sifting										
Key	?	?	?	?	?	?	?	?	?	?

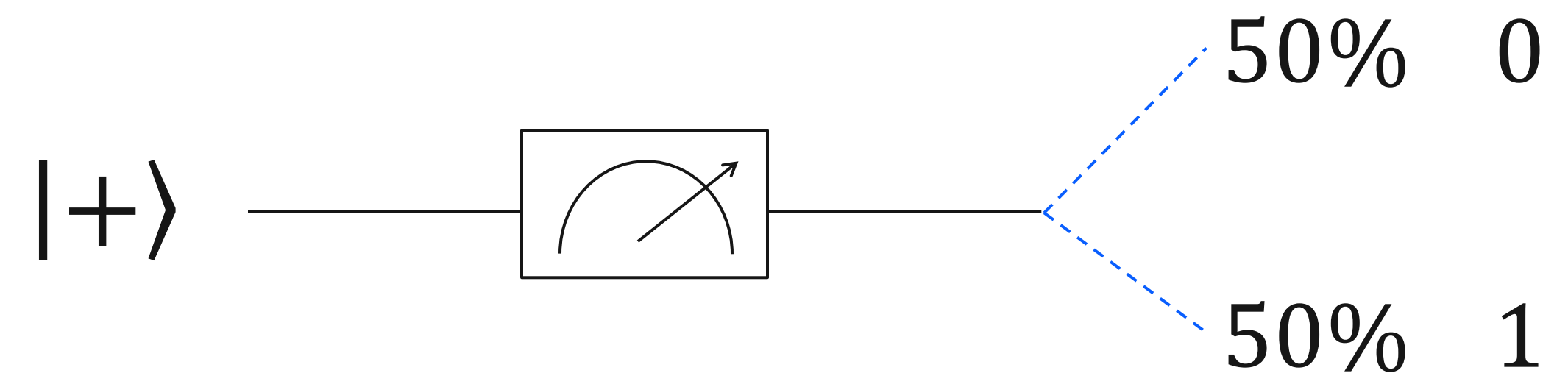
BB84 (cont.)

Alice's bit string		0
Alice's basis		$\times$
Alice's polarization		D
Bob's basis		+
Bob's polarization		V
Bob's bit string		1
Sifting		
Key		?

- Alice wants to send the bit 0.
- Alice encodes this using Hadamard basis

$$|+\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$$

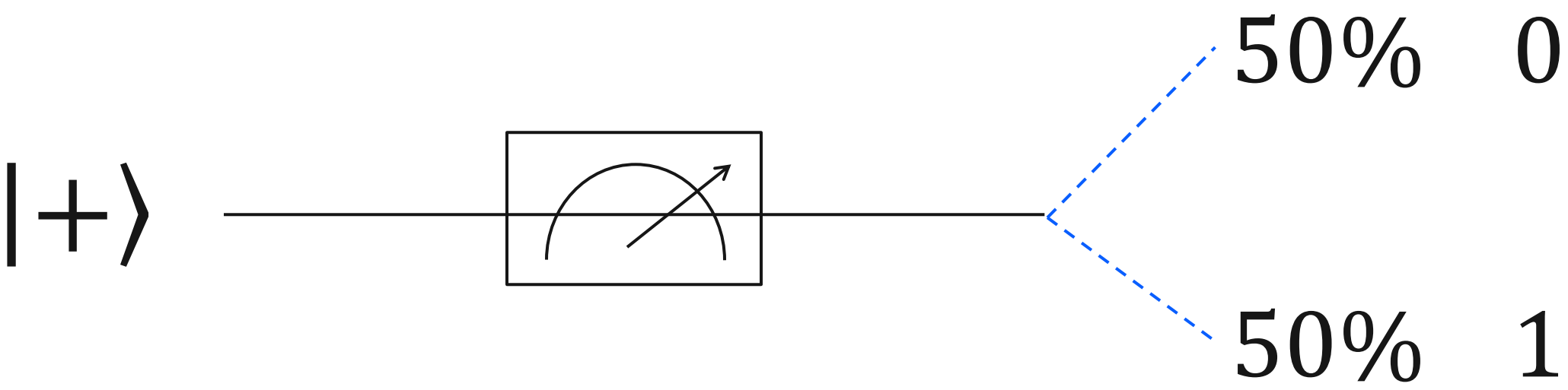
- Bob receives $|+\rangle$ measured this state using the computational basis



BB84 (cont.)

Alice's bit string		0
Alice's basis		x
Alice's polarization		D
Bob's basis		+
Bob's polarization		V
Bob's bit string		1
Sifting		
Key		?

Intuitively, if Alice sent a **diagonally polarised photon** (45°),
then that photon has an **equal probability** (50%)
of passing the horizontal or vertical filter.



BB84 (cont.)

Alice's bit string	1	0	0	1	1	0	1	0	0	1
Alice's basis	+	x	x	+	x	+	+	+	+	+
Alice's polarization	V	D	D	V	A	H	V	H	H	V
Bob's basis	+	+	x	+	+	x	+	+	+	x
Bob's polarization	V	V	D	V	H	A	V	H	H	D
Bob's bit string	1	1	0	1	1	1	1	0	0	0
Sifting										
Key	1		0	1			1	0	0	?

BB84 with Eavesdropping

Alice's bit string	1	0	0	1	1	0	1	0	0	1
Alice's basis	+	×	×	+	×	+	+	+	+	+
Alice's polarization	V	D	D	V	A	H	V	H	H	V
Eve's basis	×	×	+	×	+	+	×	+	×	+
Eve's polarization	D	D	V	A	V	H	D	H	A	V
Bob's basis	+	+	×	+	+	×	+	+	+	×
Bob's polarization	H	V	A	H	V	D	H	H	V	D
Bob's bit string	0	1	1	0	1	0	0	0	1	0
Sifting										
Key	?		?	?			?	?	?	

BB84 Error rate estimation

Sampling: Alice and Bob publicly **reveal a random subset** of their sifted bits.

Compare: They compare the values and compute the **Quantum Bit Error Rate (ϵ)** — the fraction of mismatched bits.

Decision:

- If $\epsilon \leq$ threshold (e.g. 1–3%), errors likely come from noise $\rightarrow$ continue.
- If $\epsilon >$ threshold, Eve likely interfered $\rightarrow$ abort the protocol.

$$\epsilon = \frac{\text{number of differing bits}}{\text{total bits compared}}$$

BB84 Eve's Escape Probability

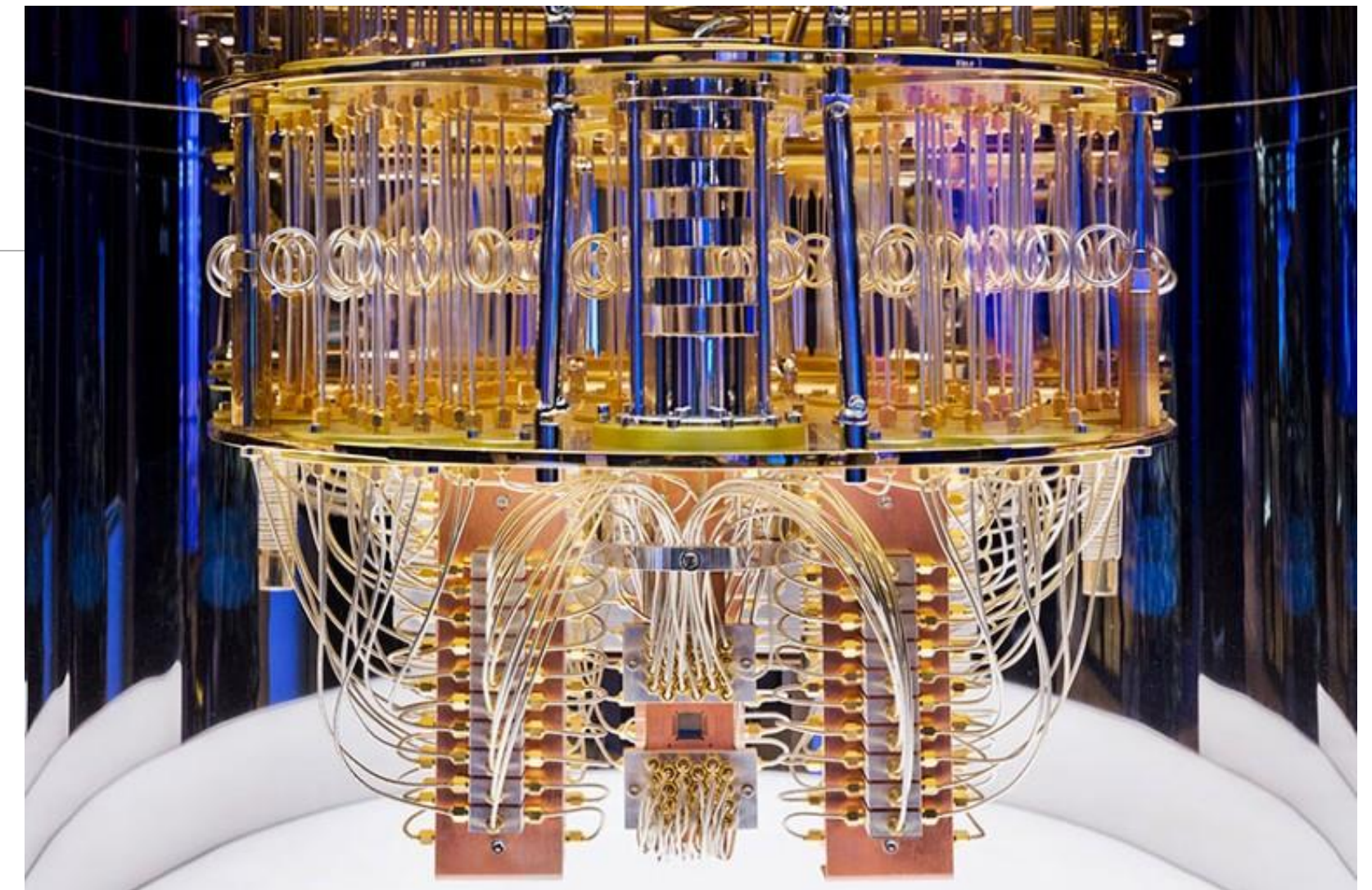
If Eve eavesdrops on n bits, she will go undetected $(\frac{3}{4})^n$ times.

$$\begin{aligned} &\text{Probability of error (Alice and Bob disagree)} \\ &= (\frac{1}{2} \text{ wrong basis}) \times (\frac{1}{2} \text{ error given wrong basis}) \\ &= \frac{1}{4} . \end{aligned}$$

$$\text{Probability of no error} = 1 - \frac{1}{4} = \frac{3}{4}$$

In our case of **10 bits**, Eve's escape probability is **0.0563**, which is very small!

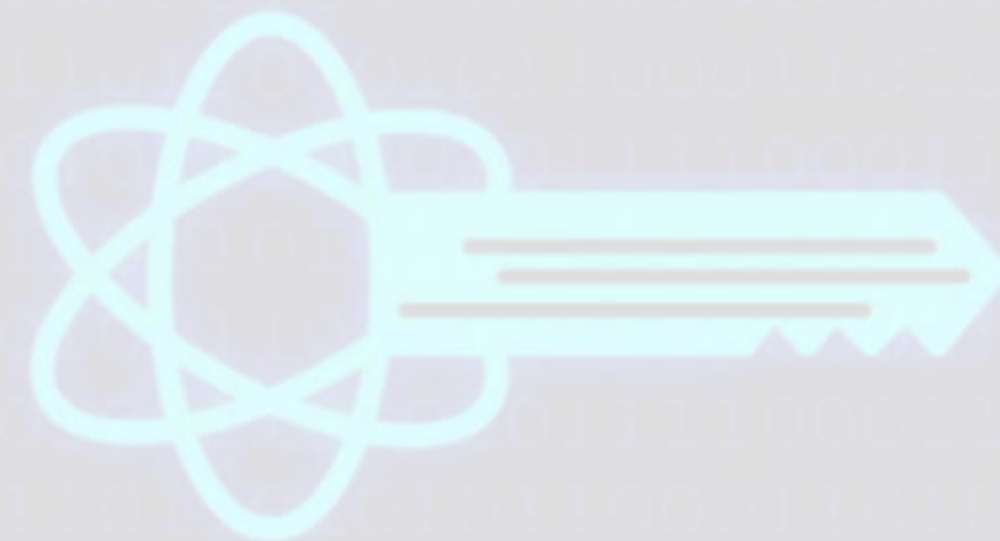
But large scale adoption is unlikely...



Post-Quantum Cryptography (PQC)

(Non-quantum) public key cryptographic algorithms designed to remain secure even if an **attacker** has a quantum computer.

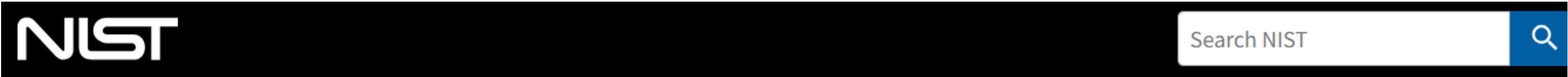
Runs on normal computers and networks, but is built on math problems believed to be hard even for quantum computers.



Main families of PQC

Family	Based on	Examples
Lattice-based	Learning With Errors (LWE), Shortest Vector Problem (SVP)	CRYSTALS-Kyber (ML-KEM), CRYSTALS-Dilithium, Falcon
Code-based	Decoding random linear error- correcting codes	Classic McEliece
Hash-based	One-way property and collision resistance of hash functions	SPHINCS+

NIST PQC Standardisation Process



NEWS

NIST Releases First 3 Finalized Post-Quantum Encryption Standards

August 13, 2024

- NIST has released a final set of encryption tools designed to withstand the attack of a quantum computer.
- These post-quantum encryption standards secure a wide range of electronic information, from confidential email messages to e-commerce transactions that propel the modern economy.
- NIST is encouraging computer system administrators to begin transitioning to the new standards as soon as possible.

 MEDIA CONTACT

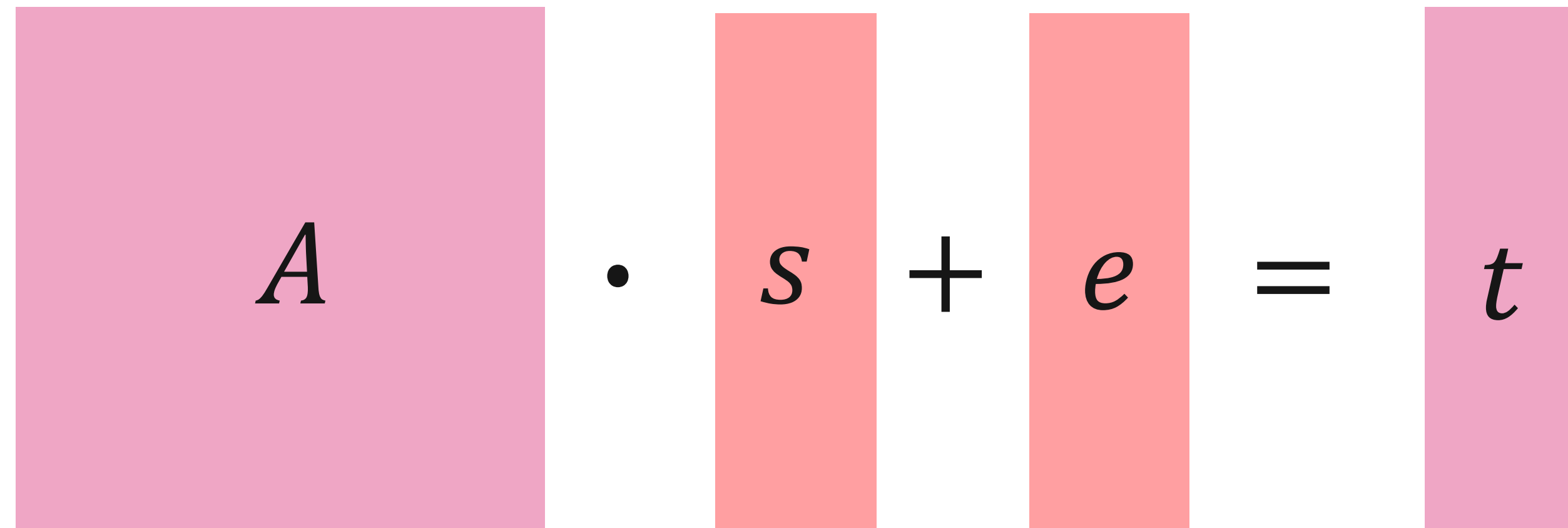
Chad Boutin
charles.boutin@nist.gov 
(301) 975-4261

<https://csrc.nist.gov/standardization>

[graphy-](#)

Learning With Errors (LWE)

Given A and t , it is computationally infeasible to find s .


$$A \cdot s + e = t$$

What does working on lattice-based PQC involve?

3.2.3 Folding: The reduction from $(\mathcal{R}_{\text{eval}}^b)^{2k}$ to $\mathcal{R}_{\text{eval}}^B$

We now describe the core protocol Π_{fold} that folds $2k$ instance-witness pairs of $\mathcal{R}_{\text{eval}}^b$ from (9) into a single instance-witness pair in $\mathcal{R}_{\text{eval}}^B$. More generally, the protocol is a reduction of knowledge from $(\mathcal{R}_{\text{hom}}^b)^{2k}$ from (10) to $\mathcal{R}_{\text{hom}}^B$ with a further restriction that the public parameter, the *sampled* homomorphism $\mathcal{L}$, is **relaxed binding** as in Section 2.3. This relaxed binding property, defined for bound $2B$ and the challenge space $\mathcal{C}_{\text{small}}$, ensures the hardness of finding two distinct weak openings for a commitment cm . Recall that a $2B$ -weak opening $(\rho \in \mathcal{C}_{\text{small}} - \mathcal{C}_{\text{small}}, \vec{x})$ of cm satisfies that $\rho \cdot \text{cm} = \mathbf{A}\vec{x} \bmod q$ with the norm constraint $\|\vec{x}\|_\infty < 2B$. Thus, $\mathcal{R}_{\text{eval}}^B$ from (9) is a special case of $\mathcal{R}_{\text{hom}}^B$ where the sampled homomorphism $\mathcal{L}$ is given by $\mathcal{L}(\vec{f}) := \mathbf{A}\vec{f}$, and the $2B$ -relaxed binding property follows from the hardness of $\text{MSIS}_{\kappa, m, 8TB}^{\infty, q}$ where $T = \|\mathcal{C}_{\text{small}}\|_{\text{op}}$ is the expansion factor of $\mathcal{C}_{\text{small}}$.

Our protocol folds the $2k$ witness vectors (with norm less than b) into a witness vector of norm less than B (where $b < B < q/2$), using *small* random scalars $[\rho_i]_{i=1}^{2k}$ sampled from a strong sampling set $\mathcal{C}_{\text{small}} \subseteq \mathcal{R}_q$. Additionally and crucially, Π_{fold} runs a sum-check protocol to enable extractions of the $2k$ witness vectors with small norms. The sum-check is for a polynomial $g(\vec{x}) := \sum_{i=1}^{2k} (\alpha_i g_{1,i}(\vec{x}) + \mu_i g_{2,i}(\vec{x}))$ with random scalars $[\alpha_i, \mu_i]_{i=1}^{2k}$. Informally, we can understand it as a batch of $4k$ separate sum-checks for polynomials $[g_{1,i}, g_{2,i}]_{i=1}^{2k}$, respectively. The sum-check for $g_{1,i}$ (defined in Eq. (15)) is used to verify that the evaluation statement $\text{mle}[\hat{\mathbf{f}}_i](\vec{r}_i) = \hat{\mathbf{v}}_i$ holds for all $i \in [2k]$. For every $i \in [2k]$, the sum-check for $g_{2,i}$ (defined in Eq. (16)) is used to verify that

$$\prod_{j=-(b-1)}^{b-1} (\text{mle}[\hat{\mathbf{f}}_i](\vec{x}) - j) = 0 \quad \text{for all } \vec{x} \in \{0, 1\}^{\log m}$$

This is the same as the Hadamard product check (the norm bound check)

$$\hat{\mathbf{f}}_i \circ \left[\bigcirc_{j=1}^{b-1} (\hat{\mathbf{f}}_i - \hat{j}) \circ (\hat{\mathbf{f}}_i + \hat{j}) \right] = \hat{0},$$

Parameters: Strong sampling sets $\mathcal{C}, \mathcal{C}_{\text{small}} \subseteq \mathcal{R}_q$ where $\mathcal{C}$ is defined as in Eq. (31) and $\mathcal{C}_{\text{small}}$ has expansion factor $\|\mathcal{C}_{\text{small}}\|_{\text{op}} \leq c \in \mathbb{N}$ as in (6).

Input: $[\mathbf{x}_i := (\vec{r}_i, \hat{\mathbf{v}}_i, y_i) \in \mathcal{R}_q^{\log m} \times \mathcal{R}_q \times \mathcal{Y}]_{i=1}^{2k}$ and $[\mathbf{w}_i := \vec{f}_i \in \mathcal{R}_q^m]_{i=1}^{2k}$

Output: $\mathbf{x}_o := (\vec{r}_o, \hat{\mathbf{v}}_o, y_o), \mathbf{w}_o := \vec{f}_o$

The protocol $\langle \text{P}(\text{pp}, \mathbf{x}; \mathbf{w}), \text{V}(\text{pp}, \mathbf{x}) \rangle$ where $\mathbf{x} = [\mathbf{x}_i]_{i=1}^{2k}$ and $\mathbf{w} = [\mathbf{w}_i]_{i=1}^{2k}$:

1. $\text{V} \rightarrow \text{P} : \text{V}$ sends P challenges $[\alpha_i, \mu_i]_{i=1}^{2k} \xleftarrow{s} (\mathcal{C} \times \mathcal{C})^{2k}$ and $\vec{\beta} \xleftarrow{s} \mathcal{C}^{\log m}$.
2. $\text{V} \leftrightarrow \text{P} : \text{P}$ and V run a sum-check protocol for the claim

$$\sum_{\vec{b} \in \{0,1\}^{\log m}} g(\vec{b}) = \sum_{i=1}^{2k} \alpha_i \hat{\mathbf{v}}_i, \quad (13)$$

where the polynomial $g(\vec{x}) \in \mathcal{R}_q^{\leq 2b}[X_1, \dots, X_{\log m}]$ is defined as

$$g(\vec{x}) := g_{\text{eval}}(\vec{x}) + g_{\text{norm}}(\vec{x}), \quad (14)$$

$$g_{\text{eval}}(\vec{x}) := \sum_{i=1}^{2k} \alpha_i \cdot g_{1,i}(\vec{x}) \quad \text{where} \quad g_{1,i}(\vec{x}) := \text{eq}(\vec{r}_i, \vec{x}) \cdot \text{mle}[\hat{\mathbf{f}}_i](\vec{x}), \quad (15)$$

$$g_{\text{norm}}(\vec{x}) := \sum_{i=1}^{2k} \mu_i \cdot g_{2,i}(\vec{x}) \quad \text{where} \quad g_{2,i}(\vec{x}) := \text{eq}(\vec{\beta}, \vec{x}) \cdot \prod_{j=-(b-1)}^{b-1} (\text{mle}[\hat{\mathbf{f}}_i](\vec{x}) - j) \quad (16)$$

The sumcheck protocol reduces checking (13) to checking the evaluation claim $g(\vec{r}_o) \stackrel{?}{=} s$, where $s \in \mathcal{R}_q$ and $\vec{r}_o \xleftarrow{s} \mathcal{C}^{\log m}$ is a sum-check challenge sampled by V .

3. $\text{P} \rightarrow \text{V} : \text{P}$ sends V values $[\theta_i := \text{mle}[\hat{\mathbf{f}}_i](\vec{r}_o)]_{i=1}^{2k}$.
4. V computes $[\mathbf{e}_i := \text{eq}(\vec{r}_i, \vec{r}_o)]_{i=1}^{2k}$ and $\mathbf{e}^* := \text{eq}(\vec{\beta}, \vec{r}_o)$ and checks that

Our PQC Team at Monash



Our PQC(IP) Team at Monash



[Home](#) | Post-Quantum Cryptography in the Indo-Pacific Program

Post-Quantum Cryptography in the Indo-Pacific Program

Proactively building regional quantum computing resilience.

From 2023 to 2025, PQCIP will engage with industry and government stakeholders to improve their post-quantum preparedness through a cycle of assessment, tailored education, planning and evaluation.



<https://www.monash.edu/it/ssc/cybersecurity/pqcip>



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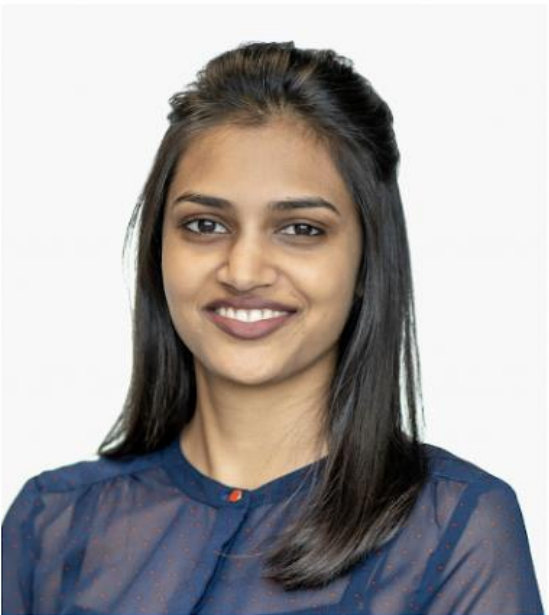
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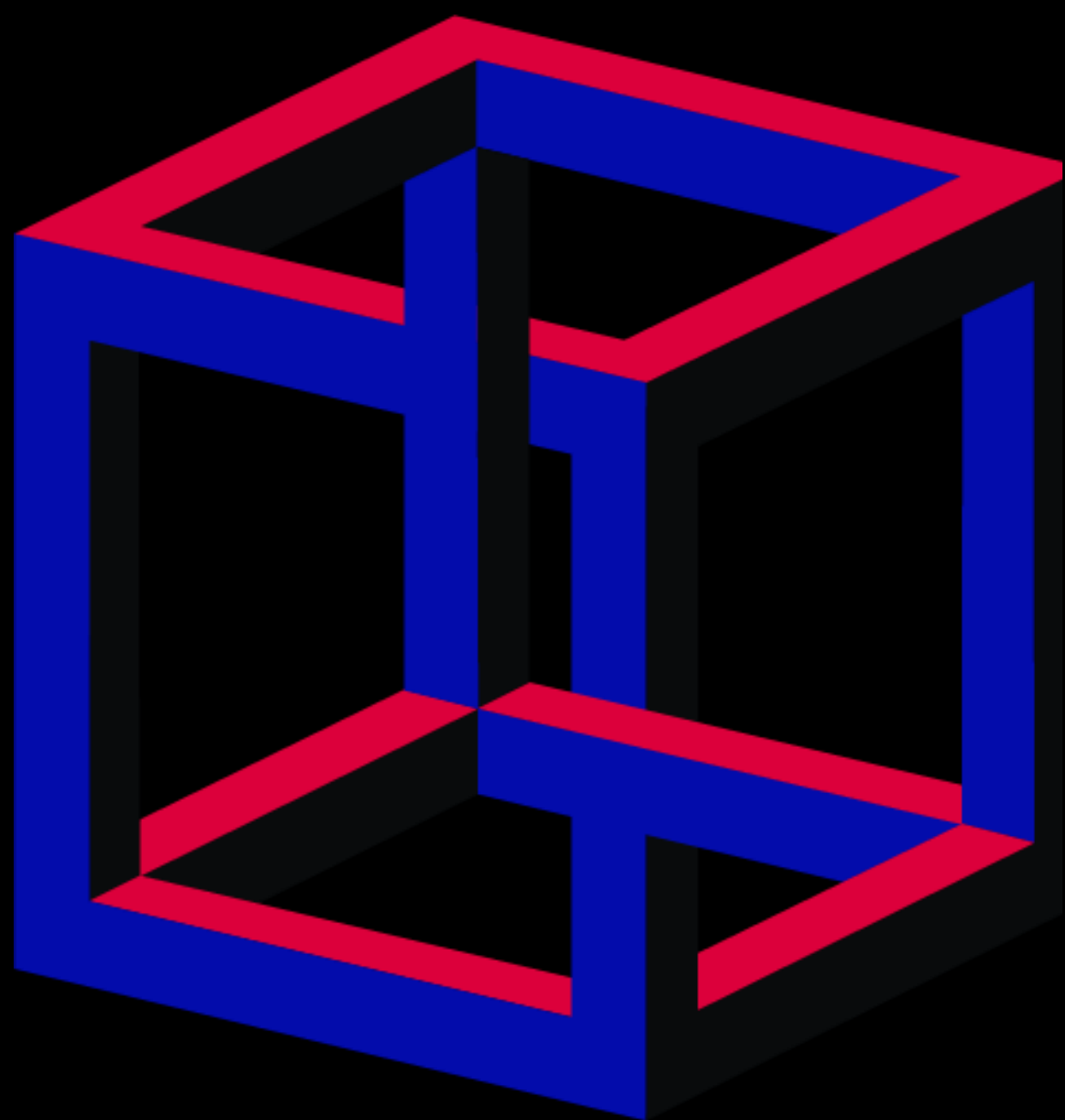
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MNET

MONASH NEXUS FOR EMERGING TECHNOLOGIES

Innovating, Collaborating and Excelling



MONASH
University

MONASH
INFORMATION
TECHNOLOGY



Who is MNET?

We are a student team intent on merging the digital and the physical realms of the world through simulation and emerging technologies.

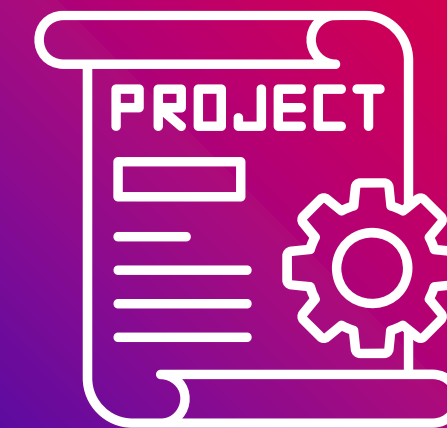
What do we do?



Competitions



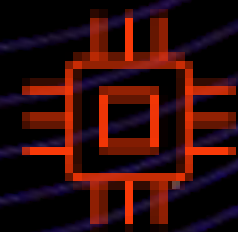
Research



Projects

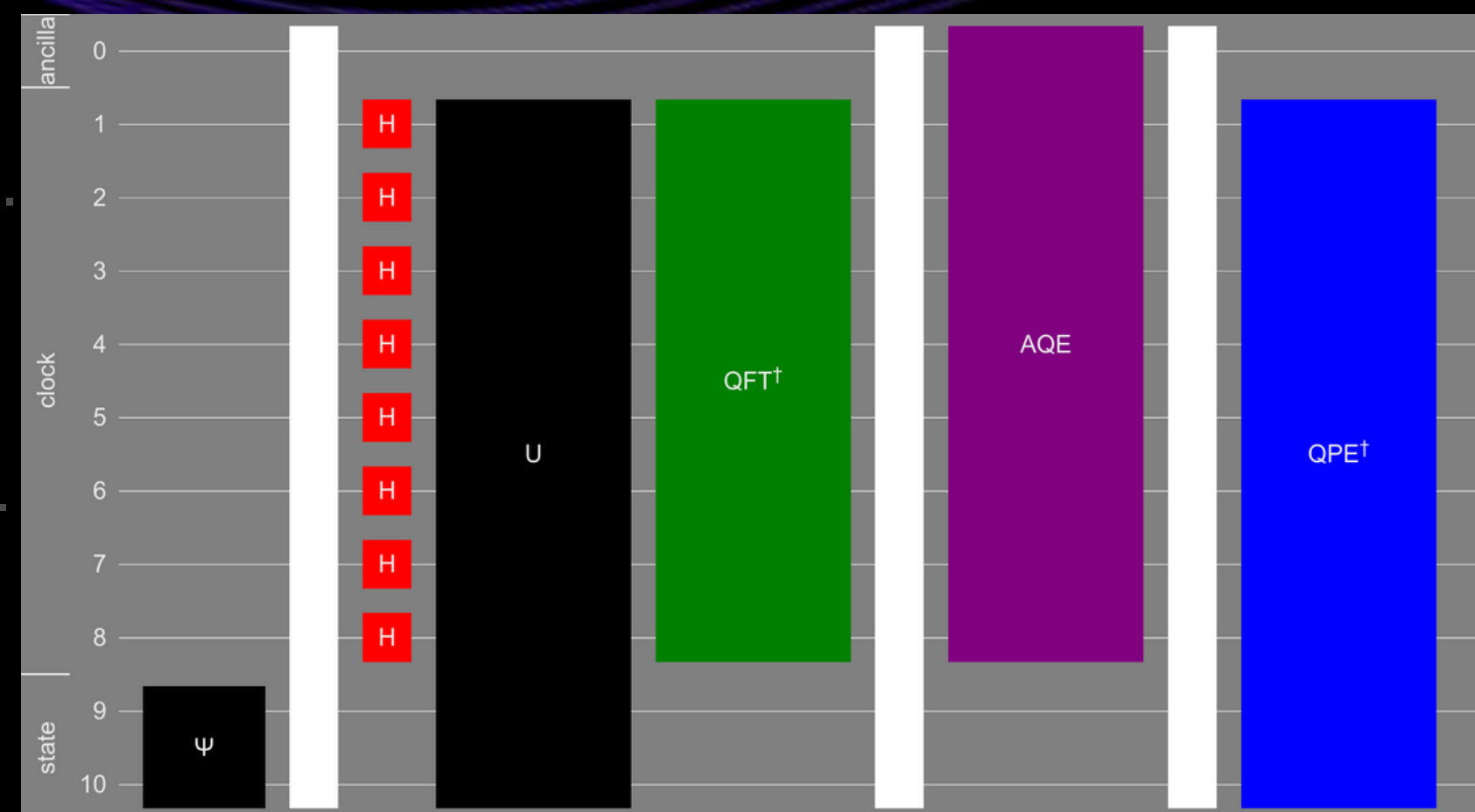


Industry



Our Quantum Portfolios

Quantum Circuit Education

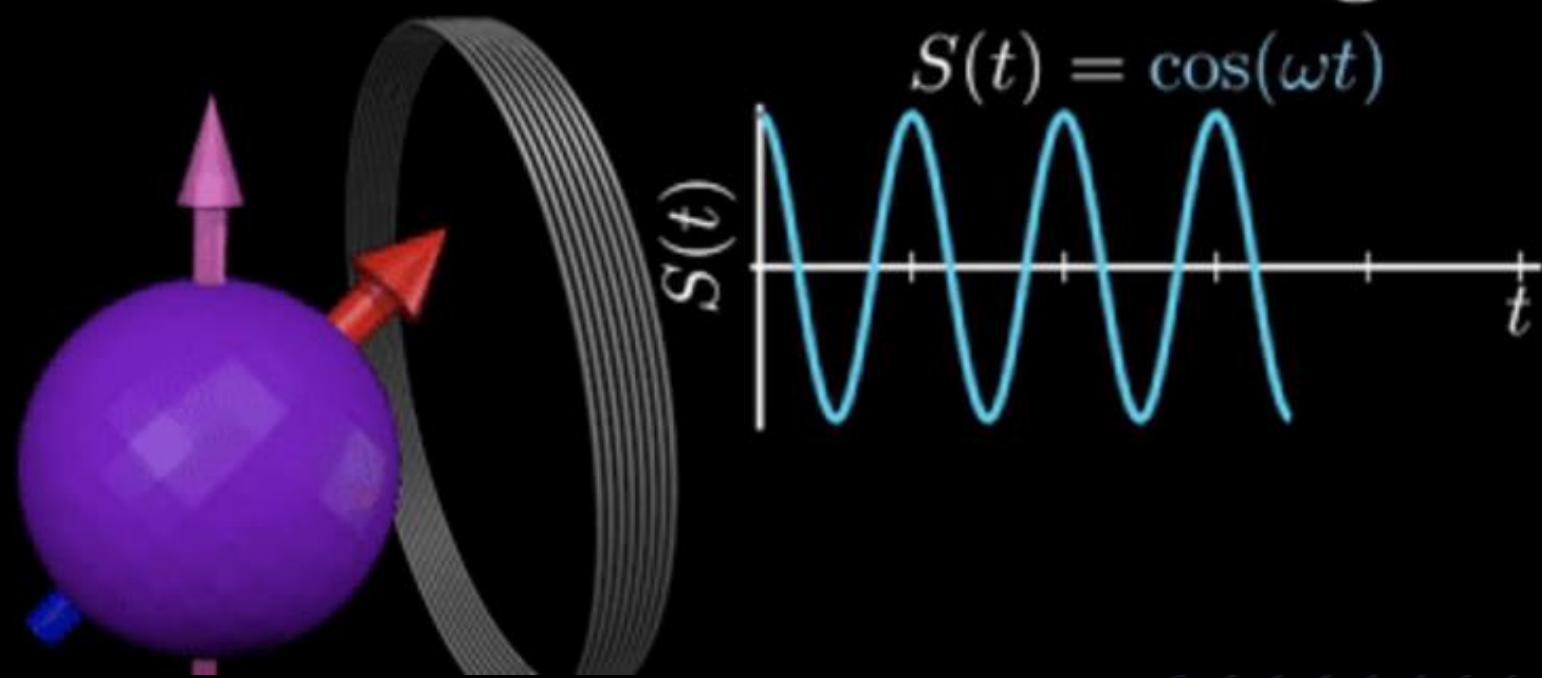


Quantum Workshop Roadmap



Detected Signal

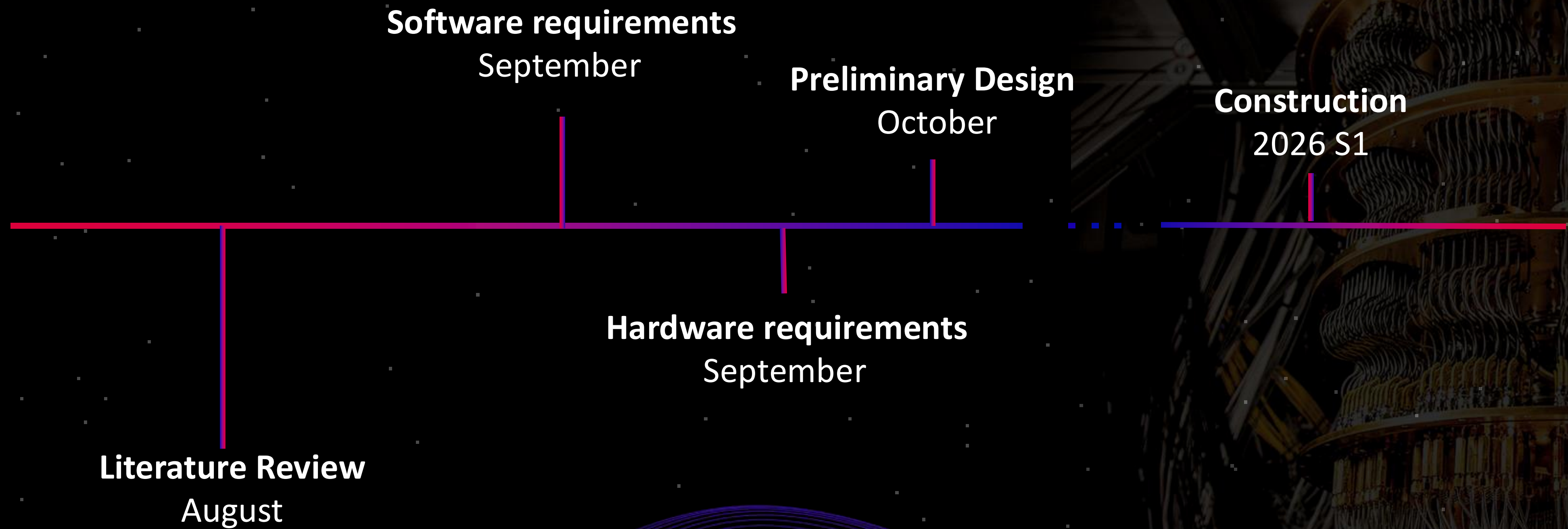
$$S(t) = \cos(\omega t)$$



Qubit

Qubit

TIMELINE

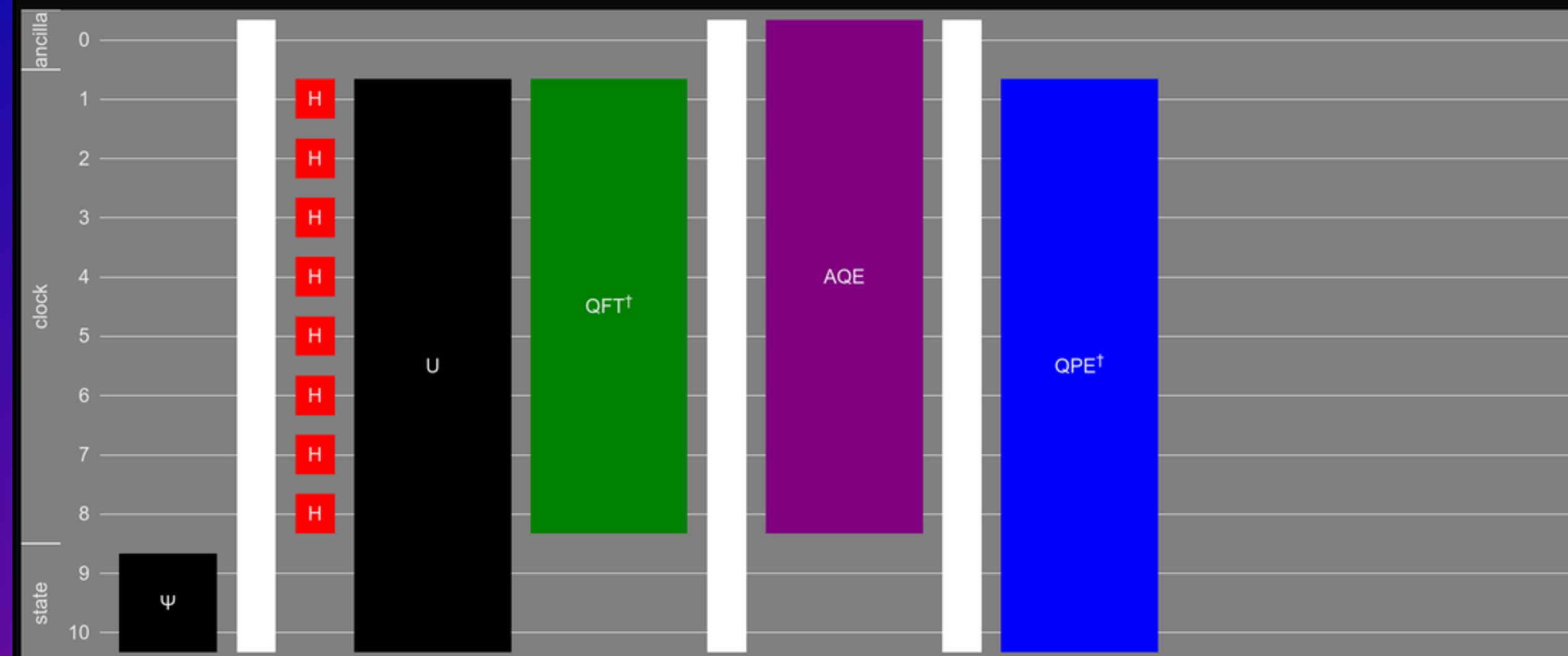


Quantum Circuit Education

Project Goals & Mission

- Make complex quantum circuits more accessible to everyone.
- Interactive quantum circuit view with collapsible subcircuits.
- References to key literature and helpful resources.
- Extra information to help with implementation.
- Mathematical explanations and proofs.

Harrow-Hassidim-Lloyd Algorithm



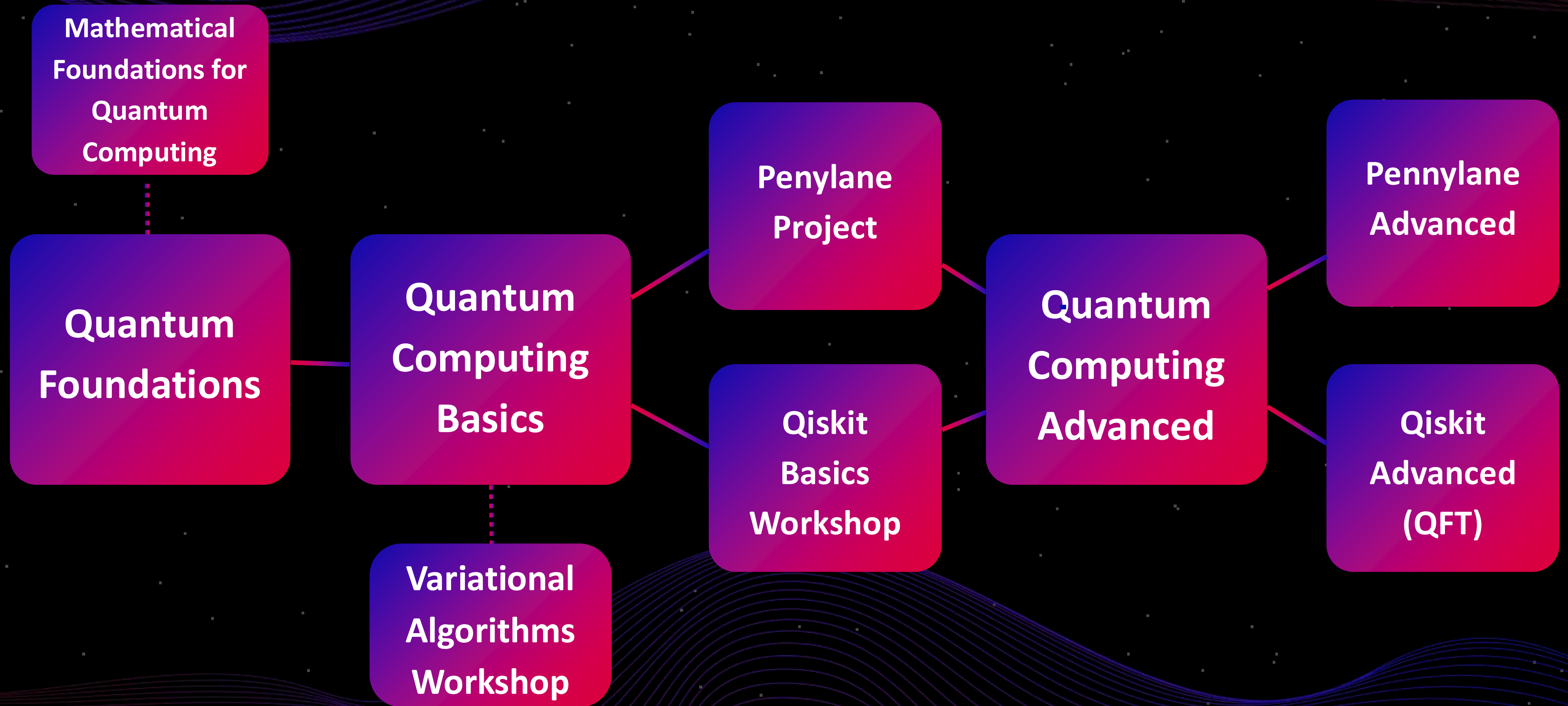
Algorithm Overview

The HHL algorithm is a quantum circuit that inverts a Hermitian matrix, A , and encodes the dot product of the inverted matrix and a vector $\vec{b}$ into a quantum state, hence finding the solution, $\vec{x}$, to the linear system of equations $A\vec{x} = \vec{b}$.

Circuit Explanation

The HHL circuit spans three quantum registers: the state register, with $n_b = \log_2 N$ qubits; the eigenvalue register, with n_l qubits; and an auxiliary register with a single qubit, also

Quantum Computing Workshop Roadmap



Thank you!

If you'd like further information or work with us:

- Come chat with us after the presentation
- Connect with us over our socials or send us an email at mnet@monash.edu



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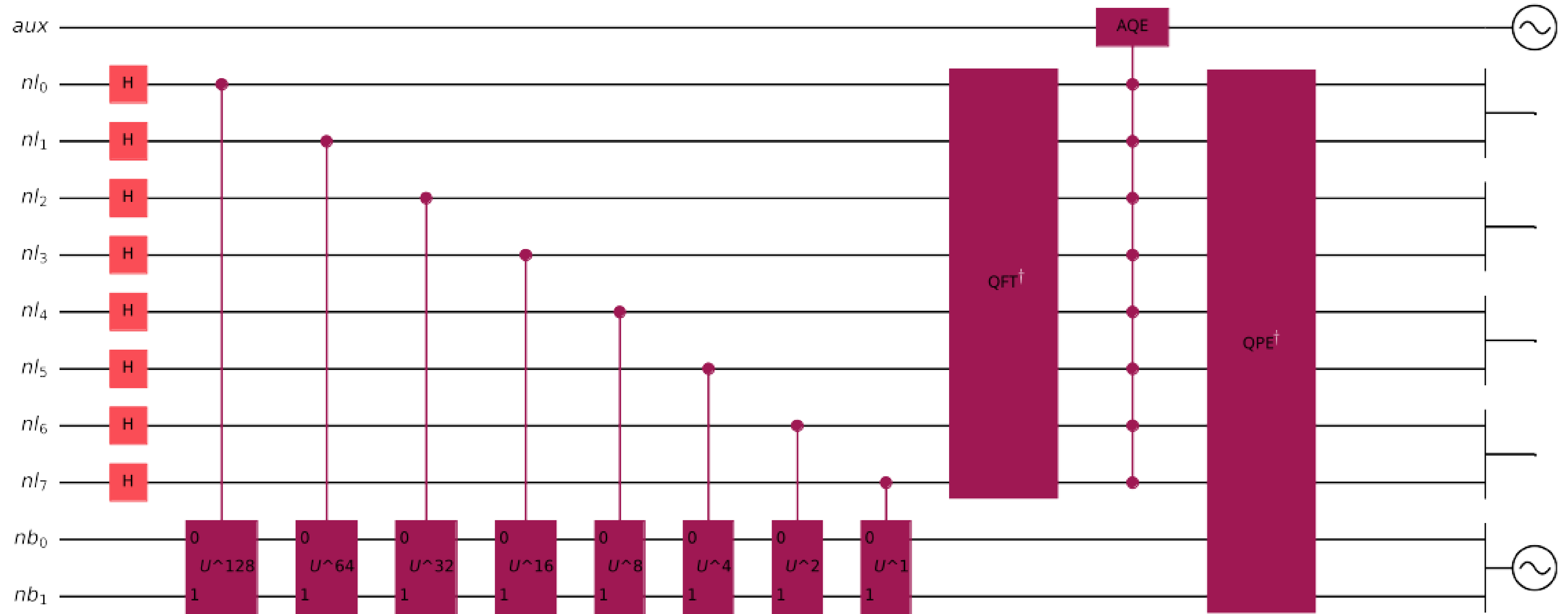


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Analysing the HHL Algorithm for Power Flow Analysis



My Quantum Journey

Followed Qiskit's [Understanding Quantum Information and Computation](#) series in 2023.



Took quantum physics units as my electives.

Chose to research [power flow with quantum computing](#) for my final year research project.

Evaluating the practicality of HHL in Fast Decoupled Power Flow

Supervised by Terrence Mak in the Data Science & AI research group.

The project is focused primarily on power systems, but much of the work was on quantum computing.

It was clear from the start that the few existing practical HHL implementations were not suitable for my purposes.

So I made my own.

The Harrow-Hassidim-Lloyd Algorithm

Quantum algorithm to solve systems of linear equations.

I found two useful implementations:

 README  Code of conduct  Contributing  Apache-2.0 license

Qiskit Aqua (*NOW DEPRECATED*)

license

Apache-2.0

 Aqua Unit Tests

no status

release

v0.9.5

downloads

2.6k/month

coverage

84%

PLEASE NOTE: As of version 0.9.0, released on 2nd April 2021, Qiskit Aqua has been deprecated with its support ending and eventual archival being no sooner than 3 months from that date. The function provided by Qiskit Aqua is not going away rather it has being split out to separate application repositories, with core algorithm and operator function moving to qiskit-terra. Please see the [Migration Guide](#) section below for more detail. We encourage you to migrate over at your earliest convenience.

 README

The HHL Quantum Algorithm Applied to Data Fitting

The HHL (Harrow-Hassidim-Lloyd) algorithm is a quantum algorithm that can solve linear systems of equations exponentially faster than classical algorithms. This repository has a sandbox for exploring the HHL algorithm as well an application of the HHL algorithm to a data-fitting problem.

Implementing HHL

I planned to read the [original paper](#) and use that knowledge to [fix the code](#).

PRL **103**, 150502 (2009)

PHYSICAL REVIEW LETTERS

week ending
9 OCTOBER 2009



Quantum Algorithm for Linear Systems of Equations

Aram W. Harrow,¹ Avinatan Hassidim,² and Seth Lloyd³

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²*Research Laboratory for Electronics, MIT, Cambridge, Massachusetts 02139, USA*

³*Research Laboratory for Electronics and Department of Mechanical Engineering, MIT, Cambridge, Massachusetts 02139, USA*
(Received 5 July 2009; published 7 October 2009)

Solving linear systems of equations is a common problem that arises both on its own and as a subroutine in more complex problems: given a matrix A and a vector $\vec{b}$, find a vector $\vec{x}$ such that $A\vec{x} = \vec{b}$. We consider the case where one does not need to know the solution $\vec{x}$ itself, but rather an approximation of the expectation value of some operator associated with $\vec{x}$, e.g., $\vec{x}^\dagger M \vec{x}$ for some matrix M . In this case, when A is sparse, $N \times N$ and has condition number κ , the fastest known classical algorithms can find $\vec{x}$ and estimate $\vec{x}^\dagger M \vec{x}$ in time scaling roughly as $N\sqrt{\kappa}$. Here, we exhibit a quantum algorithm for estimating $\vec{x}^\dagger M \vec{x}$ whose runtime is a polynomial of $\log(N)$ and κ . Indeed, for small values of κ [i.e., $\text{poly} \log(N)$], we prove (using some common complexity-theoretic assumptions) that any classical algorithm for this problem generically requires exponentially more time than our quantum algorithm.

DOI: 10.1103/PhysRevLett.103.150502

PACS numbers: 03.67.Ac, 02.10.Ud, 89.70.Eg

Introduction.—Quantum computers are devices that harness quantum mechanics to perform computations in ways that classical computers cannot. For certain problems, quantum algorithms supply exponential speedups over their classical counterparts, the most famous example being Shor’s factoring algorithm [1]. Few such exponential speedups are known, and those that are (such as the use of quantum computers to simulate other quantum systems [2]) have so far found limited use outside the domain of quantum mechanics. This Letter presents a quantum algorithm to estimate features of the solution of a set of linear equations. Compared to classical algorithms for the same task, our algorithm can be as much as exponentially faster. Linear equations play an important role in virtually all fields of science and engineering. The sizes of the data sets

However, the condition number often scales with the size of the problem, which presents a more serious limitation of our algorithm. Coping with large condition numbers has been studied extensively in the context of classical algorithms. In the discussion section, we will describe the applicability of some of the classical tools (pseudoinverses, preconditioners) to our quantum algorithm. We sketch here the basic idea of our algorithm and then discuss it in more detail in the next section. Given a Hermitian $N \times N$ matrix A , and a unit vector $\vec{b}$, suppose we would like to find $\vec{x}$ satisfying $A\vec{x} = \vec{b}$. (We discuss later questions of efficiency as well as how the assumptions we have made about A and $\vec{b}$ can be relaxed.) First, the algorithm represents $\vec{b}$ as a quantum state $|b\rangle = \sum_{i=1}^N b_i|i\rangle$. Next, we use techniques of Hamiltonian simulation [3,4] to



README

The HHL Quantum Algorithm Applied to Data Fitting

The HHL (Harrow-Hassidim-Lloyd) algorithm is a quantum algorithm that can solve linear systems of equations exponentially faster than classical algorithms. This repository has a sandbox for exploring the HHL algorithm as well as an application of the HHL algorithm to a data-fitting problem.

README Code of conduct Contributing Apache-2.0 license

Qiskit Aqua (NOW DEPRECATED)

license Apache-2.0 Aqua Unit Tests no status release v0.9.5 downloads 2.6k/month coverage 84%

PLEASE NOTE: As of version 0.9.0, released on 2nd April 2021, Qiskit Aqua has been deprecated with its support ending and eventual archival being no sooner than 3 months from that date. The function provided by Qiskit Aqua is not going away rather it has being split out to separate application repositories, with core algorithm and operator function moving to qiskit-terra. Please see the [Migration Guide](#) section below for more detail. We encourage you to migrate over at your earliest convenience.

Implementing HHL

Letter
An enhanced hybrid HHL algorithm ☆

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^b IBM Quantum, USA

ARTICLE INFO

Communicated by M.G.A. Paris

Keywords:

Solving linear systems on quantum hardware with hybrid HHL⁺⁺

Romina Yalovetzky[✉], Pierre Minssen, Dylan Herman & Marco Pistoia

The limited capabilities of current quantum hardware significantly constrain the scale of experimental demonstrations of most quantum algorithmic primitives. This makes it challenging to perform benchmarking of the current hardware using useful quantum algorithms, i.e., application-oriented benchmarking. In particular, the Harrow-Hassidim-Lloyd (HHL) algorithm is a critical quantum linear algebra primitive, but the majority of the components of HHL are far out of the reach of intermediate-scale quantum devices, which has led to the proposal of hybrid classical-quantum variants. The goal of this work is to further bridge the gap between proposed near-term friendly implementations of HHL and the kinds of quantum circuits that can be executed on noisy hardware. Our proposal adds to the existing literature of hybrid quantum algorithms for linear algebra. Our proposal is compatible with the current scale of quantum devices. Specifically, we propose two variants of the hybrid HHL algorithm proposed by Lee et al., leading to our algorithm Hybrid HHL. The first variant introduces a scaling factor for the phase estimation component, which reduces the error of Hybrid HHL by an average of 20 percent. On IBM Torino architecture, this reduces the error of Hybrid HHL by an average of 20 percent.

ABSTRACT

We present a classical enhancement to improve the accuracy of the Hybrid HHL algorithm for solving linear systems of equations proposed by Harrow et al. This is achieved by using higher precision quantum estimates of the eigenvalues of the matrix, which is guided by a classical step to guide the eigenvector estimation. The improved algorithm just two extra bits of precision result in tighter error bounds for the eigenvector estimation. On IBM Torino architecture, this reduces the error of Hybrid HHL by an average of 20 percent.

A Step-by-Step HHL Algorithm Walkthrough to Enhance Understanding of Critical Quantum Computing Concepts

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This work was supported by the National Science Foundation under Grant 2046220.

Improved circuit implementation of the HHL algorithm and its simulations on QISKIT

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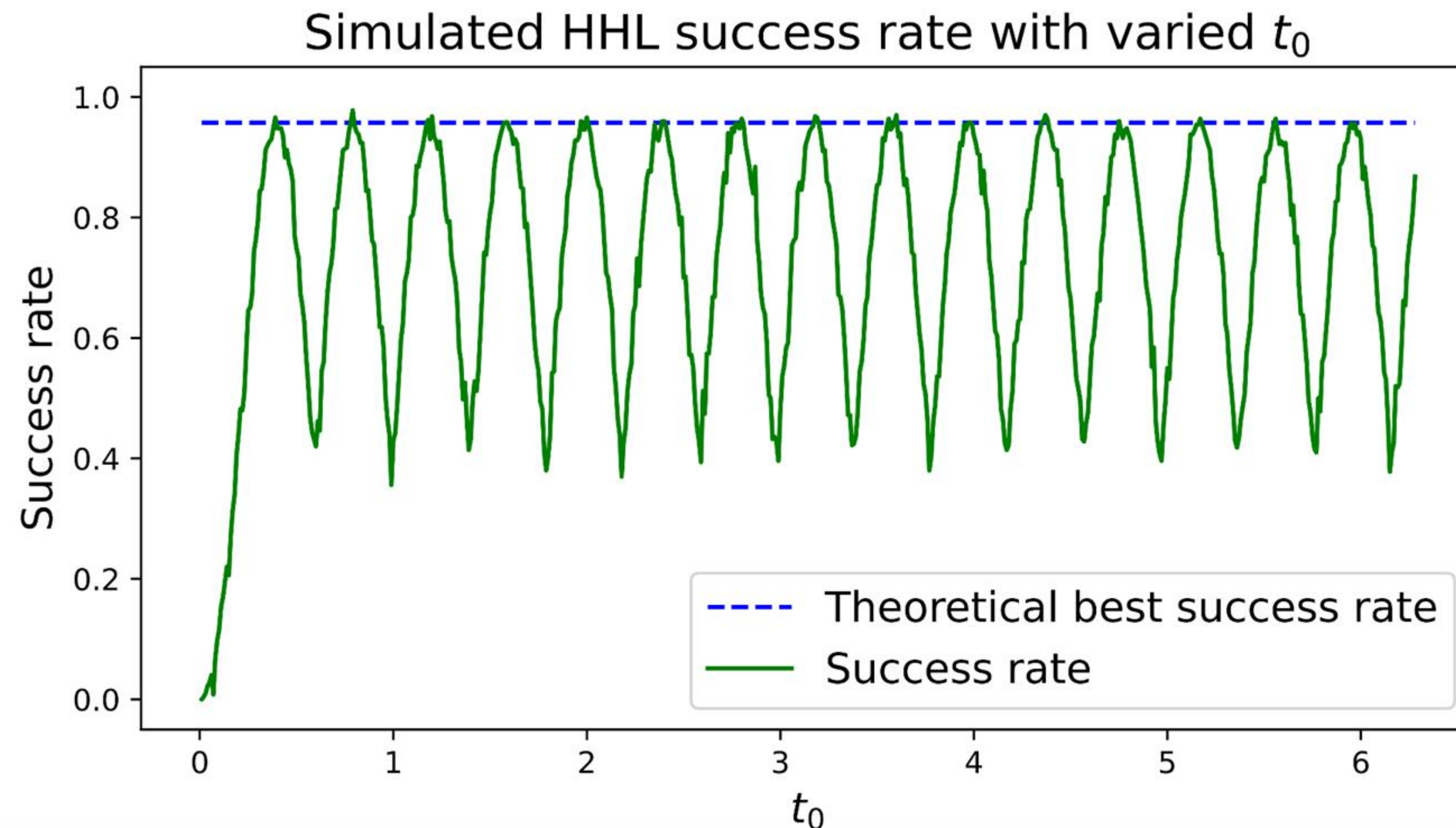
In 2019, Yonghae Lee et al. combined the circuit implementation of the Harrow-Hassidim-Lloyd (HHL) algorithm with a classical computer, and designed a hybrid HHL algorithm to reduce experimental errors caused by decoherence and so on. However, the improvement is achieved only in the auxiliary quantum coding phase, and no quantum resource reduction is done on the quantum phase estimation and inverse quantum phase estimation stages. At the same time, the circuit improvement illustration of the hybrid HHL algorithm just has the result and no specific process. In this paper, based on the idea of the HHL algorithm is proposed. The feasibility of the improved circuit implementation of the HHL algorithm is verified by IBM's qiskit. The improved circuit illustration on a 4-qubit system shows that the improved circuit implementation of the HHL algorithm can further avoid some result errors. Thus the improved circuit implementation of the HHL algorithm can further avoid some result errors.

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What's going on with t_0 ?

Since I couldn't find a clear answer on what value of t_0 to use, I plotted the performance of my HHL implementation against t_0 .



Figuring out the Mathematics

To find a good value of t_0 , I needed to be able to predict the success rate,

so I developed a **mathematical model**.

We know how the phases are rounded:

$$\phi_j = \frac{1}{2^{n_l}} \left\lfloor \frac{2^{n_l} \lambda_j t_0}{2\pi} + \frac{1}{2} \right\rfloor \pmod{1}$$

We know the theoretical success rate:

$$P_{\text{succ}} = C^2 \sum_{j=1}^N \frac{|\beta_j|^2}{\lambda_j^2}$$

$$P_{\text{pred-succ}} = C^2 \sum_{j=1}^N \frac{|\beta_j|^2}{\lambda_j^2} \left| \cos \left(\frac{2^{n_l} |\lambda_j| t_0}{2} \right) \right|$$

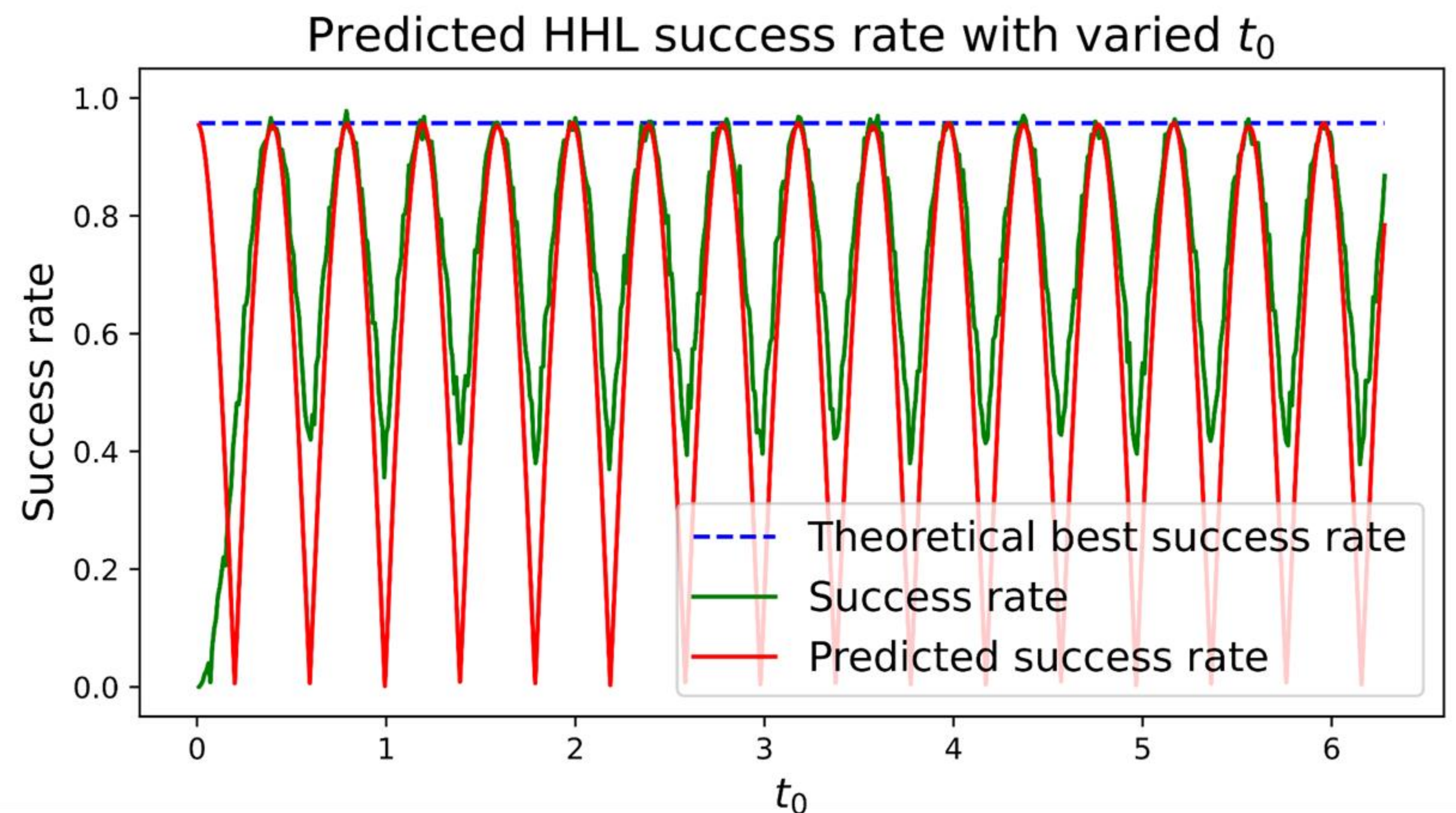
We can hence approximate the actual success rate

Finding Fun in the Challenges

Quantum computation has **huge potential**, but also **many challenges** yet to overcome.

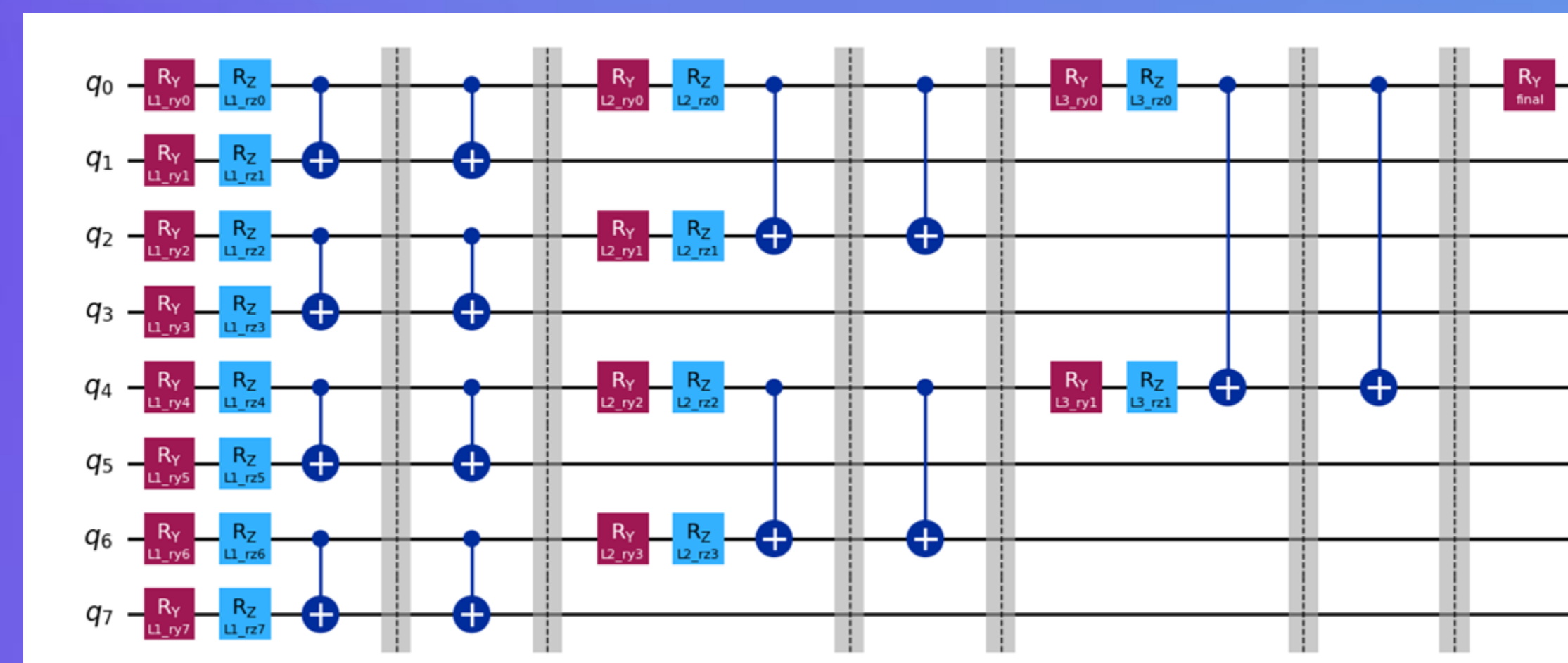
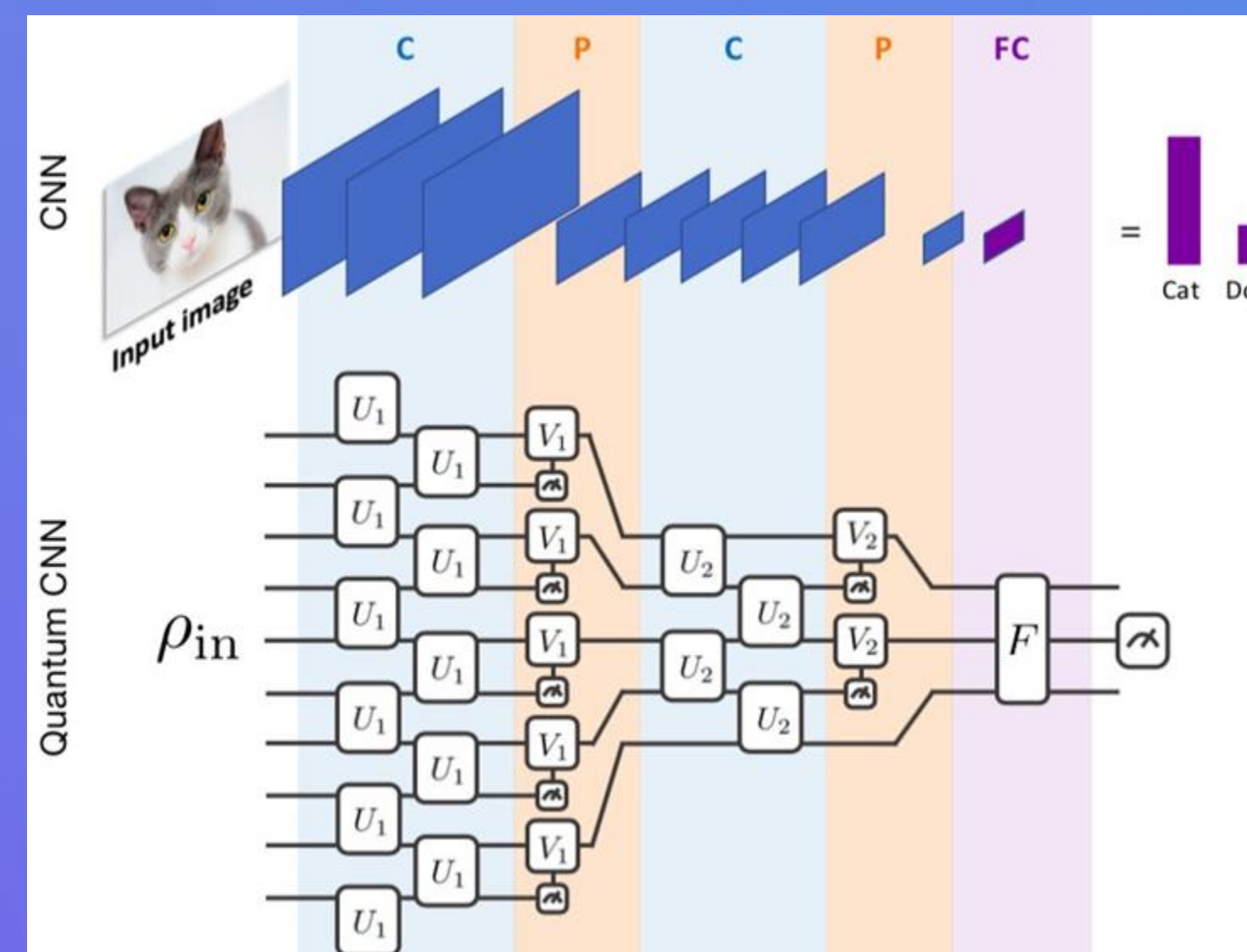
It takes people like you and me to solve them, sometimes in a big project, and sometimes in a day.

The best way to solve these hard problems is to find something fun, and **just do it**.



Quantum CNNs

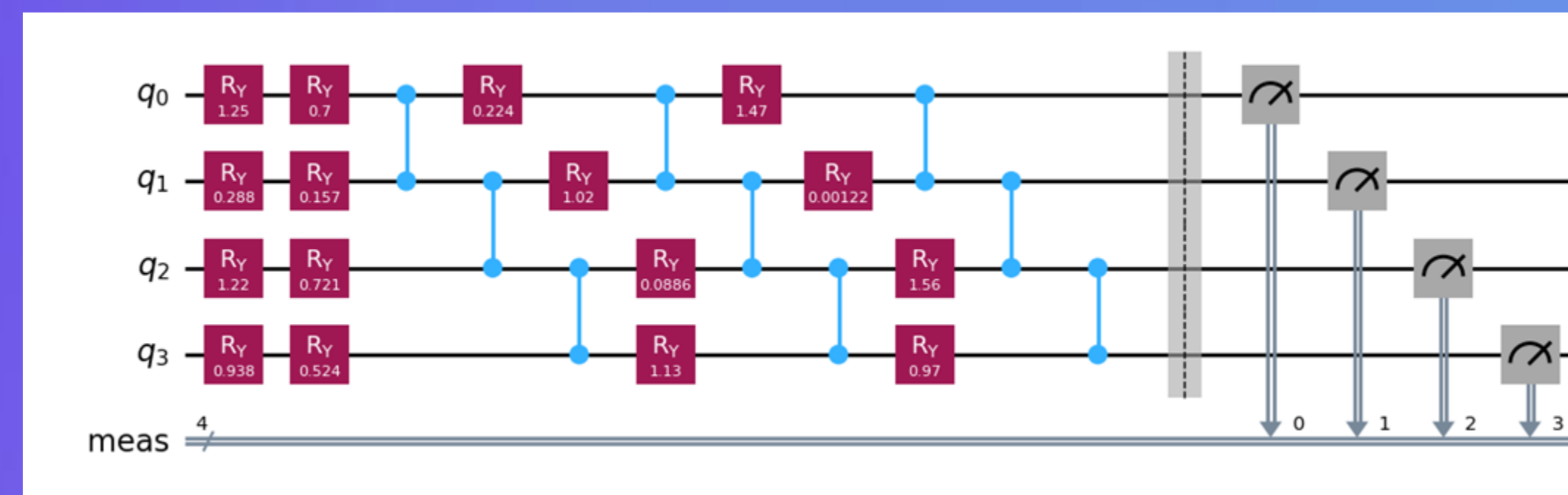
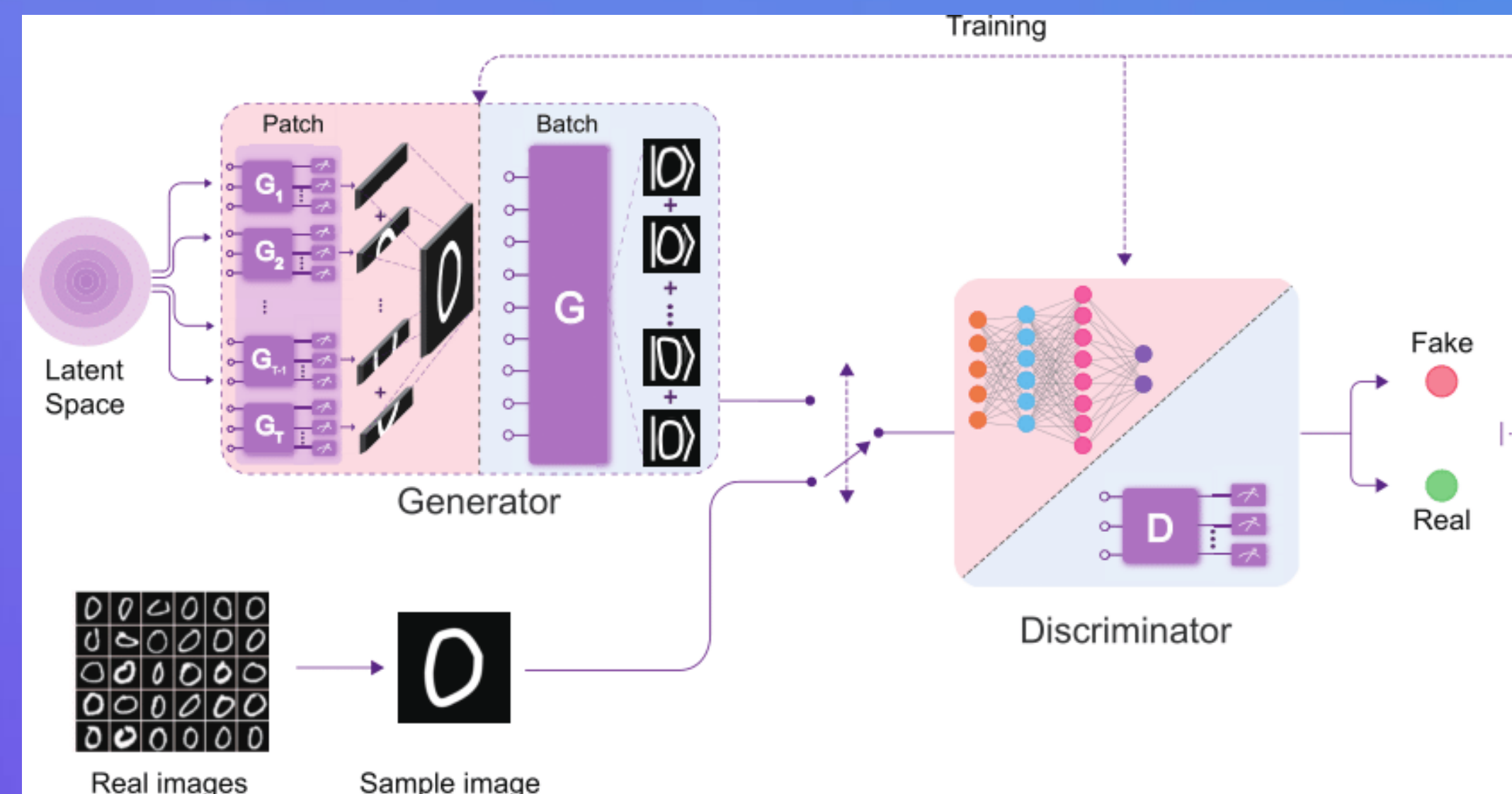
- Extend classical Convolutional Neural Networks with quantum circuits.
- Replace convolution layers with parameterized quantum gates.
- Potential exponential speedup in feature extraction for high-dimensional data.



QCNN circuit diagram using Qiskit

Quantum GANs

- Combine Generative Adversarial Networks with quantum generators/discriminators.
- Quantum circuits act as generator or discriminator.
- Generate complex quantum or classical data distributions more efficiently.



QGAN circuit diagram using Qiskit

Q&A



Check in!



Mini competition!

Thank you for joining
us today!



Quantum Australia

